

Hydrostatics

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Disclaimer

Discussions taken from Alonso Sepúlveda [1], Landau & Lifschits [2], Lautrup [3], Potter et. al [4], White [5], Clarke-Carswell [6], Shore [7] books.

1 Preliminaries

Fluids are microscopically composed of discrete particles, but hydrodynamics treats them as continuous media by averaging over sufficiently large numbers of particles so that quantities such as density, pressure, and velocity can be defined locally. The same coarse-grained idea can also be applied to systems such as stellar populations in galaxies, where the discrete constituents are stars rather than molecules. In this course, the main focus is on ordinary fluids.

1.1 Material particles

In continuum physics, matter is described in terms of *material particles*. A material particle contains a large number of molecules, but is regarded as infinitesimal or point-like at the macroscopic scale. More precisely, it represents the smallest body that can consistently be treated as part of the continuum within the required precision. The continuum description therefore does not apply at scales below that of the material particle.

The microscopic interpretation of a material particle depends on the state of matter. In a solid, it may approximately be regarded as containing a fixed collection of molecules. In liquids and gases, however, molecules continuously enter and leave a given material particle. If the molecular composition varies spatially, this molecular exchange produces *diffusion*, which tends to transport material between neighboring regions. Similarly, spatial variations of velocity lead to *momentum diffusion*, which manifests macroscopically as viscous friction.

1.1.1 The concept of a fluid element

The equations of continuum mechanics are based on the concept of a *fluid element*: a small region of matter over which macroscopic quantities such as density, pressure, temperature, and velocity can be defined locally. Although treated as infinitesimal in the continuum description, a fluid element must contain a sufficiently large number of microscopic particles.

Its characteristic size ℓ must satisfy two competing requirements. First, it must be much smaller than the macroscopic scale over which the fluid properties vary. For a quantity q , this scale can be estimated as

$$\ell_{\text{scale}} \sim \frac{q}{|\vec{\nabla} q|}, \quad (1)$$

and therefore

$$\ell \ll \ell_{\text{scale}}. \quad (2)$$

This allows the macroscopic variables to be regarded as approximately uniform within the fluid element.

Second, the element must contain a sufficiently large number of particles for statistical fluctuations to be negligible. If n is the particle number density, this requires

$$n\ell^3 \gg 1. \quad (3)$$

Thus, the fluid element must be microscopically large while remaining macroscopically small.

For a *collisional fluid*, an additional condition is required. The constituent particles must interact frequently enough to establish local thermodynamic equilibrium. If λ is the mean free path, this requires

$$\lambda \ll \ell. \quad (4)$$

Combining these conditions gives the hierarchy of scales

$$\boxed{\lambda \ll \ell \ll \ell_{\text{scale}}} \quad (5)$$

for a conventional collisional fluid, together with the requirement $n\ell^3 \gg 1$.

Frequent microscopic interactions establish a local equilibrium distribution of particle velocities, allowing thermodynamic quantities such as temperature and pressure to be defined and an equation of state

$$p = p(\rho, T) \quad (6)$$

to be introduced. Systems such as ordinary gases and liquids generally satisfy this condition.

Some astrophysical systems, such as stars orbiting within a galaxy, may be treated using continuum-like descriptions even though they are effectively collisionless. In such cases, however, local thermodynamic equilibrium and a conventional equation of state cannot in general be assumed.

The fluid element is therefore a conceptual coarse-graining scale rather than a physical object of fixed size. A continuum description is possible whenever a range of length scales exists for which the microscopic fluctuations can be averaged out while the macroscopic variations remain negligible.

1.2 Eulerian and Lagrangian descriptions

There are two complementary approaches to describing the evolution of a fluid: the *Eulerian* and *Lagrangian* descriptions.

In the *Eulerian description*, the fluid is observed at fixed positions in space. Physical quantities are therefore represented by fields such as

$$\rho = \rho(\vec{r}, t), \quad T = T(\vec{r}, t), \quad \vec{v} = \vec{v}(\vec{r}, t). \quad (7)$$

The partial derivative

$$\frac{\partial q}{\partial t} \quad (8)$$

measures the rate of change of a quantity q at a fixed position. A flow is said to be *steady* when all Eulerian quantities are independent of time, so that

$$\frac{\partial q}{\partial t} = 0. \quad (9)$$

In the *Lagrangian description*, instead, one follows an individual fluid element as it moves. Each element may be identified by a label $\vec{a}$, for example its position at some initial time, while its subsequent position is written as

$$\vec{r} = \vec{r}(\vec{a}, t). \quad (10)$$

Physical quantities are then followed along the trajectory of the element, and their rate of change is denoted by

$$\frac{Dq}{Dt}. \quad (11)$$

The distinction can be illustrated by considering a river. The Eulerian description corresponds to observing the flow from a fixed point on the bank, whereas the Lagrangian description corresponds to following the flow from a boat drifting with the water.

The two descriptions are related by the *material derivative*,

$$\frac{Dq}{Dt} = \frac{\partial q}{\partial t} + \vec{v} \cdot \vec{\nabla} q. \quad (12)$$

Thus, even in a steady flow for which

$$\frac{\partial q}{\partial t} = 0, \quad (13)$$

a fluid element may experience changes in its properties as it moves through spatial gradients,

$$\frac{Dq}{Dt} = \vec{v} \cdot \vec{\nabla} q \neq 0. \quad (14)$$

Both viewpoints are also widely used in numerical fluid dynamics. *Eulerian methods* describe the fluid on a spatial grid and calculate how the fluid variables evolve within each cell. Modern implementations often use *Adaptive Mesh Refinement* (AMR) to increase the spatial resolution where needed. In contrast, *Lagrangian methods* follow discrete representative fluid elements. An important example in astrophysics is *Smoothed Particle Hydrodynamics* (SPH), in which the fluid is represented by particles whose trajectories are followed in time.

The choice between the two descriptions depends on the physical problem. Eulerian methods are particularly natural when the spatial structure of the flow is of interest, whereas Lagrangian methods are useful when the history and trajectory of individual fluid elements are important.

1.3 Fluids in the Universe

A large fraction of the baryonic matter in the Universe is found in the form of fluids. Stars themselves are gaseous bodies, composed mainly of hydrogen and helium, with temperatures ranging from thousands of kelvin near their surfaces to millions of kelvin in their interiors. To a first approximation, a star can be regarded as a static fluid in which the inward gravitational force is balanced by an outward pressure gradient. In reality, stellar interiors also exhibit fluid motions such as convection and oscillations.

Gas is also distributed throughout the interstellar medium (ISM) and the intergalactic medium (IGM). The ISM is a highly inhomogeneous, multiphase medium, with temperatures ranging approximately from 10 K in cold molecular clouds to 10^6 K in hot gas heated by processes such as supernova explosions. Its density varies by many orders of magnitude, from very dilute hot gas to the dense cores of molecular clouds. At sufficiently low densities, however, the validity of a fluid description must be carefully examined.

Other important examples of astrophysical fluids include stellar winds, jets, accretion disks, planetary interiors, white dwarfs, and neutron stars. These systems span enormous ranges of

density, temperature, pressure, and length scale, and the microscopic properties of their constituent matter can differ substantially.

One of the remarkable features of fluid dynamics is that much of this microscopic complexity can be incorporated through an *equation of state*, which relates thermodynamic quantities such as pressure, density, and temperature,

$$p = p(\rho, T). \quad (15)$$

Once an appropriate equation of state is specified, the same general equations of continuum mechanics can be applied to systems as different as ordinary stars, giant planets, white dwarfs, and neutron stars. This universality is one of the reasons why continuum mechanics is such a powerful tool in astrophysics.

1.4 Density

Fluid is the generic name for liquids and gases. A gas completely fills a closed container, but a liquid does not. The (matter) density of a fluid is defined as

$$\rho \equiv \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V}. \quad (16)$$

In continuum physics, density is understood as a macroscopic average over a sufficiently large number of microscopic constituents. This requires a separation of scales: the microscopic length scale (molecular spacing or mean free path) must be much smaller than the macroscopic scale on which density varies appreciably.

Table 1: Densities of some common substances

Material	Density (kg/m ³)	Material	Density (kg/m ³)
Air (1 atm, 20°C)	1.20	Iron ($Z = 26$), steel	7.8×10^3
Ethanol	0.81×10^3	Brass	8.6×10^3
Benzene	0.90×10^3	Copper ($Z = 29$)	8.9×10^3
Ice	0.92×10^3	Silver ($Z = 47$)	10.5×10^3
Water	1.00×10^3	Lead ($Z = 82$)	11.3×10^3
Seawater	1.03×10^3	Mercury ($Z = 80$)	13.6×10^3
Blood	1.06×10^3	Gold ($Z = 79$)	19.3×10^3
Glycerin	1.26×10^3	Platinum ($Z = 78$)	21.4×10^3
Concrete	2.0×10^3	Iridium ($Z = 77$)	22.56×10^3
Aluminum	2.7×10^3	Osmium ($Z = 76$)	22.59×10^3

1.4.1 Density in astrophysical environments

Astrophysical systems span an extraordinary range of densities. The Sun, for example, has a mean density of approximately $1.4 \times 10^3 \text{ kg m}^{-3}$, comparable to that of water, while its central density reaches about $1.5 \times 10^5 \text{ kg m}^{-3}$. In stellar structure, the density profile $\rho(r)$ is one of the fundamental quantities entering the condition of hydrostatic equilibrium.

At much higher densities, compact stars provide extreme examples of continuous media. White dwarfs typically reach densities of order 10^9 – $10^{10} \text{ kg m}^{-3}$, while neutron-star matter can reach

densities of order $10^{18} \text{ kg m}^{-3}$. In these objects the microscopic physics is very different from that of an ordinary gas, but its effects can still be incorporated macroscopically through an appropriate equation of state.

At the opposite extreme, the interstellar medium is extraordinarily dilute by terrestrial standards. Its density varies by many orders of magnitude between hot ionized gas and dense molecular clouds. On still larger scales, the mean density of the Universe is characterized by the critical density,

$$\rho_c = \frac{3H_0^2}{8\pi G}, \quad (17)$$

which is of order $10^{-26} \text{ kg m}^{-3}$ for the present value of the Hubble constant.

Table 2: Characteristic densities of some terrestrial and astrophysical systems.

System	Density (kg/m^3)
Air (1 atm, 20°C)	~ 1
Water	1.0×10^3
Mean density of the Sun	1.4×10^3
Solar core	1.5×10^5
White dwarf	$10^9\text{--}10^{10}$
Neutron star	$\sim 10^{18}$
Interstellar medium	$\sim 10^{-21}\text{--}10^{-18}$
Critical density of the Universe	$\sim 10^{-26}$

These examples illustrate an important feature of continuum mechanics: systems with enormously different densities and microscopic compositions can nevertheless be described using the same macroscopic fields and conservation laws, provided that an appropriate equation of state is specified.

1.5 Pressure

Let us consider the forces acting between neighboring fluid elements. In a collisional fluid, these internal forces are closely related to the microscopic motion and interactions of its constituent particles.

At finite temperature, molecules undergo random motion with respect to the local rest frame of the fluid. When they cross or interact across an imaginary surface, they transport momentum. The *pressure* can therefore be interpreted microscopically as the momentum flux associated with these random motions.

In thermal equilibrium, the random microscopic motions of the particles are isotropic, so that there is no preferred spatial direction. Consequently, the force associated with pressure acts normal to any surface and has the same magnitude per unit area for all surface orientations at a given point. Thus, pressure is a scalar field,

$$p = p(\vec{r}, t), \quad (18)$$

and the pressure force acting on an oriented surface element $d\vec{S} = \hat{s} dS$ is

$$d\vec{F}_p = -p d\vec{S}. \quad (19)$$

In a more general continuum, the force per unit area acting across a surface need not be perpendicular to it. The surface force is then described by the *stress tensor* σ_{ij} . With the convention that positive normal stress represents tension, the isotropic stress associated with pressure is

$$\sigma_{ij}^{(p)} = -p \delta_{ij}. \quad (20)$$

More generally, the stress tensor of a fluid can be written as

$$\sigma_{ij} = -p \delta_{ij} + \tau_{ij}, \quad (21)$$

where τ_{ij} represents non-isotropic stresses, such as those associated with viscosity.

1.6 Conservation of mass

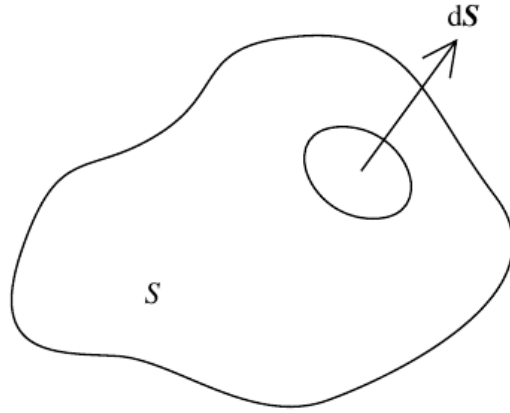


Figure 1: The surface around a fixed volume. Figure taken from Ref. [6].

Consider a fixed volume V bounded by a closed surface S (see Fig. 1). If ρ is the mass density, the total mass contained within the volume is

$$M = \int_V \rho dV, \quad (22)$$

and its rate of change is

$$\frac{dM}{dt} = \frac{\partial}{\partial t} \int_V \rho dV. \quad (23)$$

The mass flux through an infinitesimal surface element $d\vec{S}$ is $\rho \vec{v} \cdot d\vec{S}$. Since $d\vec{S}$ points outward, the net rate at which mass enters the volume is therefore

$$- \int_S \rho \vec{v} \cdot d\vec{S}. \quad (24)$$

In the absence of sources or sinks, conservation of mass requires

$$\frac{\partial}{\partial t} \int_V \rho dV = - \int_S \rho \vec{v} \cdot d\vec{S}. \quad (25)$$

Using the divergence theorem,

$$\int_S \rho \vec{v} \cdot d\vec{S} = \int_V \vec{\nabla} \cdot (\rho \vec{v}) dV, \quad (26)$$

we obtain

$$\int_V \left[\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) \right] dV = 0. \quad (27)$$

Since this relation must hold for any arbitrary volume V , the integrand must vanish. Hence, the local form of mass conservation is

$$\boxed{\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = 0}. \quad (28)$$

This is the *continuity equation* in Eulerian form.

Using

$$\vec{\nabla} \cdot (\rho \vec{v}) = \rho \vec{\nabla} \cdot \vec{v} + \vec{v} \cdot \vec{\nabla} \rho \quad (29)$$

and the definition of the material derivative,

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + \vec{v} \cdot \vec{\nabla} \rho, \quad (30)$$

the continuity equation can equivalently be written in Lagrangian form as

$$\boxed{\frac{D\rho}{Dt} + \rho \vec{\nabla} \cdot \vec{v} = 0}. \quad (31)$$

Incompressible flow

An incompressible flow is one in which each fluid element preserves its density as it moves:

$$\frac{D\rho}{Dt} = 0. \quad (32)$$

The continuity equation then implies

$$\boxed{\vec{\nabla} \cdot \vec{v} = 0}. \quad (33)$$

Thus, an incompressible velocity field is divergence-free. Notice that incompressibility does not necessarily require the density to have the same value everywhere; it requires the density of each fluid element to remain constant along its trajectory.

1.7 Momentum equation

We now derive the equation governing the motion of a fluid subject to pressure and gravitational forces (see Fig. 2). Consider a material volume V of fluid with density ρ and velocity $\vec{v}$. Newton's second law states that the rate of change of its momentum is equal to the total force acting on it.

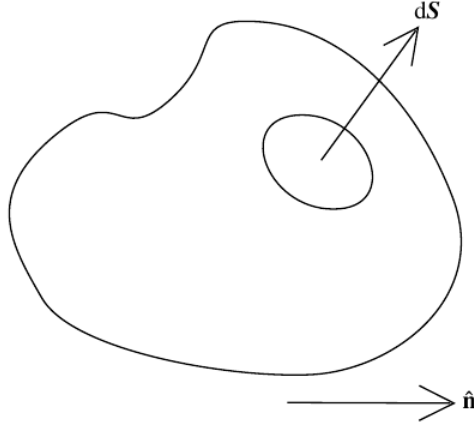


Figure 2: A fluid volume. Figure taken from Ref. [6].

The pressure exerted by the surrounding fluid on an infinitesimal surface element $d\vec{S}$ is directed inward and is therefore

$$d\vec{F}_p = -p d\vec{S}. \quad (34)$$

The total pressure force acting on the fluid volume is then

$$\begin{aligned} \vec{F}_p &= - \int_S p d\vec{S} \\ &= - \int_V \vec{\nabla} p dV, \end{aligned} \quad (35)$$

where the divergence theorem has been used. Thus, a spatially uniform pressure produces no net force; it is the *pressure gradient* that accelerates the fluid.

If $\vec{g}$ denotes the local gravitational acceleration, the gravitational force acting on the volume is

$$\vec{F}_g = \int_V \rho \vec{g} dV. \quad (36)$$

Newton's second law applied to the material volume therefore gives

$$\frac{D}{Dt} \int_V \rho \vec{v} dV = - \int_V \vec{\nabla} p dV + \int_V \rho \vec{g} dV. \quad (37)$$

For an infinitesimal material element of volume δV , conservation of mass implies

$$\frac{D}{Dt} (\rho \delta V) = 0. \quad (38)$$

Consequently,

$$\frac{D}{Dt} (\rho \vec{v} \delta V) = \rho \delta V \frac{D\vec{v}}{Dt}. \quad (39)$$

Newton's second law then becomes

$$\rho \delta V \frac{D\vec{v}}{Dt} = -\delta V \vec{\nabla} p + \rho \delta V \vec{g}. \quad (40)$$

Dividing by δV gives the momentum equation in Lagrangian form,

$$\boxed{\rho \frac{D\vec{v}}{Dt} = -\vec{\nabla} p + \rho \vec{g}}. \quad (41)$$

Using the material derivative,

$$\frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{v}, \quad (42)$$

the momentum equation can equivalently be written in Eulerian form as

$$\boxed{\rho \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{v} \right] = -\vec{\nabla} p + \rho \vec{g}}. \quad (43)$$

The two terms on the left-hand side have distinct physical meanings: $\partial \vec{v} / \partial t$ describes the local change of velocity at a fixed position, whereas $(\vec{v} \cdot \vec{\nabla}) \vec{v}$ is the *convective acceleration*, arising because a fluid element moves through regions with different velocities.

The momentum equation therefore states that a fluid element accelerates in response to pressure gradients and body forces such as gravity. In the absence of viscous stresses, this equation is the *Euler equation* of fluid dynamics.

1.8 Energy equation

The third fundamental conservation law governing a fluid is the conservation of energy. In addition to its macroscopic kinetic energy, a fluid possesses internal energy associated with the microscopic motion and interactions of its constituent particles.

Let e denote the specific internal energy, that is, the internal energy per unit mass. For an inviscid fluid (for which viscous stresses are neglected), the first law of thermodynamics applied to a fluid element may be written as

$$\rho \frac{De}{Dt} = -p \vec{\nabla} \cdot \vec{v} + \rho \dot{q}, \quad (44)$$

where $\dot{q}$ represents the net heating rate per unit mass due to processes other than mechanical compression or expansion.

The term

$$-p \vec{\nabla} \cdot \vec{v} \quad (45)$$

describes the mechanical work associated with compression or expansion of the fluid. If

$$\vec{\nabla} \cdot \vec{v} < 0, \quad (46)$$

the fluid element is compressed and its internal energy increases. Conversely, if

$$\vec{\nabla} \cdot \vec{v} > 0, \quad (47)$$

the fluid expands and performs work on its surroundings, thereby reducing its internal energy.

For an adiabatic fluid element, there is no heat exchange with its surroundings, so that

$$\dot{q} = 0. \quad (48)$$

The energy equation then reduces to

$$\rho \frac{De}{Dt} = -p \vec{\nabla} \cdot \vec{v}. \quad (49)$$

Using the continuity equation,

$$\frac{D\rho}{Dt} = -\rho \vec{\nabla} \cdot \vec{v}, \quad (50)$$

the adiabatic energy equation can equivalently be written as

$$\frac{De}{Dt} = \frac{p}{\rho^2} \frac{D\rho}{Dt}. \quad (51)$$

This relation connects the thermodynamic evolution of a fluid element with its compression or expansion.

Together with the mass and momentum equations, the energy equation forms the basic dynamical system of fluid mechanics. A constitutive relation, or equation of state, is also required in order to relate the thermodynamic variables. The simplest closed system may therefore be summarized as

$$\frac{D\rho}{Dt} + \rho \vec{\nabla} \cdot \vec{v} = 0, \quad \text{mass conservation,} \quad (52)$$

$$\rho \frac{D\vec{v}}{Dt} = -\vec{\nabla} p + \rho \vec{g}, \quad \text{momentum conservation,} \quad (53)$$

$$\rho \frac{De}{Dt} = -p \vec{\nabla} \cdot \vec{v} + \rho \dot{q}, \quad \text{energy conservation,} \quad (54)$$

$$p = p(\rho, e), \quad \text{equation of state.} \quad (55)$$

The full derivation of the energy equation, as well as the inclusion of additional transport processes such as thermal conduction, viscous dissipation, and radiative heating or cooling, will be postponed until later in the course.

2 Hydrostatic equilibrium

A fluid is in hydrostatic equilibrium when it is at rest,

$$\vec{v} = 0. \quad (56)$$

The acceleration of every fluid element therefore vanishes, and the forces acting on it must balance.

Forces acting on a fluid element can be classified into two categories. *Body forces* act throughout the volume of the fluid. Gravity is the most important example and, for a fluid element of volume dV , is given by

$$d\vec{\mathcal{F}}_g = \rho \vec{g} dV. \quad (57)$$

Thus, body forces scale with the volume of the fluid element.

In contrast, *surface forces* arise from the interaction of a fluid element with the surrounding matter and act across its boundary. Pressure is the simplest example. The pressure force acting on a surface element $d\vec{S}$ is

$$d\vec{\mathcal{F}}_p = -p d\vec{S}. \quad (58)$$

Surface forces therefore scale with the area of the fluid element.

2.1 The pressure-gradient force

As discussed in the preliminaries, the pressure force exerted on an oriented surface element $d\vec{S}$ is

$$d\vec{\mathcal{F}}_p = -p d\vec{S}, \quad (59)$$

where

$$d\vec{S} = \hat{n} dS \quad (60)$$

and, for a closed surface, $\hat{n}$ is conventionally chosen to point outward. Although pressure acts normally on each surface element, a spatially varying pressure produces a net force on a finite fluid element.

The total pressure force acting on a surface S is therefore

$$\vec{\mathcal{F}}_p = - \int_S p d\vec{S}. \quad (61)$$

A material particle in the fluid is like any other body subject to pressure forces from all sides, but being infinitesimal it is possible to derive a general expression for the resultant force. Let us consider a material particle in the shape of a small rectangular box with sides dx , dy , and dz , such that it has a volume $dV = dx dy dz$. Since the pressure is slightly different on opposite sides of the box, the z -component of total pressure force becomes

$$d\mathcal{F}_z = (p(x, y, z) - p(x, y, z + dz)) dx dy \approx -\frac{\partial p(x, y, z)}{\partial z} dx dy dz. \quad (62)$$

Doing the same for the other coordinate directions we obtain the total pressure force

$$d\vec{\mathcal{F}} = - \left(\hat{i} \frac{\partial p}{\partial x} + \hat{j} \frac{\partial p}{\partial y} + \hat{k} \frac{\partial p}{\partial z} \right) dV = -\vec{\nabla} p dV. \quad (63)$$

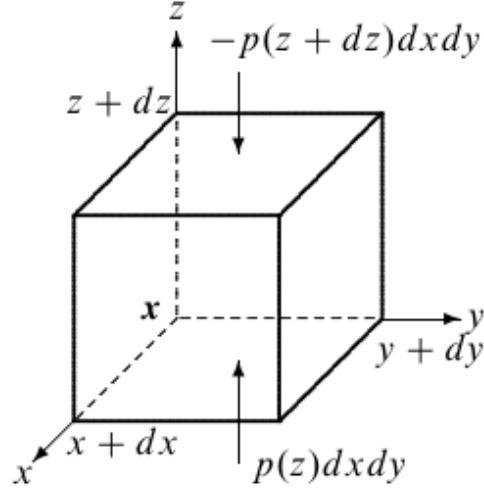


Figure 3: Pressure forces in the z -direction on a material particle in the shape of a rectangular box with sides dx, dy, dz . Figure taken from Ref. [3]

2.2 Hydrostatic equilibrium equation

The momentum equation obtained in the preliminaries is

$$\rho \frac{D\vec{v}}{Dt} = -\vec{\nabla}p + \vec{f}_e, \quad (64)$$

where $\vec{f}_e$ denotes the density of body forces. In hydrostatic equilibrium, $\vec{v} = 0$, and therefore

$$\boxed{\vec{\nabla}p = \vec{f}_e}. \quad (65)$$

For a fluid subject only to gravity,

$$\boxed{\vec{\nabla}p = \rho\vec{g}}. \quad (66)$$

2.2.1 Local hydrostatic equilibrium

In equilibrium, the net force on a differential fluid element vanishes:

$$d\vec{F}_e + d\vec{F}_p = 0, \quad (67)$$

where $d\vec{F}_e$ denotes external (volumetric) and $d\vec{F}_p$ the pressure forces.

In static fluids only normal pressures remain, since a fluid at rest cannot sustain shear stresses. Thus,

$$d\vec{F}_e - \vec{\nabla}p dV = 0. \quad (68)$$

Defining the volumetric density of external forces as

$$\vec{f}_e = \frac{d\vec{F}_e}{dV}, \quad (69)$$

and the effective force density

$$\vec{f}^* = \vec{f}_e - \vec{\nabla}p, \quad (70)$$

we obtain that in hydrostatic equilibrium $\vec{f}^* = 0$, leading to the basic equation of hydrostatics

$$\vec{f}_e = \vec{\nabla} p. \quad (71)$$

For the case of $\vec{f}_e$ being gravity force,

$$\rho \vec{g} = \vec{\nabla} p, \quad (72)$$

where $\vec{g}$ is the gravitational field.

Moreover, since $\vec{\nabla} \times \vec{\nabla} p = 0$, one has

$$\vec{\nabla} \times \vec{f}_e = 0, \quad (73)$$

which implies that $\vec{f}_e$ derives from a potential,

$$\vec{f}_e = -\vec{\nabla} \mathcal{H}. \quad (74)$$

Hence, hydrostatic equilibrium exists only when the volumetric forces acting on a fluid are conservative. This is the case for gravitational, electrostatic, and inertial forces.

2.2.2 Global hydrostatic equilibrium

Integrating the previous expression gives that the total force in mechanical equilibrium must vanish,

$$\vec{F}_e + \vec{F}_p = \int_V \vec{f}_e dV - \oint_S p d\vec{S} = 0. \quad (75)$$

When $\vec{f}_e$ is the force of gravity it corresponds to the weight of the fluid body, and the force due to pressure is called the buoyancy force.

Application: The incompressible sea In constant gravity and using flat-Earth coordinates, we have

$$\vec{g}(\vec{x}) = \vec{g}_0 = (0, 0, -g_0).$$

For symmetry reasons the sea on the flat Earth ought to have the same properties for all x and y , such that the pressure can only depend on z . Taking a rectangular vertical box with height h and cross-sectional area A , we find

$$\vec{F}_G = -\rho_0 g_0 A h \hat{e}_z, \quad \vec{F}_B = (-p(0)A + p(-h)A) \hat{e}_z, \quad (76)$$

because the pressure contributions from the sides of the box cancel. Global hydrostatic balance immediately leads to

$$p = p_0 + \rho_0 g_0 h. \quad (77)$$

2.2.3 Pressure Force and Gauss' Theorem

The total pressure force on a body of volume V can be obtained in two equivalent ways:

- By integrating the pressure forces acting on its surface S .
- By integrating the forces on all its constituent material particles.

This leads to the relation

$$\oint_S p d\vec{S} = \int_V \vec{\nabla} p dV. \quad (78)$$

This identity is a direct consequence of **Gauss' theorem**, which relates the surface integral of a scalar field $p(\mathbf{x})$ over a closed surface S to the volume integral of its gradient $\vec{\nabla} p(\vec{x})$ over the enclosed volume V . Hence Gauss' theorem allows us to convert the local equation back into the global one, showing that there is complete mathematical equivalence between the local and global formulations of hydrostatic equilibrium.

2.3 Pascal's principle

From Eq. (71), by integrating along a path from point 1 to point 2 inside a fluid, we obtain

$$\int_1^2 \vec{f}_e \cdot d\vec{r} = \int_1^2 \vec{\nabla} p \cdot d\vec{r} = \int_1^2 dp = p_2 - p_1, \quad (79)$$

so that the pressure difference between two points of a fluid in hydrostatic equilibrium depends only on the external forces, such as gravity. If there are no external forces, then $p_2 = p_1$. Moreover,

$$p_2 - p_1 = (p_2 + C) - (p_1 + C) = \int_1^2 \vec{f}_e \cdot d\vec{r}, \quad (80)$$

showing that the addition of a constant C to the pressure at all points of a fluid leaves the difference $p_2 - p_1$ invariant. That is, any pressure shift C applied at one point is accompanied by the same shift at all other points. This statement is known as **Pascal's principle**. **Pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and the walls of the containing vessel.**

This principle introduces a global gauge freedom in hydrostatics. In differential form, and taking into account that $\vec{\nabla} C = 0$, it follows that $\vec{f}_e = \vec{\nabla} p = \vec{\nabla}(p + C)$.

2.3.1 Hydraulic press

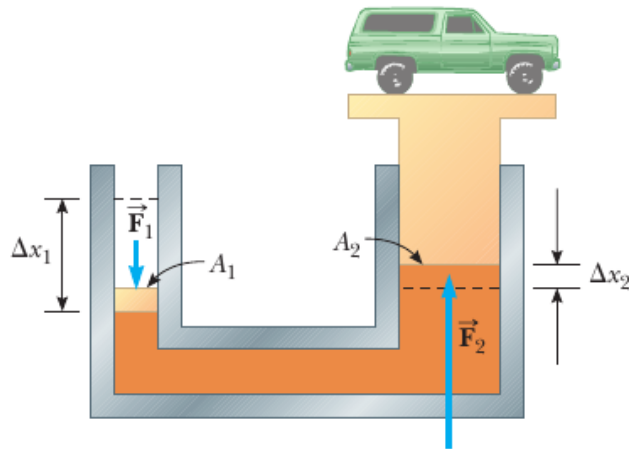


Figure 4: Diagram of a hydraulic press. Figure taken from Ref. [8]

An important application of Pascal's law is the hydraulic press (see Fig. 4). A force F_1 is applied to a small piston of surface area A_1 . The pressure is transmitted through an incompressible liquid to a larger piston of area A_2 . Since the pressure must be equal on both pistons,

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}. \quad (81)$$

Therefore, the output force is amplified by the factor A_2/A_1 .

Because the liquid is incompressible, the displaced volumes satisfy

$$A_1 \Delta x_1 = A_2 \Delta x_2,$$

so that

$$\frac{F_2}{F_1} = \frac{\Delta x_1}{\Delta x_2}.$$

Multiplying both sides by displacements shows that the work done is conserved:

$$F_1 \Delta x_1 = F_2 \Delta x_2.$$

The hydraulic press allows a small input force to generate a much larger output force, while conserving energy. This principle is widely used in hydraulic brakes, jacks and car lifts.

2.4 Hydrostatics in constant gravity

In a flat-Earth coordinate system, the constant field of gravity is $\vec{g}(\vec{r}) = (0, 0, -g_0)$ for all $\vec{r}$. If the external force is gravity, then:

$$\vec{f}_e = \frac{d\vec{F}}{dV} = \frac{dm}{dV} \vec{g} = \rho \vec{g},$$

so that,

$$\vec{\nabla} p - \rho \vec{g} = 0. \quad (82)$$

It follows that

$$\frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{\partial p}{\partial z} = -\rho g_0. \quad (83)$$

The first two equations express that the pressure does not depend on x and y but only on z . It also shows that, independently of the shape of a fluid container, the pressure will always be the same at a given depth (in constant gravity).

For the special case of constant density, $\rho(z) = \rho_0$, the last equation may immediately be integrated. The line integral between points 1 and 2 in Fig. 5, with ρ and $\vec{g}$ constant, is:

$$p_2 - p_1 = \int_1^2 \mathbf{f}_e \cdot d\mathbf{r} = \int_1^2 \rho \mathbf{g} \cdot d\mathbf{r} = -\rho g \int_1^2 dz = -\rho g(z_2 - z_1) = -\rho g h. \quad (84)$$

Thus, the pressure inside a liquid increases with depth:

$$p_1 = p_2 + \rho g h, \quad (85)$$

an expression identical to $(p_1 + C) = (p_2 + C) + \rho g h$. This equation shows that if a force is applied to the free surface of a liquid (for example, with a piston), the pressure increases equally at every point of the fluid. Therefore, the *difference in pressure* between two points depends only on their vertical separation.

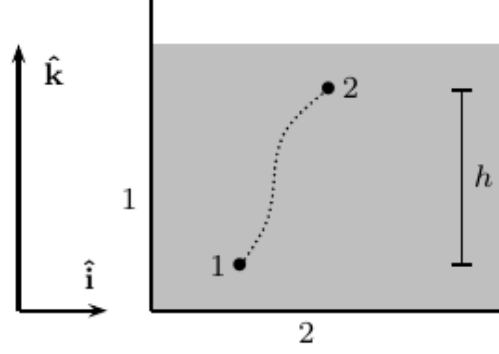


Figure 5: The pressure difference between points 1 and 2 inside a fluid in the Earth's gravity field depends only on their vertical separation. Figure taken from Ref. [1].

2.4.1 Communicating vessels

From the last expression in Eq.4 (83), it follows that $p = \rho g z + C$. At the free surface of the liquid ($z = h$) the pressure is atmospheric p_0 , so $C = p_0 + \rho g h$. Therefore

$$p = p_0 + \rho g(h - z). \quad (86)$$

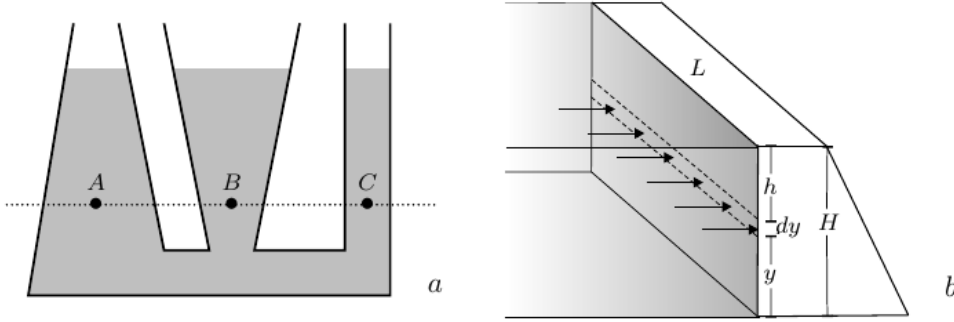


Figure 6: a) In communicating vessels, the pressure at points A , B , and C , located at the same height, is the same. b) The horizontal hydrostatic force exerted by the water in a dam on a vertical wall differential element of height dy and width L . Figure taken from Ref. [1].

It is therefore true that points at the same depth have the same pressure, independently of the shape of the container (see left panel of Fig. 6). This gives validity to the so-called *principle of communicating vessels* and resolves the so-called *hydrostatic paradox*, according to which—erroneously—the pressure inside a fluid would depend on the shape of the container. As seen here, the principle of communicating vessels is another result derived from the above expression for gravitational external forces $\vec{f}_e$.

According to the erroneous interpretation, the pressure at point B would be greater than at A , since more liquid lies above B than above A . The pressure at C would then be intermediate between those at A and B . The supposed paradox is that in fact

$$P_A = P_B = P_C.$$

This Equation shows that pressure is a scalar quantity, which ensures that *at every point inside the fluid the pressure is the same in all directions*.

2.4.2 The Force on a Dam

Water is filled to a height H behind a dam of width L (see right panel of Fig. 6). Let's determine the resultant force exerted by the water on the dam.

Because pressure increases with depth, the total force on a submerged surface cannot be calculated by simply multiplying the area by a single pressure value. Instead, we must account for the variation of pressure with depth, which requires the use of integration.

Consider a vertical y -axis with $y = 0$ at the bottom of the dam. We divide the face of the dam into narrow horizontal strips located at a height y above the bottom, as illustrated by the dashed strip in the figure.

The atmospheric pressure acting on both sides of the wall does not exert a net force, so it will not be considered. At a height y measured from the bottom of the dam, the pressure has the value

$$p = \rho gh = \rho g(H - y).$$

The horizontal force that the water exerts on the differential portion of wall of area $L dy$ is

$$dF = p dA = pL dy = \rho gL(H - y) dy,$$

so that the total horizontal net force, directed to the right, due to the water is

$$F = \rho gL \int_0^H (H - y) dy = \frac{1}{2} \rho gLH^2.$$

2.4.3 Pressure gauges

The simplest pressure gauge is the open-tube *manometer* (left panel of Fig. 7). The U-shaped tube contains a liquid of density ρ , often mercury or water. The left end of the tube is connected to the container where the pressure p is to be measured, and the right end is open to the atmosphere at pressure $p_0 = p_{\text{atm}}$. The pressure at the bottom of the tube due to the fluid in the left column is $p + \rho gy_1$, and the pressure at the bottom due to the fluid in the right column is $p_{\text{atm}} + \rho gy_2$. These pressures are measured at the same level, so they must be equal:

$$p + \rho gy_1 = p_{\text{atm}} + \rho gy_2,$$

$$p - p_{\text{atm}} = \rho g(y_2 - y_1) = \rho gh.$$

Here, p is the *absolute pressure*, and the difference $p - p_{\text{atm}}$ between absolute and atmospheric pressure is the *gauge pressure*. Thus the gauge pressure is proportional to the difference in height $h = y_2 - y_1$ of the liquid columns.

Another common pressure gauge is the *mercury barometer*. It consists of a long glass tube, closed at one end, that has been filled with mercury and then inverted in a dish of mercury (right panel of Fig. 7). The space above the mercury column contains only mercury vapor; its pressure is negligibly small, so the pressure p_0 at the top of the mercury column is practically zero. It follows that

$$p_{\text{atm}} = p = 0 + \rho g(y_2 - y_1) = \rho gh.$$

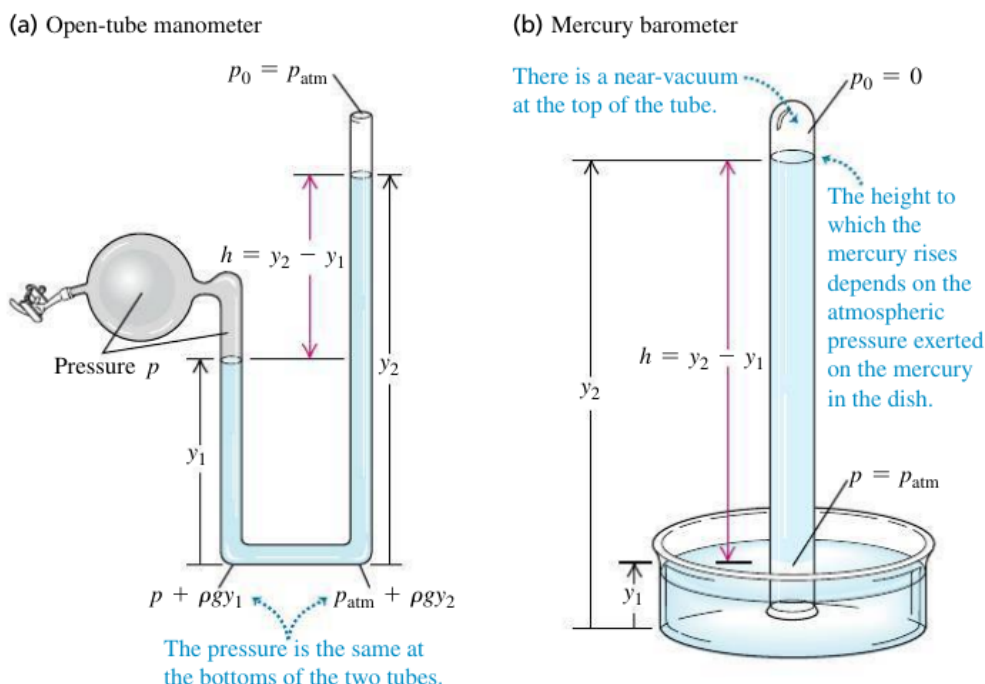


Figure 7: Two types of pressure gauge. Figure taken from Ref. [8]

So the height h of the mercury column indicates the atmospheric pressure p_{atm} .

Pressures are often described in terms of the height of the corresponding mercury column, as so many “inches of mercury” or “millimeters of mercury” (abbreviated mm Hg). A pressure of 1 mm Hg is called *1 torr*, after Evangelista Torricelli, inventor of the mercury barometer. But these units depend on the density of mercury, which varies with temperature, and on the value of g , which varies with location, so the pascal is the preferred unit of pressure.

Open Question A manometer tube is partially filled with water. Oil (which does not mix with water) is poured into the left arm of the tube until the oil-water interface is at the midpoint of the tube as shown in Fig. 8. Both arms of the tube are open to the air. Find a relationship between the heights h_{oil} and h_{water} .

Problems

1. The gate CD in the left panel of Fig. 9 can rotate about C . What horizontal force F , applied at a height h , is required to prevent the gate from rotating under the action of the water on its left, if its width is L ?
2. What is the force per unit length required to prevent the cylinder in the right panel of Fig. 9 from rolling under the action of the water located to its left?

2.5 Equation of state

The local equation of hydrostatic equilibrium with an externally given gravitational field is not sufficient by itself; we need a relation between pressure and density. Thermodynamics provides

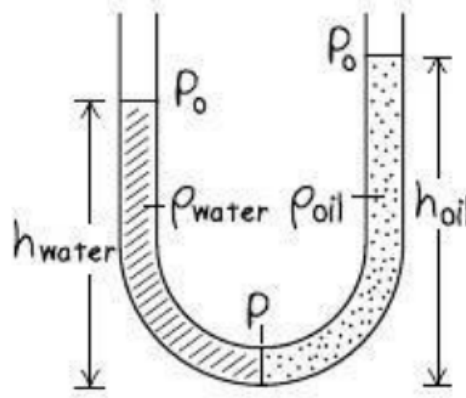


Figure 8: A manometer tube partially filled with water. Figure taken from Ref. [8].

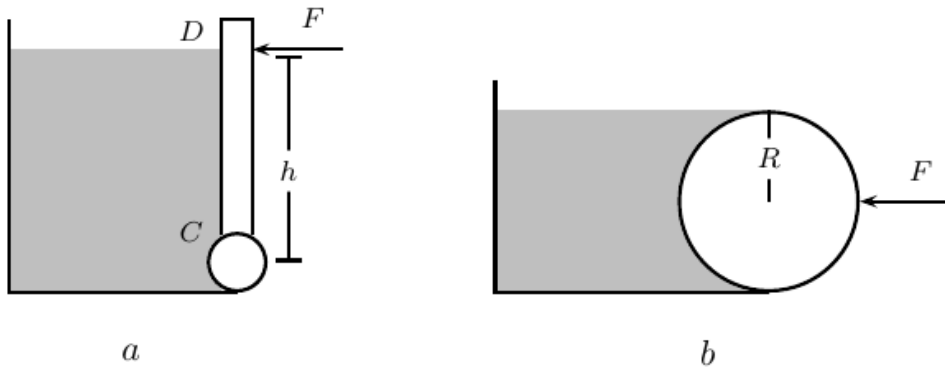


Figure 9: Hydrostatic force on a flat gate in a dam (a) and on the surface of a horizontal cylinder (b). Figure taken from Ref. [1].

this relationship through the *equation of state*, which relates density ρ , pressure p , and absolute temperature T :

$$F(\rho, p, T) = 0. \quad (87)$$

This relation is valid for any macroscopic amount of homogeneous isotropic fluid in thermodynamic equilibrium. In continuum physics, where conditions vary from point to point, it is assumed that each material particle is in thermodynamic equilibrium with its surroundings, so the equation of state holds locally:

$$F(\rho(x), p(x), T(x)) = 0. \quad (88)$$

2.5.1 The Ideal Gas Law

The *ideal gas law* is the most famous equation of state, often attributed to Clapeyron (1834). In terms of the volume V of a mass M of gas and the number of moles $n = M/M_{\text{mol}}$, it is written as

$$pV = nR_{\text{mol}}T, \quad (89)$$

where $R_{\text{mol}} = 8.31447 \text{ J K}^{-1} \text{ mol}^{-1}$ is the *universal molar gas constant*. Since $\rho = M/V = nM_{\text{mol}}/V$, the ideal gas law can be expressed in the local form:

$$p = R\rho T, \quad R = \frac{R_{\text{mol}}}{M_{\text{mol}}}. \quad (90)$$

Remarks. The specific gas constant R varies for different gases (see Table 3). Extensions of the ideal gas law include corrections for excluded molecular volume and intermolecular forces. The ideal gas law also applies to mixtures of gases if the average molar mass is used.

	$10^3 M_{\text{mol}}$ (kg mol ⁻¹)	$10^{-3} R$ (J K ⁻¹ kg ⁻¹)
H ₂	2.0	4.157
He	4.0	2.079
Ne	20.2	0.412
N ₂	28.0	0.297
O ₂	32.0	0.260
Ar	39.9	0.208
CO ₂	44.0	0.189
Air	29.0	0.287

Table 3: The molar mass, M_{mol} , and the specific gas constant, $R = R_{\text{mol}}/M_{\text{mol}}$, for a few gases.

The microscopic origin of the ideal gas law is as follows. The equipartition theorem tells us that the average kinetic energy of a molecule of mass m is

$$\frac{1}{2}m\langle v^2 \rangle = \frac{1}{2}m\langle v_x^2 \rangle + \frac{1}{2}m\langle v_y^2 \rangle + \frac{1}{2}m\langle v_z^2 \rangle = \frac{3}{2}k_B T, \quad (91)$$

from which

$$p = \frac{1}{3}\rho\langle v^2 \rangle = \rho \frac{k_B T}{m} = \rho \frac{N_A k_B T}{N_A m} = \rho \frac{R_{\text{mol}} T}{M_{\text{mol}}}. \quad (92)$$

2.5.2 Application: Isothermal atmosphere

Let us assume a constant temperature, $T(x) = T_0$, and combine the equation of hydrostatic equilibrium with the ideal gas law. Then

$$\frac{dp}{dz} = -\rho g_0 = -\frac{g_0}{RT_0} p. \quad (93)$$

This is an ordinary differential equation for the pressure, and using the initial condition $p = p_0$ for $z = 0$, we find the solution

$$p = p_0 e^{-z/h_0}, \quad h_0 = \frac{RT_0}{g_0} = \frac{p_0}{\rho_0 g_0}. \quad (94)$$

In the last step we have again used the ideal gas law at $z = 0$ to show that the expression for h_0 is identical to the incompressible atmospheric scale height.

In the isothermal atmosphere the pressure thus decreases exponentially with height on a characteristic length scale again set by $h_0 \approx 8728$ m calculated for 1 atm and 25°C. The pressure at the top of Mount Everest ($z = 8848$ m) is now finite and predicted to be 368 hPa.

The density profile becomes

$$\rho = \rho_0 e^{-z/h_0} = \rho_0 e^{-z g_0 / (p_0 / \rho_0)}. \quad (95)$$

2.5.3 Barotropic equation of state

Sometimes there exists a so-called *barotropic* relationship between density and pressure,

$$F(\rho(x), p(x)) = 0, \quad (96)$$

which does not depend on the local temperature $T(x)$.

Polytropic relation

The first law of thermodynamics in differential form reads

$$T dS = dE + p dV. \quad (97)$$

Dividing by the total mass m and introducing specific (per unit mass) quantities

$$s = \frac{S}{m}, \quad e = \frac{E}{m}, \quad v = \frac{V}{m} = \frac{1}{\rho}, \quad (98)$$

where s is the specific entropy, e the specific internal energy, p the pressure, and ρ the mass density, we obtain the specific form for the first law:

$$T ds = de + p d\left(\frac{1}{\rho}\right). \quad (99)$$

For an isentropic process we have $ds = 0$, so

$$de = -p d\left(\frac{1}{\rho}\right) = \frac{p}{\rho^2} d\rho. \quad (100)$$

For an ideal gas, $e = c_v T$ and $p = \rho R T$. Thus

$$c_v dT = \frac{p}{\rho^2} d\rho, \quad T = \frac{p}{\rho R}. \quad (101)$$

Eliminating T we obtain

$$c_v d\left(\frac{p}{\rho R}\right) = \frac{p}{\rho^2} d\rho, \quad (102)$$

or equivalently

$$\frac{c_v}{R} \left(\frac{dp}{\rho} - \frac{p}{\rho^2} d\rho \right) = \frac{p}{\rho^2} d\rho. \quad (103)$$

Now, recalling that

$$\gamma \equiv \frac{c_p}{c_v} = 1 + \frac{R}{c_v},$$

we can simplify the relation to

$$\frac{dp}{p} = \gamma \frac{d\rho}{\rho}. \quad (104)$$

Integrating both sides gives

$$\ln p = \gamma \ln \rho + \text{const}, \quad (105)$$

which leads to the **polytropic relation**

$$p = C \rho^\gamma, \quad (106)$$

where C is a constant (the *polytropic constant*), and γ is the *polytropic index* (see Table 4).

A polytropic relation arises naturally in an ideal gas when heat conduction can be completely disregarded, so that all processes take place without heat transfer. In thermodynamics such processes are called *adiabatic*, and the polytropic index is called the *adiabatic index*. Gases with diatomic molecules, like nitrogen and oxygen, have $\gamma \approx 7/5$, whereas monatomic noble gases have $\gamma \approx 5/3$ and multiatomic gases, like water vapor or carbon dioxide, have $\gamma \approx 4/3$. Since the local entropy stays constant during reversible, adiabatic processes, they are also said to be *isentropic*.

	γ	$10^{-3}c_p$
H ₂	1.41	14.30
He	1.63	5.38
Ne	1.64	1.05
N ₂	1.40	1.04
O ₂	1.40	0.91
Ar	1.67	0.52
CO ₂	1.30	0.82
Air	1.40	1.00

Table 4: Table of the adiabatic index and the specific heat at constant pressure ($c_p = \gamma R/(\gamma - 1)$) for a few nearly ideal gases. Units: J K⁻¹ kg⁻¹.

2.6 Isobars, isopycnals, and Potentials

For a fluid of density ρ —in general dependent on position—in a gravitational field $\vec{g}$:

$$-\rho \vec{g} + \vec{\nabla} p = 0 \quad \text{or:} \quad \rho \vec{\nabla} \Phi + \vec{\nabla} p = 0. \quad (107)$$

Taking the curl of the second equation,

$$\rho \vec{\nabla} \times \vec{\nabla} \Phi + \vec{\nabla} \rho \times \vec{\nabla} \Phi + \vec{\nabla} \times \vec{\nabla} p = 0. \quad (108)$$

Since $\vec{\nabla} \times \vec{\nabla} \Phi \equiv 0$ and $\vec{\nabla} \times \vec{\nabla} p \equiv 0$, it follows that, in a fluid of variable density, this condition holds:

$$\vec{\nabla} \rho \times \vec{\nabla} \Phi = 0 \quad \text{or:} \quad \vec{\nabla} \rho \times \vec{g} = 0. \quad (109)$$

This means that, point by point, the direction of the gravitational field lines coincides with that of the density gradient, or, equivalently, that the surfaces of equal density (isopycnals) and the gravitational equipotentials are coincident.

Considering

$$\vec{\nabla} \Phi + \frac{\vec{\nabla} p}{\rho} = 0,$$

and its curl is taken, it follows that for a compressible fluid

$$\vec{\nabla} \rho \times \vec{\nabla} p = 0, \quad (110)$$

from which it is concluded that the surfaces of constant pressure (isobars) and those of constant density coincide. From this equation it follows that the pressure can be expressed as *functional* of density: $p = p(\rho)$. Indeed,

$$\vec{\nabla}\rho \times \vec{\nabla}p = \vec{\nabla}\rho \times \left(\frac{dp}{d\rho} \vec{\nabla}\rho \right) = \frac{dp}{d\rho} \vec{\nabla}\rho \times \vec{\nabla}\rho = 0. \quad (111)$$

Fluids for which it is true that $p = p(\rho)$ are known as barotropes.

In an analogous way, it can be shown that Φ is a functional of ρ : $\Phi = \Phi(\rho)$. Thus, the surfaces of Φ , ρ and p constant are the same. This result is valid regardless of the form of the gravitational field. In particular, if $\vec{g} = -\hat{k}g$, the surfaces Φ , ρ and p will be horizontal planes. Outside the Earth's mass, assumed spherical, it is true that

$$\mathbf{g} = -\frac{GM\hat{\mathbf{r}}}{r^2}, \quad \Phi = -\frac{GM}{r},$$

so the three surfaces are spherical and concentric.

2.7 Pressure potential

Since the gravitational field is conservative and can be obtained from the gradient of the gravitational potential $\Phi(x)$, it follows that

$$\vec{g}(x) = -\vec{\nabla}\Phi(x). \quad (112)$$

In terms of the potential, the hydrostatics equilibrium equation ($\vec{\nabla}p = \rho\vec{g}$) may now be written as

$$\nabla\Phi + \frac{\nabla p}{\rho} = 0. \quad (113)$$

In the case of a constant density, $\rho(x) = \rho_0$, we obtain that

$$\Phi^* \equiv \Phi + \frac{p}{\rho_0} \quad (114)$$

obeys

$$\nabla\Phi^* = 0, \quad (115)$$

meaning Φ^* is also a constant, independent of $\vec{x}$. Φ^* is dubbed the *effective potential*. In flat-Earth gravity, where $\Phi = g_0z$, the constancy of

$$\Phi^* = g_0z + \frac{p}{\rho_0}$$

leads directly to the linear pressure field in the incompressible sea.

It is always possible to integrate the hydrostatic equation for any barotropic fluid with $\rho = \rho(p)$. Introducing the *pressure potential*,

$$w(p) = \int \frac{dp}{\rho(p)}, \quad (116)$$

it turns out that hydrostatic equilibrium may be written as

$$\Phi^* = \Phi + w(p). \quad (117)$$

The requirement $\Phi^*(\vec{x}) = \text{const}$ is completely equivalent to the equation of hydrostatic equilibrium for barotropic fluids.

2.7.1 Isothermal gas

Under isothermal conditions, the pressure potential of an ideal gas is calculated by means of the ideal gas law:

$$w = \int \frac{RT_0}{p} dp = RT_0 \log p. \quad (118)$$

2.7.2 Polytropic fluid

For fluids obeying a polytropic relation, the pressure potential becomes

$$w = \int C\gamma\rho^{\gamma-1} \frac{d\rho}{\rho} = C \frac{\gamma}{\gamma-1} \rho^{\gamma-1} = \frac{\gamma}{\gamma-1} \frac{p}{\rho}. \quad (119)$$

When the fluid is an ideal gas with $p = R\rho T$, this takes the simpler form

$$w = \frac{\gamma}{\gamma-1} RT = c_p T, \quad c_p = \frac{\gamma}{\gamma-1} R, \quad (120)$$

where c_p is the specific heat at constant pressure.

2.8 Bulk modulus

All fluids compress if the pressure increases, resulting in a decrease in volume or an increase in density. A common way to describe the compressibility of a fluid is by the bulk modulus of elasticity. It is defined the *bulk modulus* as the pressure increase dp per *fractional decrease* in volume, $-dV/V$, or

$$K = \frac{dp}{-dV/V} = \frac{dp}{d\rho/\rho} = \rho \frac{dp}{d\rho}. \quad (121)$$

In the second step we have used the constancy of the mass $M = \rho V$ of the fluid in the volume to derive that $dM = \rho dV + V d\rho = 0$, from which we get $-dV/V = d\rho/\rho$.

The above definition makes immediate sense for a barotropic fluid, where $p = p(\rho)$ is a function of density. For general fluid states it is necessary to specify the conditions under which the bulk modulus is defined, for example whether the temperature is held constant (isothermal) or whether there is no heat transfer (adiabatic or isentropic). Thus, the equation of state for an ideal gas implies that the isothermal bulk modulus is

$$K_T = \left(\rho \frac{\partial p}{\partial \rho} \right)_T = p, \quad (122)$$

where the index—as commonly done in thermodynamics—indicates that the temperature T is held constant. Similarly, for an isentropic ideal gas obeying the polytropic relation that the *isentropic bulk modulus* becomes

$$K_S = \left(\rho \frac{dp}{d\rho} \right)_S = \gamma p, \quad (123)$$

where the index indicates that the entropy S is held constant. It is larger than the isothermal bulk modulus by a factor of $\gamma > 1$, because adiabatic compression also increases the temperature of the gas, which further increases the pressure.

The definition of the bulk modulus (and the above equation) shows that it is measured in the same units as pressure, for example pascals, bars, or atmospheres. The bulk modulus is actually

a measure of *incompressibility*, because the larger it is, the greater is the pressure increase that is needed to obtain a given fractional increase in density. The inverse bulk modulus $\beta = 1/K$ may be taken as a measure of *compressibility*.

The bulk modulus for water at standard conditions is approximately 2100 MPa, or 21 000 times the atmospheric pressure. For air at standard conditions, K is equal to 1 atm. In general, K for a gas is equal to the pressure of the gas. To cause a 1% change in the density of water a pressure of 21 MPa (210 atm) is required. This is an extremely large pressure needed to cause such a small change; thus liquids are often assumed to be incompressible. For gases, if significant changes in density occur, say 4%, they should be considered as compressible; for small density changes under 3% they may also be treated as incompressible. This occurs for atmospheric airspeeds under about 100 m/s (220 mph), which includes many airflows of engineering interest: air flow around automobiles, landing and take-off of aircraft, and air flow in and around buildings.

The bulk modulus can also be used to calculate the speed of sound in a liquid:

$$c = \sqrt{\left(\frac{dp}{d\rho}\right)_T} = \sqrt{\frac{K_T}{\rho}} \quad (124)$$

This yields approximately 1450 m/s (4800 ft/s) for the speed of sound in water at standard conditions.

2.9 Earth's homentropic atmosphere

A process that takes place without exchange of heat between a system and its environment is said to be adiabatic. If furthermore the process is reversible, it will conserve the entropy and is called isentropic.

2.9.1 Homentropic equation of state

From the first law:

$$dU = dQ - p dV. \quad (125)$$

Adiabaticity implies $dQ = 0$, and for an ideal gas $U = nC_V T$:

$$nC_V dT = -p dV. \quad (126)$$

Using the ideal-gas law $pV = nRT \Rightarrow dT = \frac{1}{nR}(p dV + V dp)$, it follows that

$$\frac{C_V}{R}(p dV + V dp) = -p dV. \quad (127)$$

Rearrange and use $C_P = C_V + R$:

$$\frac{C_V}{R} V dp + \frac{C_P}{R} p dV = 0. \quad (128)$$

Dividing by pV and using $\gamma \equiv C_P/C_V$:

$$\frac{dp}{p} + \gamma \frac{dV}{V} = 0. \quad (129)$$

Finally

$$\ln p + \gamma \ln V = \text{const} \Rightarrow p V^\gamma = \text{const}. \quad (130)$$

In other words,

$$p V^\gamma = p_0 V_0^\gamma. \quad (131)$$

Expressed in terms of the density $\rho = M/V$, an isentropic process that locally changes density ρ_0 and pressure p_0 to ρ and p must obey the local *polytropic* relation:

$$p \rho^{-\gamma} = p_0 \rho_0^{-\gamma}. \quad (132)$$

2.9.2 Homentropic solution

The hydrostatic equilibrium implies

$$\Phi^* = \Phi + w(p) = g_0 z + c_p T = \text{const}. \quad (133)$$

Defining this constant as $\Phi^* = c_p T_0$ (with T_0 the sea-level temperature), the temperature profile is

$$T(z) = T_0 - \frac{g_0}{c_p} z. \quad (134)$$

Introducing the scale height for the isothermal atmosphere $h_0 = p_0/(\rho_0 g_0)$, this expression takes the form

$$T = T_0 \left(1 - \frac{z}{h_2}\right), \quad h_2 = \frac{c_p T_0}{g_0} = \frac{\gamma}{\gamma - 1} h_0, \quad (135)$$

For $\gamma = 7/5$ and $T_0 = 25^\circ\text{C}$, the isentropic scale height is $h_2 \approx 31$ km.

Combining the ideal-gas law with the homentropic (isentropic) relation gives that $T \rho^{1-\gamma}$ and $p^{1-\gamma} T^\gamma$ are constants, so the density and pressure become

$$\rho = \rho_0 \left(1 - \frac{z}{h_2}\right)^{1/(\gamma-1)}, \quad p = p_0 \left(1 - \frac{z}{h_2}\right)^{\gamma/(\gamma-1)}, \quad (136)$$

where ρ_0 and p_0 are the density and pressure at sea level. Note that T , ρ , and p formally vanish at $z = h_2$, which is unphysical; the real atmosphere is more complicated than this simple model.

2.9.3 The atmospheric temperature lapse rate

The negative vertical temperature gradient, called the *atmospheric temperature lapse rate*, is

$$-\frac{dT}{dz} = \frac{g_0}{c_p}. \quad (137)$$

Its value is 9.8 K km^{-1} for dry air. At the top of Mount Everest ($z = 8.848$ km), the model with $T_0 = 25^\circ\text{C}$ predicts a temperature of -61°C , a density of 0.50 kg m^{-3} , and a pressure of 306 hPa .

Water vapor is always present in the atmosphere and condenses in rising air. The latent heat released during condensation warms the air, making the lapse rate smaller than in dry air, typically $6\text{--}7 \text{ K km}^{-1}$, which corresponds to an effective adiabatic index $\gamma \approx 1.23$. Using this value with $T_0 = 25^\circ\text{C}$, the scale height increases to 46 km, and the predicted conditions at the summit of Mount Everest are $p \approx 329 \text{ hPa}$ and $T \approx -33^\circ\text{C}$, close to the measured average of about -25°C .

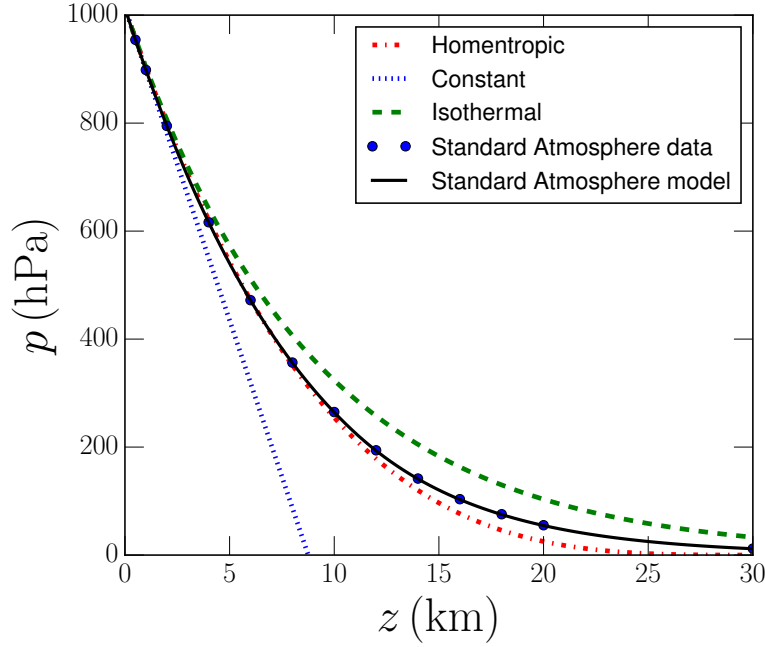


Figure 10: The pressure as a function of altitude (for $T_0 = 25^\circ\text{C}$) in three different atmospheric models analyzed in this chapter: constant density (dotted), isothermal (dashed), and homentropic (dot-dashed). The solid curve is the Standard Atmosphere (1976) model.

2.10 Standard Atmosphere

The atmosphere is divided into four layers (see Fig. 11):

1. the *troposphere* (nearest Earth),
2. the *stratosphere*,
3. the *mesosphere*, and
4. the *ionosphere*.

Because conditions vary with time and latitude and the layers are thicker at the equator and thinner at the poles, calculations are based on the *standard atmosphere*, taken at 40° latitude.

In the standard atmosphere the temperature in the troposphere varies linearly with elevation,

$$T(z) = T_0 - \alpha z, \quad (138)$$

where the lapse rate is $\alpha = 0.0065 \text{ K/m}$ and $T_0 = 288 \text{ K}$. In the portion of the stratosphere between 11 and 20 km the temperature is constant at about -56.5°C . Above this, the temperature increases again and reaches a maximum near 50 km, then decreases toward the edge of the ionosphere. Because the air density in the ionosphere is so low, it is possible for satellites to orbit the Earth within this layer.

To determine the pressure variation in the troposphere we combine the ideal-gas law $p = \rho RT$ with the hydrostatic relation $dp = -\rho g dz$. Eliminating ρ gives

$$\frac{dp}{p} = -\frac{g}{RT} dz. \quad (139)$$

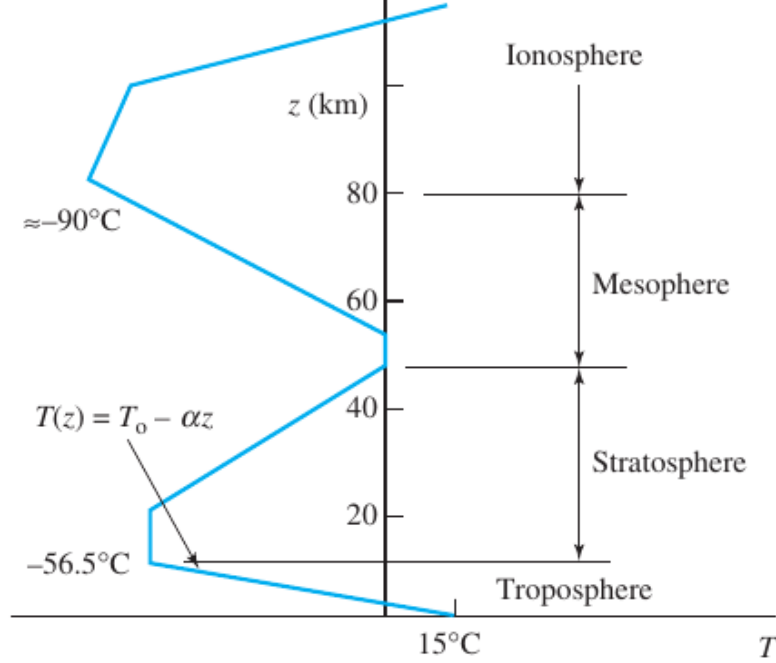


Figure 11: Standard atmosphere. Figure taken from Ref. [4].

With the standard tropospheric lapse $T(z) = T_0 - \alpha z$, integrating from sea level ($z = 0$, $p = p_{\text{atm}}$) to an elevation z

$$\int_{p_{\text{atm}}}^p \frac{dp}{p} = -\frac{g}{R} \int_0^z \frac{dz}{T_0 - \alpha z}, \quad (140)$$

leads to

$$\ln\left(\frac{p}{p_{\text{atm}}}\right) = \frac{g}{\alpha R} \ln\left(\frac{T_0 - \alpha z}{T_0}\right), \quad (141)$$

or, equivalently,

$$p = p_{\text{atm}} \left(\frac{T_0 - \alpha z}{T_0}\right)^{g/(\alpha R)}. \quad (142)$$

In the lower stratosphere the temperature is (approximately) constant, $T = T_s$. Integrating $\frac{dp}{p} = -\frac{g}{RT_s} dz$ from the tropopause ($z = z_s$, $p = p_s$) to altitude z gives

$$\int_{p_s}^p \frac{dp}{p} = -\frac{g}{RT_s} \int_{z_s}^z dz \Rightarrow \ln\left(\frac{p}{p_s}\right) = -\frac{g}{RT_s} (z - z_s), \quad (143)$$

or

$$p = p_s \exp\left[\frac{g}{RT_s} (z_s - z)\right]. \quad (144)$$

2.11 The Sun's convective envelope

Stars, like the Sun, are self-gravitating, gaseous, and almost perfectly spherical bodies that generate heat by thermonuclear processes in a fairly small region close to the center. The heat is transported to the surface by radiation, conduction, and convection and is eventually released into

Altitude	$h(\text{m})$	$T(\text{K})$	$p(\text{kPa})$	$\rho(\text{kg/m}^3)$	$v_s(\text{m/s})$
	0	288.2	101.3	1.225	340
	500	284.9	95.43	1.167	338
	1000	281.7	89.85	1.112	336
	2000	275.2	79.48	1.007	333
	4000	262.2	61.64	0.8194	325
	6000	249.2	47.21	0.6602	316
	8000	236.2	35.65	0.5528	308
	10000	223.3	26.49	0.4136	300
	12000	216.7	19.40	0.3119	295
	14000	216.7	14.17	0.2278	295
	16000	216.7	10.35	0.1665	295
	18000	216.7	7.563	0.1216	295
	20000	216.7	5.528	0.0889	295
	30000	226.5	1.196	0.0184	302
	40000	250.4	0.287	4.00×10^{-3}	317
	50000	270.7	0.0798	1.03×10^{-3}	330
	60000	255.8	0.0225	3.06×10^{-4}	321
	70000	219.7	0.00551	8.75×10^{-5}	297
	80000	180.7	0.00103	2.00×10^{-5}	269

Table 5: Properties of the Standard Atmosphere. Data taken from Ref. [4].

space as radiation. Like planets, stars also have a complex structure with several layers differing in chemical composition and other physical properties.

Our Sun consists—like the rest of the universe—of roughly 75% hydrogen and 25% helium (by mass), plus small amounts of other elements (usually called “metals”). The radius of its photosphere is $a = 700,000$ km, its mass $M_\odot = 2 \times 10^{30}$ kg, and its total luminosity (power) 3.8×10^{26} W. The gases making up the Sun are almost completely ionized, forming a plasma consisting of positively charged hydrogen ions H^+ , doubly charged helium ions He^{2+} , and negatively charged electrons e^- . The mean molar mass of the plasma is $M_{\text{mol}} = 0.59$ g mol $^{-1}$. It is smaller than unity because the nearly massless electrons make up about 52% of all the particles in the plasma.

The density in the convective zone is so low that the zone contains only about 3% of the Sun’s mass, so that we may approximate the gravitational potential in this region by that of a point particle with the Sun’s mass,

$$\Phi = -G \frac{M_0}{r}, \quad (145)$$

where $G = 6.674 \times 10^{-11}$ N m 2 kg $^{-2}$ is the universal gravitational constant. The pressure potential for a homentropic gas is as before given by Eq. (120). Since the solar plasma consists of monatomic “molecules”, the adiabatic index is $\gamma = 5/3$, leading to a specific heat capacity at constant pressure $c_p = 35.1$ J K $^{-1}$ g $^{-1}$. This is 35 times that of the Earth’s atmosphere, mainly because of the low average molar mass of the solar plasma.

Hydrostatic equilibrium requires the effective potential $\Phi^* = \Phi + c_p T$ to be constant. Fixing this constant by means of the surface temperature T_0 , we obtain the expression

$$T(r) = T_0 + T_1 \left(\frac{a}{r} - 1 \right), \quad \text{where} \quad T_1 = \frac{GM_0}{c_p a}. \quad (2.57)$$

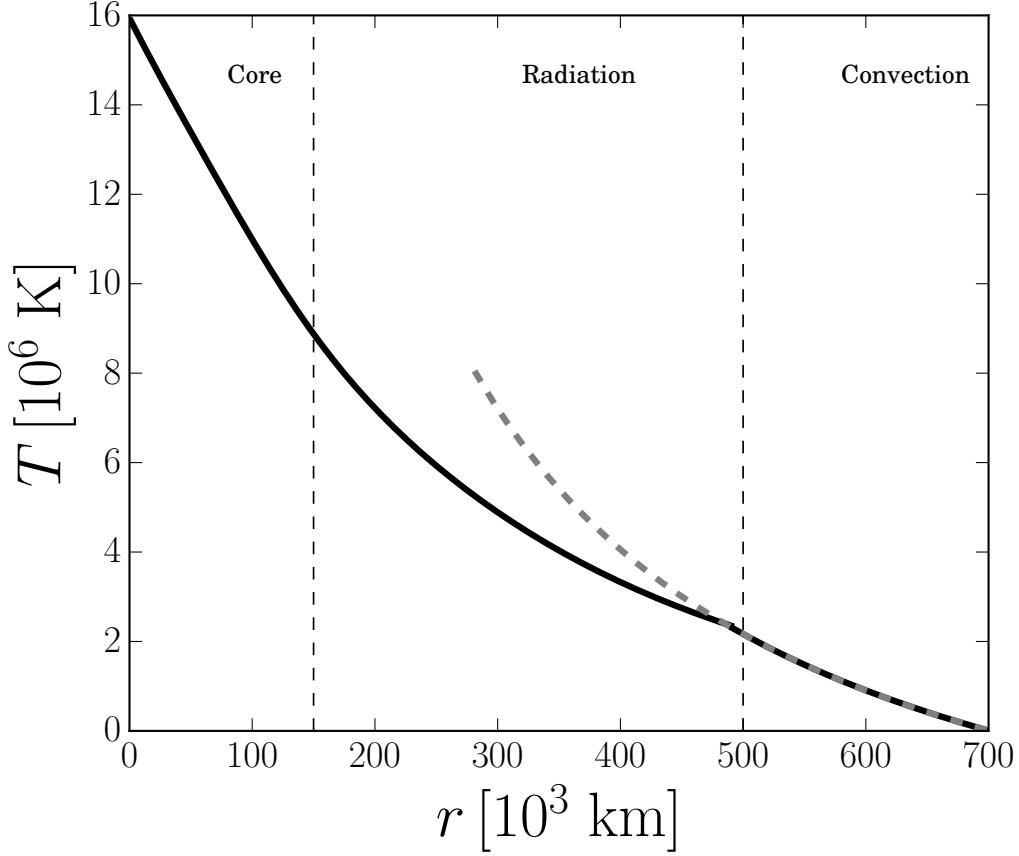


Figure 12: The temperature distribution in the Sun as a function of the distance from the center. The vertical lines are boundaries between the three major layers of the Sun. The fully drawn curve is an approximate fit to the solution from the Standard Sun model. The dashed curve is the approximative homentropic solution for the convective envelope.

With the Sun's parameters we find $T_1 = 5.4 \times 10^6$ K, a value that also sets the scale for the Sun's central temperature. The precise surface of the Sun is not too well defined, but normally chosen to be that spherical surface at radius $r = a$ where the plasma turns into neutral gas and becomes transparent. The transition to complete transparency actually takes place over a few hundred kilometers. The surface temperature is about $T_0 \approx 6,000$ K, which is about 1,000 times smaller than T_1 and can mostly be ignored. At the surface of the Sun the temperature lapse rate becomes

$$-\frac{dT}{dr} = \frac{T_1}{a} = 7.8 \text{ K km}^{-1},$$

which accidentally is nearly the same as in the Earth's troposphere.

In Figure 12 the solution is plotted together with the temperature distribution from the Standard Sun model. The agreement is nearly perfect in the convective zone. The density and pressure can as in the preceding section be determined from the adiabatic law.

2.12 Helioseismology

Hydrostatics plays a central role in the theoretical description of the solar interior that underlies helioseismology [9]. In stellar structure theory the Sun is assumed to be in *hydrostatic equilibrium*, meaning that the inward gravitational force is balanced by the pressure gradient of the plasma. This balance determines the radial profiles of pressure, density, and temperature that define the background solar model.

Helioseismic studies then analyze small perturbations—observed as oscillations of the solar surface—around this equilibrium state. The frequencies of these oscillations depend sensitively on quantities such as the sound speed and density profile, which themselves follow from the hydrostatic structure of the star. Consequently, comparisons between observed oscillation spectra and theoretical predictions allow one to test and refine models of stellar interiors, providing detailed constraints on the Sun’s internal structure, composition, and rotation.

Hydrostatic equilibrium therefore forms the essential baseline from which the dynamics probed by helioseismology are interpreted.

2.13 Questions

1. A rubber hose is attached to a funnel, and the free end is bent around to point upward. When water is poured into the funnel, it rises in the hose to the same level as in the funnel, even though the funnel has a lot more water in it than the hose does. Why? What supports the extra weight of the water in the funnel?
2. Suppose the door of a room makes an airtight but frictionless fit in its frame. Do you think you could open the door if the air pressure on one side were standard atmospheric pressure and the air pressure on the other side differed from standard by 1%? Explain.
3. At a certain depth in an incompressible liquid, the absolute pressure is p . At twice this depth, will the absolute pressure be equal to $2p$, greater than $2p$, or less than $2p$? Justify your answer.
4. A meteorologist states that the barometric pressure is 28.5 inches of mercury. Convert this pressure to kilopascals.
(A) 98.6 kPa (B) 97.2 kPa (C) 96.5 kPa (D) 95.6 kPa
5. If the pressure in the air shown in Fig. 13 is increased by 10 kPa, the magnitude of H will be nearest (initially $H = 16$ cm):
(A) 8.5 cm (B) 10.5 cm (C) 16 cm (D) 24.5 cm

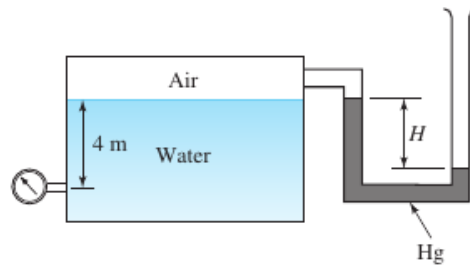


Figure 13: Question 5.

6. The rectangular gate shown in Fig. 14 is 3 m wide. The force P needed to hold the gate in the position shown is nearest:
- (A) 24.5 kN (B) 32.7 kN (C) 98 kN (D) 147 kN

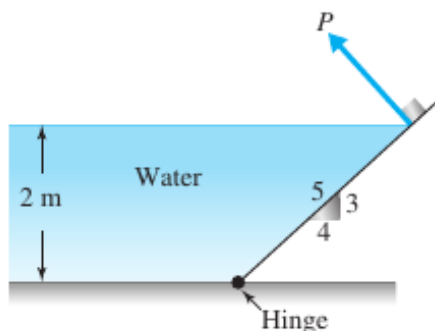


Figure 14: Question 6.

7. A force $P = 300$ kN is needed to just open the gate of Fig. 15 with $R = 1.2$ m and $H = 4$ m. How wide is the gate?
- (A) 2.98 m (B) 3.67 m (C) 4.32 m (D) 5.16 m

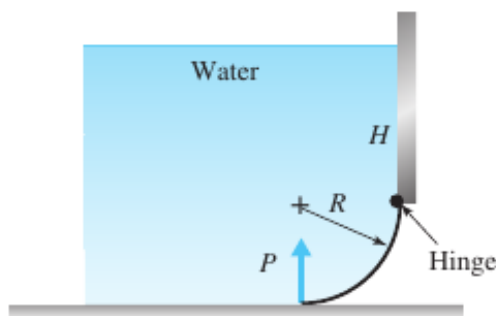


Figure 15: Question 7.

2.14 Problems

1. What is the volume change of 2 m^3 of water at 20°C due to an applied pressure of 10 MPa?
2. Two engineers wish to estimate the distance across a lake. One pounds two rocks together under water on one side of the lake and the other submerges his head and hears a small sound 0.62 s later, as indicated by a very accurate stopwatch. What is the distance between the two engineers?
3. A pressure is applied to 20 L of water. The volume is observed to decrease to 18.7 L. Calculate the applied pressure.
4. **Black Smokers.** Black smokers are hot volcanic vents that emit smoke deep in the ocean floor. Many of them teem with exotic creatures, and some biologists think that life on earth

may have begun around such vents. The vents range in depth from about 1500 m to 3200 m below the surface. What is the gauge pressure at a 3200-m deep vent, assuming that the density of water does not vary? Express your answer in pascals and atmospheres.

5. **Oceans on Mars.** Scientists have found evidence that Mars may once have had an ocean 0.500 km deep. The acceleration due to gravity on Mars is 3.71 m/s^2 . (a) What would be the gauge pressure at the bottom of such an ocean, assuming it was freshwater? (b) To what depth would you need to go in the earth's ocean to experience the same gauge pressure?
6. (a) Calculate the difference in blood pressure between the feet and top of the head for a person who is 1.65 m tall. (b) Consider a cylindrical segment of a blood vessel 2.00 cm long and 1.50 mm in diameter. What additional outward force would such a vessel need to withstand in the person's feet compared to a similar vessel in her head?
7. In intravenous feeding, a needle is inserted in a vein in the patient's arm and a tube leads from the needle to a reservoir of fluid (density 1050 kg/m^3) located at height h above the arm. The top of the reservoir is open to the air. If the gauge pressure inside the vein is 5980 Pa, what is the minimum value of h that allows fluid to enter the vein? Assume the needle diameter is large enough that you can ignore the viscosity of the fluid.
8. You are designing a diving bell to withstand the pressure of seawater at a depth of 250 m. a) What is the gauge pressure at this depth? (You can ignore changes in the density of the water with depth.) b) At this depth, what is the net force due to the water outside and the air inside the bell on a circular glass window 30.0 cm in diameter if the pressure inside the diving bell equals the pressure at the surface of the water? (Ignore the small variation of pressure over the surface of the window.)
9. There is a maximum depth at which a diver can breathe through a snorkel tube because as the depth increases, so does the pressure difference, which tends to collapse the diver's lungs. Since the snorkel connects the air in the lungs to the atmosphere at the surface, the pressure inside the lungs is atmospheric pressure. What is the external-internal pressure difference when the diver's lungs are at a depth of 6.1 m (about 20 ft)? Assume that the diver is in freshwater. (A scuba diver breathing from compressed air tanks can operate at greater depths than can a snorkeler, since the pressure of the air inside the scuba diver's lungs increases to match the external pressure of the water.)
10. The lower end of a long plastic straw is immersed below the surface of the water in a plastic cup. An average person sucking on the upper end of the straw can pull water into the straw to a vertical height of 1.1 m above the surface of the water in the cup. a) What is the lowest gauge pressure that the average person can achieve inside his lungs? b) Explain why your answer in part (a) is negative.
11. An electrical short cuts off all power to a submersible diving vehicle when it is 30 m below the surface of the ocean. The crew must push out a hatch of area 0.75 m^2 and weight 300 N on the bottom to escape. If the pressure inside is 1.0 atm, what downward force must the crew exert on the hatch to open it?
12. A tall cylinder with a cross-sectional area 12.0 cm^2 is partially filled with mercury; the surface of the mercury is 8.00 cm above the bottom of the cylinder. Water is slowly poured in on

top of the mercury, and the two fluids don't mix. What volume of water must be added to double the gauge pressure at the bottom of the cylinder?

13. A closed container is partially filled with water. Initially, the air above the water is at atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$) and the gauge pressure at the bottom of the water is 2500 Pa. Then additional air is pumped in, increasing the pressure of the air above the water by 1500 Pa. (a) What is the gauge pressure at the bottom of the water? (b) By how much must the water level in the container be reduced, by drawing some water out through a valve at the bottom of the container, to return the gauge pressure at the bottom of the water to its original value of 2500 Pa? The pressure of the air above the water is maintained at 1500 Pa above atmospheric pressure.
14. **Exploring Venus.** The surface pressure on Venus is 92 atm, and the acceleration due to gravity there is $0.894g$. In a future exploratory mission, an upright cylindrical tank of benzene is sealed at the top but still pressurized at 92 atm just above the benzene. The tank has a diameter of 1.72 m, and the benzene column is 11.50 m tall. Ignore any effects due to the very high temperature on Venus. (a) What total force is exerted on the inside surface of the bottom of the tank? (b) What force does the Venusian atmosphere exert on the outside surface of the bottom of the tank? (c) What total inward force does the atmosphere exert on the vertical walls of the tank?
15. **The Great Molasses Flood.** On the afternoon of January 15, 1919, an unusually warm day in Boston, a 17.7-m-high, 27.4-m-diameter cylindrical metal tank used for storing molasses ruptured. Molasses flooded into the streets in a 5-m-deep stream, killing pedestrians and horses and knocking down buildings. The molasses had a density of 1600 kg/m^3 . If the tank was full before the accident, what was the total outward force the molasses exerted on its sides? (Hint: Consider the outward force on a circular ring of the tank wall of width dy and at a depth y below the surface. Integrate to find the total outward force. Assume that before the tank ruptured, the pressure at the surface of the molasses was equal to the air pressure outside the tank.)
16. What is the pressure in the water pipe shown in the figure Fig. 16?

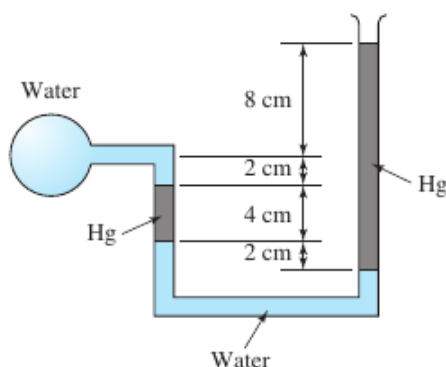


Figure 16: Problem.

17. Determine the force P needed to hold the 4-m wide gate in the position shown in Fig. 17.

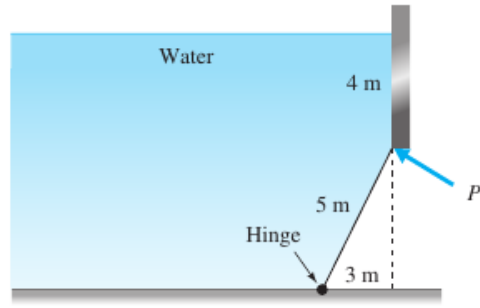


Figure 17: Problem.

18. Find the force P needed to hold the 10-m-long cylindrical object in position as shown in Fig. 18. Here $S \equiv \rho/\rho_{\text{water}}$.

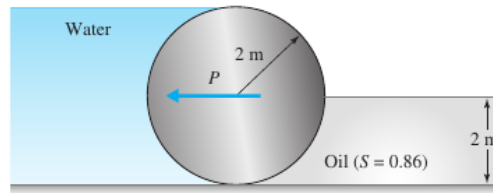


Figure 18: Problem.

19. Find the force P if the parabolic gate shown in Fig. 19 has a width w .

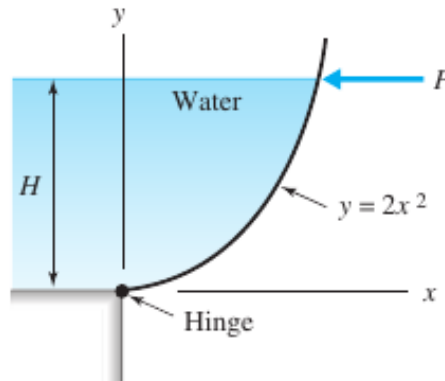


Figure 19: Problem.

20. **Hydrostatic atmospheres** For this exercise you will need the following constants: Boltzmann's constant $k_B = 1.380658 \times 10^{-16} \text{ erg K}^{-1}$, the atomic mass unit $m = 1.6605 \times 10^{-24} \text{ g}$, the solar radius $R_{\odot} = 6.9599 \times 10^{10} \text{ cm}$, the solar mass $M_{\odot} = 1.98892 \times 10^{33} \text{ g}$, the gravitational constant $G = 6.6743 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$, the earth radius $R_{\oplus} = 6.378 \times 10^8 \text{ cm}$ and the earth mass $M_{\oplus} = 5.974 \times 10^{27}$. You will calculate characterizations of the sun and earth atmospheres, assuming that the ideal gas law to be valid $p = \frac{\rho k_B T}{\mu m}$.

- i) In addition to the ideal gas assumption, we further assume that pressure and density are related according to

$$p = K\rho^{1+\frac{1}{n}},$$

where K and n are two constants. If $n \rightarrow \infty$, what does this imply for the temperature profiles in the atmospheres?

- ii) Let's start with the case of the sun's photosphere. You can assume a planar atmosphere and constant T . For the purpose of computing the effect of gravity, you may further assume that the atmosphere has a total mass content that is negligible relative to the mass of the sun and an equally negligible radial extent relative to the radius of the sun (which also justifies the planar assumption). Starting from momentum conservation in the hydrostatic case,

$$\nabla p = f,$$

show that the pressure profile obeys

$$p = p_0 \exp \left[-(z - z_0) \frac{\mu m g}{k_B T} \right],$$

where g is gravitational acceleration.

- iii) Identify a measure of “scale height” in this expression for pressure and interpret the meaning of a “scale height” in an atmosphere.
- iv) Compute the scale heights for the solar photosphere at temperature $T_\odot = 5770$ K and the earth at $T_\oplus = 300$ K. The solar photosphere is highly ionized and has a mean molecular weight $\mu = 0.6$, while the earth atmosphere consists largely of dinitrogen (N_2) and has a mean molecular weight $\mu = 28$. The scale heights for sun and earth should come out as $H = 2.9 \times 10^7$ cm and 9.0×10^5 cm respectively.
- v) Mauna Kea in Hawaii hosts a number of the world's most powerful telescopes requiring astronomers to work at an altitude of 4.2 km. What is the implication for the atmospheric *density* at this elevation relative to sea level?

3 Self-gravitating continuous media

3.1 Newtonian gravitation

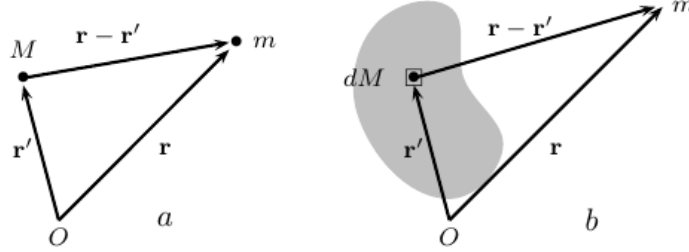


Figure 20: Geometry for describing gravitational attraction, a. between point masses; b. between a mass distribution and a point mass

According to Newton, the gravitational force that a point mass M exerts on another point mass m (see Fig. 20) is given by

$$\vec{F}_{M \rightarrow m} = - \frac{G M m (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}. \quad (146)$$

The gravitational force that a mass distribution with density $\rho(\vec{r}')$ exerts on a point mass m located at $\vec{r}$ is computed as the force produced by the collection of point masses making up the distribution:

$$\vec{F}(\vec{r}) = - G m \int \rho(\vec{r}') \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} d^3 \vec{r}'. \quad (147)$$

The corresponding gravitational fields are

$$\vec{g} = \vec{F}_{M \rightarrow m} / m = - \frac{G M (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}. \quad (148)$$

and

$$\vec{g}(\vec{r}) = \vec{F}(\vec{r}) / m = - G \int \rho(\vec{r}') \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} d^3 \vec{r}'. \quad (149)$$

The corresponding gravitational potentials are

$$\Phi = - \frac{G M}{|\vec{r} - \vec{r}'|}. \quad (150)$$

and

$$\Phi(\vec{r}) = - G \int \rho(\vec{r}') \frac{1}{|\vec{r} - \vec{r}'|} d^3 \vec{r}'. \quad (151)$$

The gravitational potential satisfies

$$\nabla^2 \Phi(\vec{r}) = 4\pi G \rho(\vec{r}), \quad (152)$$

while the gravitational field

$$\vec{\nabla} \times \vec{g} = 0. \quad (153)$$

The Poisson's equation tells us what the local impact is of a self-gravitating system on its gravitational potential. It is valid in general, even for the isothermal atmosphere. However, for that case Poisson's equation collapses into the trivial $0 = 0$. After all, g was approximated as constant, so $\nabla^2 \Phi = 0$. At the same time, ρ on the right-hand side of Poisson's equation was taken to be zero: the particles of the atmosphere were assumed to only *experience* gravity and not to be counted as mass charges in their own right.

3.1.1 Problems

1. A static infinite slab of incompressible self-gravitating fluid of density ρ occupies the region $|z| < a$.
 - (a) Find the gravitational field everywhere and the pressure distribution within the slab.
 - (b) If a galactic disc is approximated by a uniform density slab with density $10^{-18} \text{ kg m}^{-3}$ and $a = 10^{18} \text{ m}$, determine the velocity of a star at the mid-plane if it starts from rest at $z = a$, and the period of its oscillation.
2. A particle is released at rest at radius R_0 from the centre of a body of mass M . Compute
 - (a) its initial acceleration,
 - (b) the time it takes to reach the centre of the body

for the two cases

- (i) that the body is a point mass,
- (ii) that the body is a uniform sphere of radius R_0 .

A cluster consists initially of stars at rest, distributed in a uniform sphere. Find how long it takes a star to reach the centre as a function of its initial radius in the cluster and comment on your results.

3. A planet consists of two layers with constant mass density

$$\rho(r) = \begin{cases} \rho_1 & r \leq a_1 \\ \rho_2 & a_1 < r \leq a \\ 0 & r > a \end{cases} \quad (154)$$

- (a) Calculate the strength of gravity and the potential.
- (b) Show that the strength of gravity at the boundary between the layers is greater than at the surface when

$$\frac{\rho_1 - \rho_2}{\rho_2} > \frac{a^2}{a_1(a + a_1)}. \quad (155)$$

Verify that this is fulfilled for the Earth.

4. A spherical planet has mass distribution of the form

$$\rho(r) = Ar^\alpha, \quad r \leq a. \quad (156)$$

- (a) Calculate the gravitational field strength and the potential inside the planet for this distribution.
- (b) For what values of α is the problem solvable with finite planet mass?
- (c) For what value of α does gravity grow stronger towards the centre?

5. An “exponential star” has a mass density

$$\rho = \rho_0 e^{-r/a}, \quad (157)$$

where ρ_0 is the central mass density and a is the “radius”. Calculate the gravitational field and potential.

- 6. (a) Calculate the gravitational potential and field from a mass distribution shaped like a very thin line (a model of a cosmic string) of length $2a$ with uniform mass λ per unit length.
- (b) Calculate the behaviour of the potential at infinity orthogonally to the line.
- (c) Discuss what happens in the limit $a \rightarrow \infty$.
- 7. (a) Write an expression for the gravitational potential from a mass distribution shaped like a very thin circular plate of radius a with uniform mass σ per unit area (a model of the luminous matter of a galaxy).
- (b) Calculate the value of the potential along the central normal of the plate.
- (c) Calculate its form far from the disk.
- (d) What happens for $a \rightarrow \infty$?

3.2 Self-gravitating mass

Gravitation, in this case, is that of a body acting on itself. This is the case for the Earth, stars, galaxies, and clusters of galaxies—objects whose global behavior, to some extent, can be approximated by that of a fluid.

From hydrostatics equilibrium condition and taking the divergence, we obtain

$$\vec{\nabla} \cdot \left(\frac{\vec{\nabla} p}{\rho} \right) + \nabla^2 \Phi = 0, \quad (158)$$

and using the Poisson’s equation it follows that

$$\vec{\nabla} \cdot \left(\frac{\vec{\nabla} p}{\rho} \right) + 4\pi G \rho = 0. \quad (159)$$

In this equation, in general, $\rho = \rho(\vec{r})$ and $p = p(\vec{r})$, so it is valid even for compressible fluids. With the above equation and an equation of state one can build simple models of the Earth, stars, or galaxies.

3.3 An isothermal slab model

The hydrostatic plane-parallel atmosphere model discussed above involves a number of simplifying assumptions that will not be generally applicable. It assumes the atmosphere to be:

1. isothermal,
2. plane-parallel,
3. with negligible gravitational pull of its own, and
4. obeying a single equation of state (both on account of being isothermal and consisting of the same particles throughout).

Devising a stellar-structure model will quickly prompt us to drop at least assumptions 1–3, even while retaining the still massively simplifying assumption (4), and we will return to this later. Meanwhile, let us relax only assumption (3), as a stepping stone to stellar-structure modeling, and construct a slightly more sophisticated atmospheric profile model (the “isothermal slab”).

We consider an infinite (in x and y) static isothermal slab, symmetric about $z = 0$, supported by gas pressure and under its own self-gravity. No other forces are acting. An example of the sort of situation where this might arise in an astronomical context is if two clouds collide and generate a shocked slab of gas between them. Of course we are never in reality dealing with infinite systems but for our purposes here we just require that the slab diameter is very large compared with the slab thickness and hence that we can consider all gravitational forces to act in the z direction.

Because we deal with an isothermal ideal gas, the equation of state $pV = nRT = (N/N_A)RT = Nk_B T$ can be written as

$$p = \frac{N}{V} k_B T = \frac{Nm}{V} \frac{k_B T}{m} = \frac{M}{V} \frac{k_B T}{m} = \rho \frac{k_B T}{m}. \quad (160)$$

This means that for constant T it may be written as

$$p = A\rho, \quad (161)$$

where A is a constant. The geometry is plane-parallel, so all quantities depend only on z and $\nabla = \partial/\partial z$. Hydrostatic equilibrium then gives

$$A \frac{1}{\rho} \nabla \rho = -\nabla \Phi. \quad (162)$$

Since the variation is only with z ,

$$A \frac{d}{dz} \ln \rho = -\frac{d\Phi}{dz} \quad \Rightarrow \quad \Phi = -A \ln \left(\frac{\rho}{\rho_0} \right) + \Phi_0, \quad (163)$$

with $\rho_0 = \rho|_{z=0}$ and $\Phi_0 = \Phi|_{z=0}$. Hence

$$\rho(z) = \rho_0 \exp \left[-\frac{\Phi - \Phi_0}{A} \right]. \quad (164)$$

The profile for $\Phi(z)$ for plane-parallel layers can be determined now that we have a complementary expression for density in terms of gravitational potential that we can apply to Poisson's equation:

$$\frac{d^2 \Phi}{dz^2} = 4\pi G \rho_0 \exp \left[-\frac{\Phi - \Phi_0}{A} \right]. \quad (165)$$

Introduce the dimensionless variables

$$\chi \equiv -\frac{\Phi - \Phi_0}{A}, \quad Z \equiv \left(\frac{2\pi G \rho_0}{A} \right)^{1/2} z, \quad (166)$$

to obtain the ODE

$$\frac{d^2 \chi}{dZ^2} = -2e^\chi, \quad (167)$$

with boundary conditions $\chi(0) = 0$ and, by symmetry, $d\chi/dZ|_0 = 0$.

Multiply by $d\chi/dZ$ and integrate:

$$\frac{1}{2} \frac{d}{dZ} \left(\frac{d\chi}{dZ} \right)^2 = -2 \frac{d}{dZ} [e^\chi] \quad \Rightarrow \quad \left(\frac{d\chi}{dZ} \right)^2 = c_1 - 4e^\chi. \quad (168)$$

The boundary conditions at $Z = 0$ give $c_1 = 4$, hence

$$\frac{d\chi}{dZ} = 2\sqrt{1 - e^\chi}. \quad (169)$$

Integrate this by letting $s^2 = e^\chi$:

$$\int \frac{d\chi}{\sqrt{1 - e^\chi}} = \int \frac{2 ds}{s\sqrt{1 - s^2}} = 2 \int \frac{d\theta}{\sin \theta} = 2 \int \frac{dt}{t}, \quad (170)$$

where $s = \sin \theta$ and $t = \tan(\theta/2)$. Therefore

$$2 \ln t = 2Z + c_2. \quad (171)$$

With $\chi(0) = 0 \Rightarrow s(0) = 1 \Rightarrow \theta(0) = \frac{\pi}{2} \Rightarrow t(0) = 1$, we get $c_2 = 0$ and thus $t = e^Z$. Back-substituting,

$$s = \frac{2t}{1 + t^2} = \frac{1}{\cosh Z} \quad \Rightarrow \quad e^{\chi/2} = \frac{1}{\cosh Z} \quad \Rightarrow \quad \chi = -2 \ln(\cosh Z). \quad (172)$$

Returning to the original variables,

$$\Phi - \Phi_0 = 2A \ln \left[\cosh \left(\sqrt{\frac{2\pi G \rho_0}{A}} z \right) \right]. \quad (173)$$

Finally, the isothermal-slab density profile is

$$\rho(z) = \frac{\rho_0}{\cosh^2 \left(\sqrt{\frac{2\pi G \rho_0}{A}} z \right)}. \quad (174)$$

3.4 General isothermal slab density profiles

The general solution to the differential equation Eq. (167) is

$$\chi = -2 \ln \left[c_1 \cosh \left(\frac{Z}{c_1} + c_2 \right) \right]. \quad (175)$$

with the corresponding density profile being

$$\rho = \frac{\rho_0}{c_1^2 \cosh^2 \left(\sqrt{\frac{2\pi G \rho_0}{A}} \left(\frac{z}{c_1} + c_2 \right) \right)}. \quad (176)$$

Problem: show the previous expressions.

3.4.1 A fully self-gravitating plane

Let us consider a fully self-gravitating plane that is not resting on top of a massive foundation (e.g., a galactic disc rather than an atmosphere on top of a planet). Picking $z = 0$ to be the plane of symmetry, we set the gravitational potential at $z = z_0 = 0$ equal $\Phi = \Phi_0$, such that $\chi(Z = 0) = 0$. On account of the symmetry, $z = 0$ represents an extremum of the gravitational potential, so $d\chi/dZ = 0$ at $z = 0$. When implementing these boundary conditions in Eqs. (175) and (176), it turns out that $c_1 = 1$ and $c_2 = 0$, which is the result reported in Eqs. (173) and (174).

3.4.2 A planetary atmosphere

In the case of a planetary atmosphere, $\Phi(z) = gz$ such that $\Phi(z = 0) = \Phi_0 = 0$. Hence $\chi_0 = 0$. The gravitational field for a planet of mass M and radius R is given by $g = GM/R^2$. We also have that

$$\left. \frac{d\chi}{dZ} \right|_0 = -(2\pi G\rho_0 A)^{-1/2} g = -(2\pi G\rho_0 A)^{-1/2} \frac{GM}{R^2}. \quad (177)$$

The general expressions for the constants are

$$c_1 = (x^2 + 1)^{-1/2}, \quad c_2 = \ln \left[x + \sqrt{x^2 + 1} \right], \quad (178)$$

with

$$x \equiv \frac{1}{\sqrt{2\pi G\rho_0 A}} \frac{MG}{2R^2}. \quad (179)$$

Limiting behaviour. The ad-hoc variable x scales linearly with the planet mass M , so for the galactic disc case where this mass is absent, we can take the limit $M \rightarrow 0$ and find:

$$C_1 = (0 + 1)^{-1/2} = 1, \quad C_2 = \ln [0 + (0 + 1)^{1/2}] = 0, \quad x = 0. \quad (180)$$

The corresponding density profile Eq. (176) becomes

$$\rho = \frac{\rho_0}{\cosh^2 \left(\sqrt{\frac{2\pi G\rho_0}{A}} z \right)}, \quad (181)$$

all of which matching our earlier statements for this scenario.

For a large mass M we have

$$c_1 \approx x^{-1}, \quad c_2 \approx \ln(2x), \quad x \gg 1. \quad (182)$$

Substituting these into the density profile Eq. (176) and using $\cosh y = (e^y + e^{-y})/2 \sim e^y/2$ for large y , we obtain

$$\rho \approx \rho_0 x^2 4 \left[\exp\left(\frac{zMG}{2AR^2} + \ln(2x)\right) + \exp\left(-\frac{zMG}{2AR^2} - \ln(2x)\right) \right]^{-2} \quad (183)$$

$$\approx \rho_0 x^2 4 \left[2x \exp\left(\frac{zMG}{2AR^2}\right) \right]^{-2} \quad (184)$$

$$\approx \rho_0 \exp\left(-\frac{zMG}{AR^2}\right). \quad (185)$$

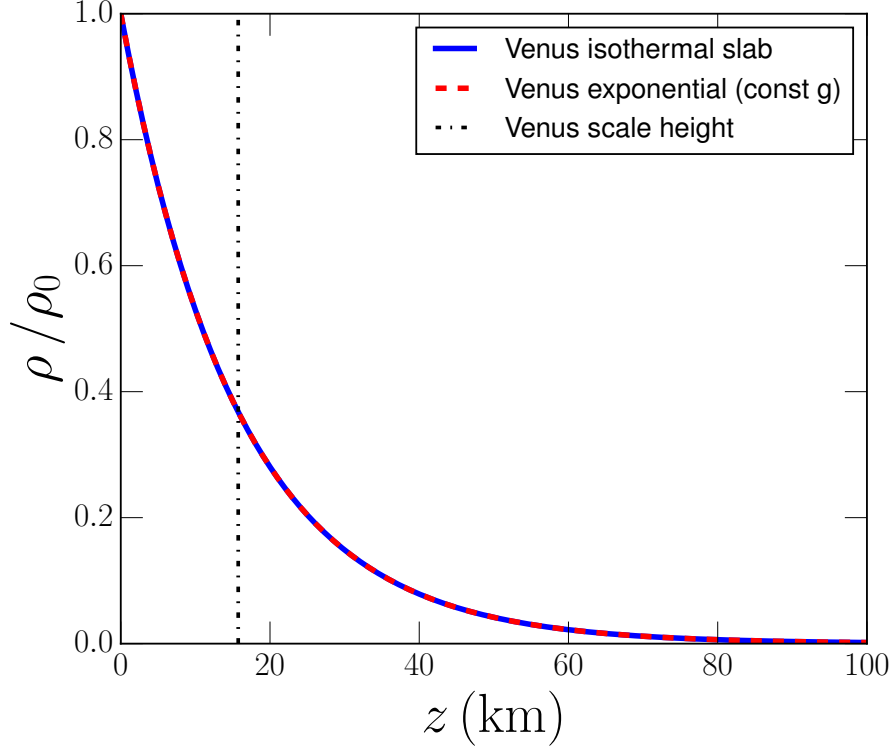


Figure 21: Isothermal slab model applied to Venus. The difference between the full expression and an exponential profile is negligible. In this case, $M \approx 0.815 M_{\oplus}$, $R \approx 0.95 R_{\oplus}$, and $p_0 \approx 93 \times 10^5$ Pa, leading to $x \approx 70 \gg 1$.

Thus, in the large- M limit we recover the isothermal-atmosphere result (Eq. (95)) in which the self-gravity of the atmosphere is negligible.

The general expressions for constants c_1 and c_2 suggest that this limit applies if we can safely assume $x \gg 1$. Other than on G and some mathematical constants, x depends on M , R and surface pressure $p_0 = \rho_0 A$. It should therefore be simple to verify this assumption, for example in the case of the Earth’s atmosphere (assuming for the sake of argument that we are not already completely convinced by the fact that the total mass of the Earth’s atmosphere is six orders of magnitude smaller than the total mass of the Earth). In this case, $M = M_{\oplus} = 5.974 \times 10^{27}$ g, $R = R_{\oplus} = 6.378 \times 10^8$ cm and $p_0 = 1$ atm $= 1.013 \times 10^6$ Ba (the latter being the cgs unit ‘Barye’ for pressure). This indeed results in $x \approx 752 \gg 1$. Even for Venus, with its extremely high atmospheric pressure, this limit safely applies. In this case, $M \approx 0.815 M_{\oplus}$, $R \approx 0.95 R_{\oplus}$ and $p_0 \approx 93 \times 10^6$ Ba, leading to $x \approx 70 \gg 1$ (see Fig. 21).

3.4.3 A curved isothermal atmosphere without self-gravity

Here we study what happens if we no longer neglect curvature but retain the assumptions of an isothermal atmosphere and no self-gravity.

1. Assume a gravitational acceleration of magnitude $g = MG/r^2$ and an isothermal atmosphere with $p = A\rho$. Neglect self-gravity (i.e. treat M as a constant throughout the atmosphere).

Show that the density profile obeys

$$\rho = \rho_0 \exp\left(\frac{MG}{Ar} - \frac{MG}{Ar_0}\right), \quad (186)$$

where ρ_0 is the mass density at a reference radius r_0 .

2. Show how a series approximation of the previous result reduces (to lowest order) to the case of an isothermal plane-parallel atmosphere without self-gravity when considered only at small distances beyond the reference radius r_0 .

3.5 Spherical bodies

If there is spherical symmetry, i.e., density and pressure depend only on the radial coordinate, it follows that

$$\vec{g}(r) = -\frac{G_N M(r)}{r^2} \hat{e}_r, \quad (187)$$

and

$$\vec{\nabla} p(r) = \frac{dp(r)}{dr} \vec{\nabla} r = \frac{dp(r)}{dr} \hat{e}_r, \quad (188)$$

from which the hydrostatic equilibrium equation takes the form

$$\frac{dp(r)}{dr} = \rho(r)g(r) = -\rho(r)\frac{G_N M(r)}{r^2}. \quad (189)$$

Multiplying by r^2/ρ and differentiating after with respect to r we arrive to

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dp}{dr} \right) + 4\pi G \rho = 0. \quad (190)$$

3.5.1 Boundary conditions

In principle, a second-order differential equation requires two boundary values (or integration constants), for example the central pressure $p_c = p(0)$ and its first derivative dp/dr for $r = 0$. We shall make the reasonable assumption that the density ρ_c at the centre of the body is finite. Then for ‘small’ r we have $M(r) \approx \frac{4}{3}\pi r^3 \rho_c$ and equation (189) becomes for $r \rightarrow 0$,

$$\frac{dp}{dr} \approx -\frac{4}{3}\pi G \rho_c^2 r, \quad (191)$$

which integrates to

$$p(r) \approx p_c - \frac{2}{3}\pi G \rho_c^2 r^2. \quad (192)$$

Thus, under the assumption of finite central density, the pressure is parabolic near the centre with $dp/dr = 0$ for $r = 0$. This shows that under reasonable physical assumptions the hydrostatic equation requires in fact only one boundary condition, for example the central pressure. Knowing p_c together with the equation of state (which also determines ρ_c), the pressure may be calculated throughout the body.

The central pressure and density are, of course, not known for planets and stars, objects that are only accessible from the outside. Most such bodies have a well-defined surface radius, $r = a$, at which the pressure vanishes. We shall arbitrarily call a body a planet, if the density jumps abruptly to zero at the surface, and a star if the density vanishes along with the pressure at the surface. Such a convention makes the gaseous giant planets, Jupiter and Saturn, count as stars even though they probably do not burn much hydrogen.

The requirement of zero pressure at $r = a$ will determine the central pressure. The solutions to the hydrostatic equation can be expressed entirely in terms of the radius of the body and the parameters in the equation of state. In particular the mass M_0 of the body is—as we shall see below—calculable in terms of a (and the state parameters). Conversely, if the mass and radius are known, one of the other unknown parameters may be determined.

3.5.2 Planet with constant density

For a planet with constant density, ρ_0 , the assumption of finite central density is exactly valid throughout the planet,

$$p = p_c - \frac{2}{3}\pi G \rho_0^2 r^2. \quad (193)$$

At the surface of the planet where the pressure has to vanish this leads to

$$p_c = \frac{2}{3}\pi G \rho_0^2 a^2. \quad (194)$$

If the mass and radius are known, the density is obtained from

$$M_0 = \frac{4}{3}\pi a^3 \rho_0. \quad (195)$$

3.5.3 The Moon with constant density

Let us consider the Moon. The Moon's mass is 7.3×10^{22} kg and its radius is 1738 km, making the average density $\rho_0^{\text{Moon}} = 3.34$ g/cm³. From this the central pressure is predicted to be 46 500 atm.

3.6 Pressure distribution in the Earth

The four main layers of the Earth, from outermost to innermost, are the **crust**, **mantle**, **outer core**, and **inner core**. The crust is the thin, rocky shell we live on; the mantle is a thick, dense layer of rock; the outer core is a liquid layer of iron and nickel; and the inner core is a solid, extremely hot ball of iron and nickel at Earth's center.

3.6.1 A two-layer model

Let us consider that the Earth consists of two layers with constant mass density,

$$\rho(r) = \begin{cases} \rho_1, & 0 \leq r \leq a_1, \\ \rho_2, & a_1 < r \leq a, \\ 0, & r > a. \end{cases}$$

Let $M(r)$ be the mass enclosed within radius r and $g(r)$ the (inward) gravitational field strength. By spherical symmetry,

$$g(r) = \frac{GM(r)}{r^2}, \quad \frac{d\Phi}{dr}(r) = -g(r), \quad \Phi(\infty) = 0,$$

so that $\Phi(r) = -\int_{\infty}^r g(r') dr'$.

Gravitational field and potential. The enclosed mass is

$$M(r) = \frac{4\pi}{3} \times \begin{cases} \rho_1 r^3, & 0 \leq r \leq a_1, \\ \rho_1 a_1^3 + \rho_2 (r^3 - a_1^3), & a_1 < r \leq a, \\ \rho_1 a_1^3 + \rho_2 (a^3 - a_1^3) \equiv M, & r > a. \end{cases}$$

Hence

$$g(r) = \frac{GM(r)}{r^2} = \begin{cases} \frac{4\pi G}{3} \rho_1 r, & 0 \leq r \leq a_1, \\ \frac{4\pi G}{3} \left(\rho_2 r + \frac{(\rho_1 - \rho_2) a_1^3}{r^2} \right), & a_1 < r \leq a, \\ \frac{GM}{r^2}, & r > a. \end{cases}$$

With a zero potential at infinity $\Phi(\infty) = 0$ one gets the potential

$$\Phi(r) = \begin{cases} -\frac{GM}{a} + \frac{2\pi G \rho_2}{3} (a^2 - a_1^2) + \frac{4\pi G}{3} (\rho_1 - \rho_2) a_1^2 \left(1 - \frac{a_1}{a}\right) + \frac{2\pi G \rho_1}{3} (a_1^2 - r^2), & 0 \leq r \leq a_1, \\ -\frac{GM}{a} + \frac{2\pi G \rho_2}{3} (a^2 - r^2) + \frac{4\pi G}{3} (\rho_1 - \rho_2) a_1^3 \left(\frac{1}{r} - \frac{1}{a}\right), & a_1 < r \leq a, \\ -\frac{GM}{r}, & r > a. \end{cases}$$

Where is gravity stronger: at $r = a_1$ or at $r = a$? At the interface $r = a_1$,

$$g(a_1) = \frac{4\pi G}{3} \rho_1 a_1,$$

while at the surface

$$g(a) = \frac{GM}{a^2} = \frac{4\pi G}{3a^2} [\rho_1 a_1^3 + \rho_2 (a^3 - a_1^3)].$$

The condition $g(a_1) > g(a)$ is therefore equivalent (after cancelling $\frac{4\pi G}{3}$) to

$$\rho_1 a_1 > \frac{\rho_1 a_1^3 + \rho_2 (a^3 - a_1^3)}{a^2} \iff \frac{\rho_1 - \rho_2}{\rho_2} > \frac{a^2}{a_1(a + a_1)}.$$

Numerical check for Earth. Take $a \simeq 6371$ km, $a_1 \simeq 3480$ km (core radius), $\rho_1 \simeq 11$ – 13 g cm⁻³ (core), $\rho_2 \simeq 4.5$ – 5.5 g cm⁻³ (mantle). Then

$$\frac{\rho_1 - \rho_2}{\rho_2} \approx 1.2\text{--}1.9, \quad \frac{a^2}{a_1(a + a_1)} \approx 1.18,$$

so the inequality is satisfied, and the gravitational field is indeed stronger at the core–mantle boundary than at the surface.

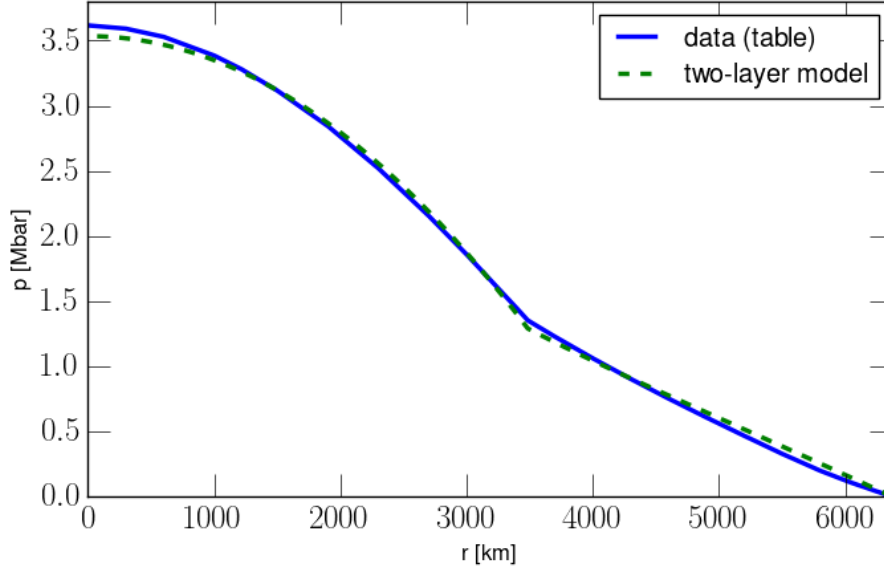


Figure 22: Pressure distribution in the Earth. Solid line data from Table 3.6.1 and dashed: the two-layer model.

Hydrostatic pressure Hydrostatic balance and the enclosed mass are

$$\frac{dp}{dr} = -\rho \frac{GM(r)}{r^2}, \quad M(r) = 4\pi \int_0^r \rho(r') r'^2 dr'.$$

We use the boundary condition $p(a) = 0$ and integrate inward.

(i) *Mantle*, $a_1 < r \leq a$. Here $\rho = \rho_2$ and

$$M(r) = \frac{4\pi}{3} [\rho_1 a_1^3 + \rho_2 (r^3 - a_1^3)].$$

Hence

$$p(r) = \int_r^a \rho_2 \frac{GM(r')}{r'^2} dr' = \frac{4\pi G \rho_2}{3} \left[\frac{\rho_2}{2} (a^2 - r^2) + (\rho_1 - \rho_2) a_1^3 \left(\frac{1}{r} - \frac{1}{a} \right) \right].$$

(ii) *Core*, $0 \leq r \leq a_1$. Here $\rho = \rho_1$ and $M(r) = \frac{4\pi}{3} \rho_1 r^3$. Demanding continuity at $r = a_1$ gives

$$p(r) = p(a_1) + \int_r^{a_1} \rho_1 \frac{GM(r')}{r'^2} dr' = p(a_1) + \frac{2\pi G}{3} \rho_1^2 (a_1^2 - r^2),$$

with

$$p(a_1) = \frac{4\pi G \rho_2}{3} \left[\frac{\rho_2}{2} (a^2 - a_1^2) + (\rho_1 - \rho_2) a_1^2 \left(1 - \frac{a_1}{a} \right) \right].$$

Central pressure. At $r = 0$,

$$p_c = \frac{2\pi G}{3} \left[\rho_1^2 a_1^2 + \rho_2^2 (a^2 - a_1^2) + 2\rho_2(\rho_1 - \rho_2) a_1^3 \left(\frac{1}{a_1} - \frac{1}{a} \right) \right].$$

Numerical estimate for Earth. Taking

$$a = 6.371 \times 10^6 \text{ m}, \quad a_1 = 3.48 \times 10^6 \text{ m}, \quad \rho_1 \simeq 1.2 \times 10^4 \text{ kg m}^{-3}, \quad \rho_2 \simeq 5.0 \times 10^3 \text{ kg m}^{-3},$$

the above expression yields

$$p_c \approx 3.9 \times 10^{11} \text{ Pa} \quad (\approx 390 \text{ GPa}),$$

consistent with accepted seismological values ($\sim 3.6 \times 10^{11} \text{ Pa}$). See Fig. 22.

3.6.2 Problems

1. Show that for a planet with constant density and fixed mass, the central pressure falls as a^{-4} .
2. Consider a spherically symmetric self-gravitating astrophysical object—for example, a galaxy—with density $\rho(r) = \rho_0 \left(1 - \frac{r^2}{a^2}\right)$ for $0 \leq r \leq a$, where a is the sphere's radius. Compute dp/dr and $p(r)$ by imposing $p = 0$ at $r = a$ and regularity (no divergences). What is the total mass?
3. Evaluate the pressure distribution $p(r)$ inside a uniform self-gravitating spherical fluid of constant density ρ , radius R , and mass M .
4. Evaluate $p(r)$ for a sphere of self-gravitating gas of radius R and mass M (e.g., a star) if its equation of state is $p(r) = \alpha \rho(r)$.

3.7 Idealized stellar structure models

If we wish to model stellar structure from the centre of a star to its outer edge, we cannot assume the profile to be plane-parallel and neither can we neglect self-gravity. A full discussion of stellar structure lies beyond the scope of this course, but isothermal and polytropic stellar-structure models provide us with nice and insightful applications of hydrostatics. Considering a polytropic fluid, the resulting expression, the Lane-Emden equation, is a famous cornerstone of stellar-structure theory.

3.7.1 Stars as self-gravitating polytropes

We assume a spherical system in hydrostatic equilibrium (no rotation that would break spherical symmetry). As always,

$$\nabla p = -\rho \nabla \Phi.$$

In spherical coordinates this reads

$$\frac{dp}{dr} = -\rho \frac{d\Phi}{dr}.$$

Since $\rho > 0$ within the star, p is a monotonic function of Φ ; hence we may write $p = p(\Phi)$. Then

$$\frac{dp}{dr} = \frac{dp}{d\Phi} \frac{d\Phi}{dr} = -\rho \frac{d\Phi}{dr} \quad \Rightarrow \quad \rho = -\frac{dp}{d\Phi} \quad \Rightarrow \quad \rho = \rho(\Phi).$$

Because $p = p(\Phi)$ and $\rho = \rho(\Phi)$, it follows that $p = p(\rho)$; non-rotating stars are therefore *barotropes* (surfaces of constant p, ρ, Φ coincide).

A convenient barotropic equation of state is the *polytrope*

$$p = K \rho^{1+\frac{1}{n}},$$

where n is the polytropic index. Substituting this EOS into hydrostatic equilibrium gives

$$-\nabla\Phi = \frac{1}{\rho} \nabla \left(K \rho^{1+\frac{1}{n}} \right) = (n+1) \nabla \left(K \rho^{\frac{1}{n}} \right).$$

Therefore

$$\rho = \left(\frac{\Phi_T - \Phi}{(n+1)K} \right)^n,$$

where Φ_T is the value of the potential at the stellar surface ($\rho = 0$).

Using Poisson's equation,

$$\nabla^2\Phi = 4\pi G \rho = 4\pi G \left(\frac{\Phi_T - \Phi}{(n+1)K} \right)^n.$$

Let ρ_c and Φ_c denote the density and potential at the centre. Then

$$\rho = \rho_c \left(\frac{\Phi_T - \Phi}{\Phi_T - \Phi_c} \right)^n, \quad \nabla^2\Phi = 4\pi G \rho_c \left(\frac{\Phi_T - \Phi}{\Phi_T - \Phi_c} \right)^n.$$

Define the dimensionless function

$$\theta = \frac{\Phi_T - \Phi}{\Phi_T - \Phi_c} \quad \Rightarrow \quad \Phi = -(\Phi_T - \Phi_c) \theta + \Phi_T.$$

Then

$$\nabla^2\theta = -\frac{4\pi G \rho_c}{\Phi_T - \Phi_c} \theta^n.$$

In spherical polars,

$$\nabla^2\theta = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\theta}{dr} \right).$$

Centrifugal deformation of a spherical planet

Introduce the dimensionless radius

$$\xi = \left(\frac{4\pi G \rho_c}{\Phi_T - \Phi_c} \right)^{1/2} r,$$

to obtain the *Lane–Emden equation of index n* :

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n.$$

Boundary conditions at the centre:

- $\theta = 1$ at $\xi = 0$ (from the definition of θ),
- $\frac{d\theta}{d\xi} = 0$ at $\xi = 0$. This implies zero force at $\xi = 0$, which is true for gas spheres because the enclosed mass $\rightarrow 0$ as $\xi \rightarrow 0$. It is not true if there is a point mass in the middle of the sphere.

This equation has closed-form solutions for $n = 0, 1, 5$, but generally must be integrated numerically.

It was the first equation to succeed in describing the internal structure of a self-gravitating polytropic body. For nearly a century (from about 1867, when Lane first derived it, to the early 1950s), it was the primary model for the discussion of stellar interiors.

3.8 Solutions for the Lane–Emden equation

3.8.1 Solution for $n = 0$

For $n = 0$,

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -1. \quad (196)$$

Integrating once,

$$\xi^2 \frac{d\theta}{d\xi} = -\frac{1}{3} \xi^3 - C, \quad (197)$$

so

$$\theta = D + \frac{C}{\xi} - \frac{1}{6} \xi^2, \quad (198)$$

with constants C, D . Regularity at the origin together with $\theta(0) = 1$ and $d\theta/d\xi|_{\xi=0} = 0$ require $C = 0$, $D = 1$, hence

$$\theta_0 = 1 - \frac{\xi^2}{6}. \quad (199)$$

The surface of the polytrope is where the function θ is zero, i.e.

$$\xi_1 = \sqrt{6}.$$

(The subscript 0 is just to highlight that this is the solution only for $n = 0$.)

3.8.2 Solution for $n = 1$

For $n = 1$,

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta. \quad (200)$$

Let $\theta = \chi/\xi$. The Lane–Emden equation becomes

$$\frac{d^2\chi}{d\xi^2} = -\chi. \quad (201)$$

Thus $\chi = A \sin(\xi + B)$ and

$$\theta = \frac{A \sin(\xi + B)}{\xi}. \quad (202)$$

Regularity at $\xi = 0$ imposes $A = 1$, $B = 0$, giving

$$\theta_1 = \frac{\sin \xi}{\xi}. \quad (203)$$

This has its first zero at $\xi = \pi$, and is monotonically decreasing over the range $(0, \pi)$.

3.8.3 Solution for $n = 5$

For $n = 5$,

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^5. \quad (204)$$

Set $x = 1/\xi$. Then the equation can be written

$$x^4 \frac{d^2\theta}{dx^2} = -\theta^5. \quad (205)$$

Introduce $t = \ln x$ and

$$\theta = \left[\frac{2(n-3)}{(n-1)^2} \right]^{1/(n-1)} x^{\frac{2}{n-1}} z(t). \quad (206)$$

This yields

$$\frac{d^2z}{dt^2} + \frac{5-n}{n-1} \frac{dz}{dt} - \frac{2(n-3)}{(n-1)^2} z(1-z^{n-1}) = 0. \quad (207)$$

For $n = 5$ this reduces to

$$\frac{d^2z}{dt^2} = \frac{1}{4} z(1-z^4). \quad (208)$$

Multiplying by dz/dt and integrating,

$$\frac{1}{2} \frac{d}{dt} \left(\frac{dz}{dt} \right)^2 = \frac{1}{4} z(1-z^4) \frac{dz}{dt} \Rightarrow \frac{1}{2} \left(\frac{dz}{dt} \right)^2 = \frac{1}{8} z^2 - \frac{1}{24} z^6 + D. \quad (209)$$

Boundary conditions imply $D = 0$. With the substitution $\frac{1}{3} z^4 = \sin^2 \phi$ one finds $\tan(\phi/2) = Ce^{-t}$, leading finally to

$$\theta_5 = \frac{1}{\left(1 + \frac{1}{3} \xi^2\right)^{1/2}} \quad (210)$$

which vanishes only as $\xi \rightarrow \infty$; the equilibrium configuration thus extends to infinity.

3.8.4 Exercises

1. The isothermal ($n = \infty$) self-gravitating sphere

Consider a spherically symmetric, self-gravitating gas in hydrostatic equilibrium with the isothermal equation of state

$$p = K \rho, \quad K = \text{const.}$$

The hydrostatic balance can be written as

$$K \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{1}{\rho} \frac{d\rho}{dr} \right) = -4\pi G \rho.$$

Let the central density be ρ_c , and introduce the usual isothermal nondimensionalization

$$\rho(r) = \rho_c e^{-\psi(\xi)}, \quad r = a \xi, \quad a \equiv \sqrt{\frac{K}{4\pi G \rho_c}}.$$

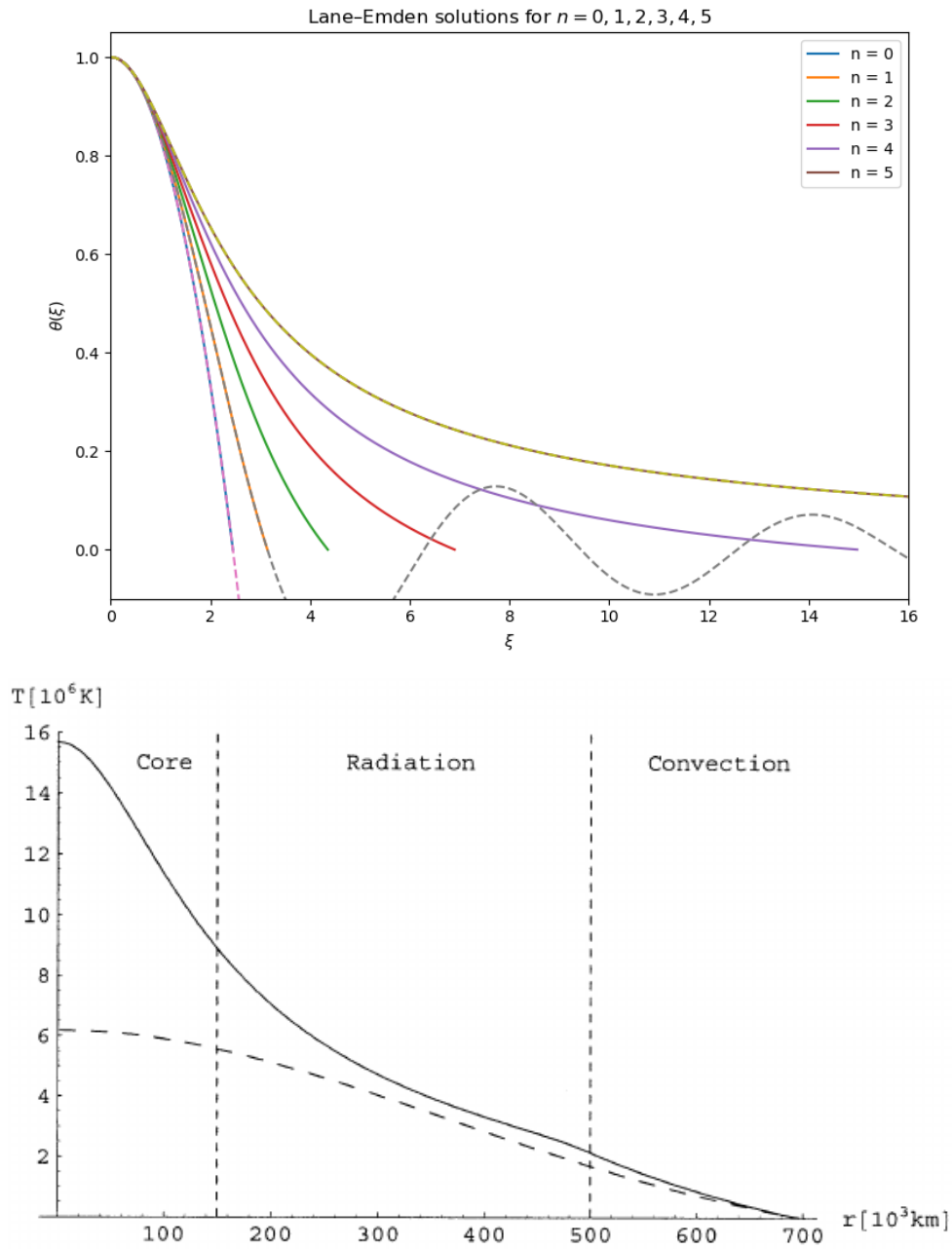


Figure 23: Top: Solutions to the Lane-Emden equation for various values of n . The higher n , the larger the ξ value for which the solution crosses $\theta = 0$. Bottom: The temperature distribution in the Sun as a function of the distance from the centre. The fully drawn curve is from the ‘standard’ Sun model and the dashed curve is the Lane-Emden solution for $\gamma = 5/3$.

- (a) Derive the isothermal Lane–Emden equation.
Show that the dimensionless function $\psi(\xi)$ satisfies

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\psi}{d\xi} \right) = e^{-\psi},$$

with regularity (central) boundary conditions

$$\psi(0) = 0, \quad \psi'(0) = 0.$$

- (b) Central series expansion.

Seek a power-series (Frobenius) solution about $\xi = 0$, $\psi(\xi) = \sum_{k=1}^{\infty} a_{2k} \xi^{2k}$. Obtain the first two nonzero coefficients and verify that

$$\psi(\xi) = \frac{\xi^2}{6} - \frac{\xi^4}{120} + \mathcal{O}(\xi^6), \quad \rho(\xi) = \rho_c \left(1 - \frac{\xi^2}{6} + \frac{\xi^4}{180} + \cdots \right).$$

- (c) Asymptotic behaviour at large radius.

Assume that ψ grows slowly for $\xi \gg 1$ and show that a consistent leading-order solution satisfies

$$\psi(\xi) \sim 2 \ln \xi + C, \quad \text{as } \xi \rightarrow \infty,$$

hence

$$\rho(r) \propto \frac{1}{r^2} \quad (\text{for large } r).$$

- (d) Mass inside a finite radius.

Show that the dimensional mass within $R = a \xi_b$ may be written purely in terms of ψ and its derivative at the boundary:

$$M(R) = 4\pi a^3 \rho_c \int_0^{\xi_b} e^{-\psi(\xi)} \xi^2 d\xi = 4\pi a^3 \rho_c \xi_b^2 \psi'(\xi_b).$$

Use the asymptotic result from part (3) to argue that $M(R)$ diverges linearly with R , i.e. an isothermal sphere of infinite extent has infinite mass.

- (e) Scaling and a numerical estimate (optional).

For an isothermal gas with temperature T and mean molecular weight μ , $K = k_B T / \mu$. Taking $T = 10^4$ K, $\mu = 0.6 m_p$, and $\rho_c = 10^{-19}$ kg m⁻³:

- i. Compute the scale length $a = \sqrt{K / (4\pi G \rho_c)}$.
- ii. Using either a numerical integration of the isothermal Lane–Emden equation (or the large- ξ asymptotic form), estimate the mass enclosed within $R = 10 a$.

Hints.

- To obtain the identity in part (d), integrate the isothermal Lane–Emden equation once to get $(\xi^2 \psi')' = \xi^2 e^{-\psi}$.
- For part (c), try an ansatz $\psi'(\xi) \sim A/\xi$ as $\xi \rightarrow \infty$ and determine A by balancing the dominant terms.

3.9 Exercises

- 1) (a) If the Earth's atmosphere can be approximated by a perfect static gas at constant temperature of 300 K subject to a uniform gravitational field, find the variation in number density (molecules per cubic metre) with height above the Earth's surface. If the number density of molecules at the Earth's surface is $3 \times 10^{25} \text{ m}^{-3}$, estimate the height above the Earth where the fluid approximation breaks down, and compare it with the height at which the assumption of constant gravity breaks down.
- (b) The Earth runs into a cloud which is stationary with respect to the Sun. Estimate the number density the cloud would have to have in order that it seriously disturbed the Earth's atmosphere.
(Some data possibly relevant to this question: $T \sim 300 \text{ K}$, $\mu \sim 30$, $R_* = 8300$, $g = 10 \text{ m s}^{-2}$, $M_\odot = 2 \times 10^{30} \text{ kg}$.)

- 2) Estimate the temperature in the core of the Sun if the Sun is supported by gas pressure. Assume all quantities vary over a radial scale length of order the radius of the Sun. Estimate the corresponding temperature if the Sun is supported by radiation pressure. Which is likely to be the case?
(Radiation pressure is $\frac{1}{3}aT^4$ where $a = 7.6 \times 10^{-14} \text{ J m}^{-3} \text{ K}^{-4}$. The radius of the Sun is $6 \times 10^8 \text{ m}$. $M_\odot = 2 \times 10^{30} \text{ kg}$.)

- 3) The structure of a star in hydrostatic equilibrium is described by

$$M(r) = \frac{M_0}{\left[1 + \left(\frac{r_0}{r}\right)^2\right]^{3/2}}, \quad (211)$$

where $M(r)$ is the mass enclosed within radius r , and M_0 and r_0 are constants. Determine

- (a) the density distribution, and
- (b) the pressure distribution within the star.

Show that the star can be described as a polytrope, and determine the polytropic index n .

- 4) A planet is composed of a material that is incompressible, density ρ , at pressures $\leq p_0$. Show that the maximum mass of such a planet that is incompressible throughout its interior is given by

$$M_{\text{max}} = \frac{2}{3\rho^2} \sqrt{\frac{1}{2\pi}} \left(\frac{3p_0}{G}\right)^{3/2}. \quad (212)$$

- 5) An equilibrium ring of isothermal fluid orbits a star at radius R . In the plane of the ring, mechanical equilibrium results from a balance of centrifugal force and the gravitational force of the central object; normal to the ring (i.e. vertically) equilibrium is between the vertical component of the gravitational force of the central object and vertical pressure gradients in the ring gas.

Show that in the limit that the ring thickness $H \ll R$, the vertical density stratification in the ring is a Gaussian and determine its e -folding length in terms of the gas temperature and the angular velocity at the ring, Ω . Hence determine an upper limit to the temperature such that the ring is thin ($H \ll R$) and calculate this temperature if the ring's radius is that of the Earth's orbit around the Sun.

6) Show that if

$$\Phi = -\frac{GM_s}{\sqrt{r^2 + b^2}} \quad (213)$$

is the gravitational potential for a spherical distribution of matter then its density $\rho \propto \Phi^5$. Deduce the pressure and hence show that the equation of state is barotropic with $n = 5$. Find the total internal energy U as a function of K , M_s and b where

$$K = \frac{P}{\rho^{6/5}}.$$

7) Sketch the density distribution for an isothermal slab and discuss the asymptotic limits $z \rightarrow 0$, $z \rightarrow \infty$.

A galactic disc can be well approximated in its vertical structure by an isothermal slab of gas, temperature T , central density ρ_0 . If a star falls from rest from a height z_0 , show that its vertical velocity at height z is given by

$$\dot{z}^2 = \frac{4R_*T}{\mu} \ln \left(\frac{\cosh(\alpha z_0)}{\cosh(\alpha z)} \right).$$

8) Explain (for the case of a polytrope, index n) why the internal energy per kg, $\mathcal{E}$, is equal to

$$\mathcal{E} = \int_0^\rho \frac{P}{\rho'^2} d\rho'$$

if and only if $\gamma = 1 + \frac{1}{n}$.

Calculate how the total internal energy of a polytropic star varies with stellar mass.

9) Derive the mass-radius relation for polytropic stars (equation of state $p = K\rho^{1+\frac{1}{n}}$) on the assumption that K varies with stellar mass in such a way as to maintain a constant central temperature independent of mass.

10) Show that the polytropic equation $\nabla^2\theta = -\theta^n$ has power law solutions $\theta \propto r^{-\zeta}$ provided $\zeta = \frac{2}{n-1}$. Give the corresponding density distribution and show that in these solutions the mass at small r is small provided $n > 3$ while the total mass at large r is small provided $1 < n < 3$.

For what values of n and ζ is the contribution to the total gravitational potential binding energy $|V|$ from material close to the origin small and when is it large?

$$|V| = G \int \frac{m dm}{r}.$$

4 Hydrostatics in non-inertial systems: hydrostatic shapes

Disregarding surface tension, in hydrostatic equilibrium with gravity, an interface between two fluids of different densities, for example the sea and the atmosphere, must coincide with a surface of constant potential, an equipotential surface. Otherwise, if an interface crosses an equipotential surface, there will arise a tangential component of gravity which can only be balanced by shear contact forces which a fluid at rest is unable to supply.

4.1 Fluid interfaces in hydrostatic equilibrium

For a fluid of density ρ —in general dependent on position— in a gravitational field $\vec{g}$:

$$-\rho\vec{g} + \vec{\nabla}p = 0 \quad \text{or:} \quad \rho\vec{\nabla}\Phi + \vec{\nabla}p = 0. \quad (214)$$

Taking the curl of the second equation,

$$\rho\vec{\nabla} \times \vec{\nabla}\Phi + \vec{\nabla}\rho \times \vec{\nabla}\Phi + \vec{\nabla} \times \vec{\nabla}p = 0. \quad (215)$$

Since $\vec{\nabla} \times \vec{\nabla}\Phi \equiv 0$ and $\vec{\nabla} \times \vec{\nabla}p \equiv 0$, it follows that, in a fluid of variable density, this condition holds:

$$\vec{\nabla}\rho \times \vec{\nabla}\Phi = 0 \quad \text{or:} \quad \vec{\nabla}\rho \times \vec{g} = 0. \quad (216)$$

This means that, point by point, the direction of the gravitational field lines coincides with that of the density gradient, or, equivalently, that the surfaces of equal density (isopycnics) and the gravitational equipotentials are coincident.

Considering

$$\vec{\nabla}\Phi + \frac{\vec{\nabla}p}{\rho} = 0,$$

and its curl is taken, it follows that for a compressible fluid

$$\vec{\nabla}\rho \times \vec{\nabla}p = 0, \quad (217)$$

from which it is concluded that the surfaces of constant pressure (isobars) and those of constant density coincide. hydrostatic equilibrium, and therefore with the isobars. *Note that the coincidence of isobars and isopycnics in hydrostatic equilibrium is valid in general, not only for barotropic fluids where pressure and density are simply related.*

Thus, the surfaces of Φ , ρ and p constant are the same. This result is valid regardless of the form of the gravitational field. In particular, if $\vec{g} = -\hat{k}g$, the surfaces Φ , ρ and p will be horizontal planes. Outside the Earth's mass, assumed spherical, it is true that

$$\vec{g} = -\frac{GM\hat{r}}{r^2}, \quad \Phi = -\frac{GM}{r},$$

so the three surfaces are spherical and concentric.

4.2 Motion in Accelerated Frames: inertial (fictitious) forces

Newton's Second Law of motion is only valid in *inertial* reference frames, where free particles move on straight lines with constant velocity. In classical mechanics a particle is commonly said

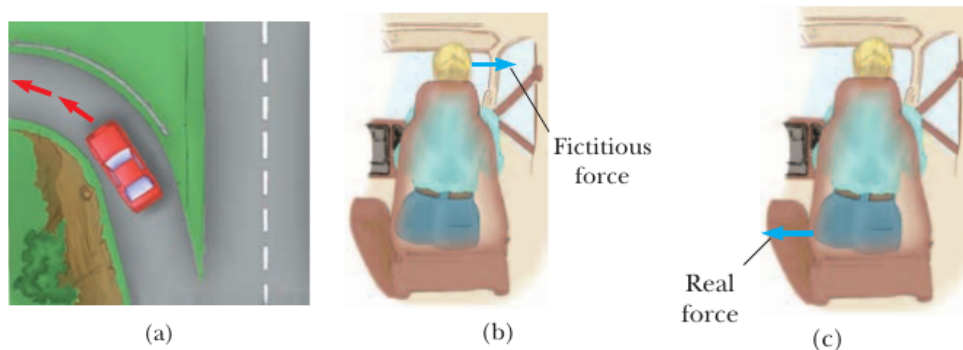


Figure 24: (a) A car approaching a curved exit ramp. What causes a passenger in the front seat to move toward the right-hand door? (b) From the passenger's frame of reference, a force appears to push her toward the right door, but it is a fictitious force. (c) Relative to the reference frame of the Earth, the car seat applies a real force toward the left on the passenger, causing her to change direction along with the rest of the car. Figure taken from Ref. [10].

to be free if it is not subject to forces caused by identifiable sources, for example the mass or electric charge of a material body. In reference frames that are accelerated or rotating relative to inertial frames, one may formally write the Second Law in its usual form, but the price to pay is the inclusion of certain force-like terms that do not have any obvious sources, but only depend on the motion of the reference frame. Such terms are called *fictitious forces*, although they are by no means pure fiction, as one becomes painfully aware when standing in a bus that suddenly stops. A more reasonable name might be *inertial forces*, since they arise as a consequence of the inertia of material bodies.

Review question. Consider the passenger in the car making a left turn in Figure 24. Which of the following is correct about forces in the horizontal direction if she is making contact with the right-hand door?

- The passenger is in equilibrium between real forces acting to the right and real forces acting to the left.
- The passenger is subject only to real forces acting to the right.
- The passenger is subject only to real forces acting to the left.
- None of these statements is true.

Review question. A small sphere of mass m hangs by a cord from the ceiling of a boxcar that is accelerating to the right as shown in Figure 25. The noninertial observer in the bottom panel claims that a force, which we know to be fictitious, causes the observed deviation of the cord from the vertical. How is the magnitude of this force related to the boxcar's acceleration measured by the inertial observer in the top panel?

- **Inertial vs. noninertial frames.** Newton's laws apply in inertial frames (no acceleration). In noninertial (accelerating or rotating) frames, objects appear to have unexplained accelerations that are opposite the frame's acceleration.

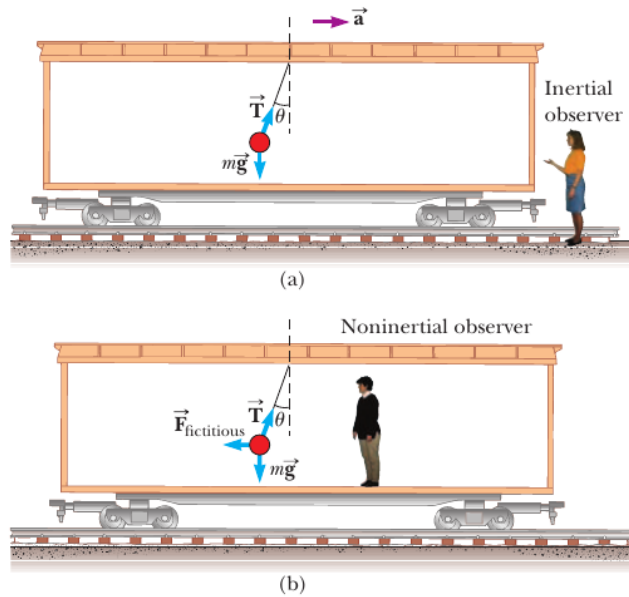


Figure 25: A small sphere suspended from the ceiling of a boxcar accelerating to the right is deflected as shown. (a) An inertial observer at rest outside the car claims that the acceleration of the sphere is provided by the horizontal component of $\vec{T}$. (b) A noninertial observer riding in the car says that the net force on the sphere is zero and that the deflection of the cord from the vertical is due to a fictitious force $\vec{T}_{\text{fictitious}}$ that balances the horizontal component of $\vec{T}$. Figure taken from Ref. [10].

- **Fictitious (inertial) forces.** To use Newton’s second law in a noninertial frame, one introduces “fictitious” forces that do not arise from interactions with other bodies but from the frame’s acceleration/rotation. They act as if real in that frame but cannot be attributed to a second interacting object.
- **Linear acceleration example (train).** On an accelerating train, a puck at rest (relative to the car) appears to accelerate toward the rear with no applied force; this is modeled by a fictitious force opposite the train’s acceleration.
- **Turning car example (centripetal vs. “centrifugal”).** Entering a curved exit ramp, a passenger tends to keep a straight-line path (inertia) and slides toward the door because friction with the seat is insufficient to supply the required *centripetal* force. The commonly cited “centrifugal force” is the fictitious force used in the car’s noninertial frame.
- **Friction’s role.** If friction is large enough, it provides the inward (toward the center of curvature) force that bends the passenger’s path to match the car’s curve; otherwise the passenger continues straight and strikes the door.
- **Coriolis force (rotation).** In rotating frames, motion relative to the frame experiences an apparent sideways deflection (Coriolis force). For example, a ball thrown across a rotating platform appears to curve to an observer riding the platform, despite being a straight line in an inertial frame.
- **Real effects of fictitious forces.** Although not interaction forces, fictitious forces have observable consequences in accelerating/rotating frames: objects slide on dashboards when

a car accelerates; riders feel pushed outward on a merry-go-round; on Earth's rotating frame, the Coriolis force helps organize hurricanes and influences large-scale ocean currents.

4.3 Linearly accelerated containers

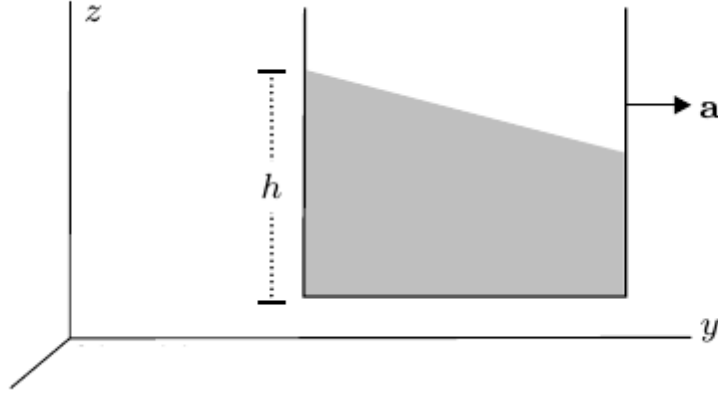


Figure 26: Linearly accelerating container. Figure taken from [1].

Near Earth's surface, a container of liquid is subjected to a constant horizontal acceleration in the y -direction, $\vec{a} = a \hat{j}$ (see Fig. 26). The resulting external force density is

$$\vec{f}_e = \rho \vec{g} + \vec{f}_{\text{fict}} = \rho \vec{g} - \rho \vec{a} = \rho \vec{\nabla}(-gz - ay) = -\rho \vec{\nabla}\Phi_{\text{eff}} = -\rho \vec{\nabla}(gz + ay), \quad (218)$$

where $\Phi_{\text{eff}} = gz + ay$ is the *effective potential*. Thus the hydrostatic balance in the accelerated frame becomes

$$\vec{\nabla}p + \rho \vec{\nabla}\Phi_{\text{eff}} = \vec{0}. \quad (219)$$

4.3.1 Constant-density case

If the density is constant, integrating gives

$$p + \rho(gz + ay) = C. \quad (220)$$

If at the point $(y, z) = (0, h)$ the pressure equals p_0 , then

$$p = p_0 + \rho g(h - z) - \rho a y. \quad (221)$$

The free surface $p = p_0$ is therefore an inclined plane,

$$z = h - \frac{a}{g} y. \quad (222)$$

4.3.2 Acceleration in a plane

If, instead of a pure y -acceleration, the accelerated frame has components $\vec{a} = \hat{j} a_y + \hat{k} a_z$, then

$$\Phi_{eff} = \vec{a} \cdot \vec{r} + gz. \quad (223)$$

Taking into account that

$$\vec{\nabla}(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \vec{\nabla})\vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} + \vec{A} \times (\vec{\nabla} \times \vec{B}) + \vec{B} \times (\vec{\nabla} \times \vec{A}), \quad (224)$$

it follows that

$$\vec{\nabla}(\vec{a} \cdot \vec{r}) = (\vec{a} \cdot \vec{\nabla})\vec{r} = \left(a_y \frac{\partial}{\partial y} + a_z \frac{\partial}{\partial z} \right) (\hat{i}x + \hat{j}y + \hat{k}z) = \hat{j}a_y + \hat{k}a_z. \quad (225)$$

Then

$$\vec{\nabla}\Phi_{eff} = \hat{j}a_y + \hat{k}(a_z + g). \quad (226)$$

The hydrostatic equilibrium condition now reads

$$\vec{\nabla}p + \rho\vec{\nabla}\Phi_{eff} = \vec{\nabla}p + \rho(\hat{j}a_y + \hat{k}(a_z + g)) = 0. \quad (227)$$

In components

$$-\rho[\hat{k}(g + a_z) + \hat{j}a_y] = \hat{i} \frac{\partial p}{\partial x} + \hat{j} \frac{\partial p}{\partial y} + \hat{k} \frac{\partial p}{\partial z}, \quad (228)$$

$$\therefore \frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = -\rho a_y, \quad \frac{\partial p}{\partial z} = -\rho(g + a_z). \quad (229)$$

Using the total differential then gives

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz = -\rho a_y dy - \rho(g + a_z) dz. \quad (230)$$

Integrating between two arbitrary points 1 and 2 results in

$$p_2 - p_1 = -\rho a_y (y_2 - y_1) - \rho(g + a_z) (z_2 - z_1). \quad (231)$$

If points 1 and 2 lie on a constant-pressure line (e.g., the free surface in Fig. 27), then $p_2 - p_1 = 0$ and

$$\frac{z_1 - z_2}{y_2 - y_1} = \tan \alpha = \frac{a_y}{g + a_z}, \quad (232)$$

where α is the angle that the constant-pressure line makes with the horizontal.

In solving problems involving liquids, we often use conservation of mass and equate the volumes before and after the acceleration is applied. After the acceleration is first applied, sloshing may occur. Our analysis will assume that sloshing is not present: either sufficient time has passed to damp out time-dependent motions, or the acceleration is applied in such a way that such motions are minimal.

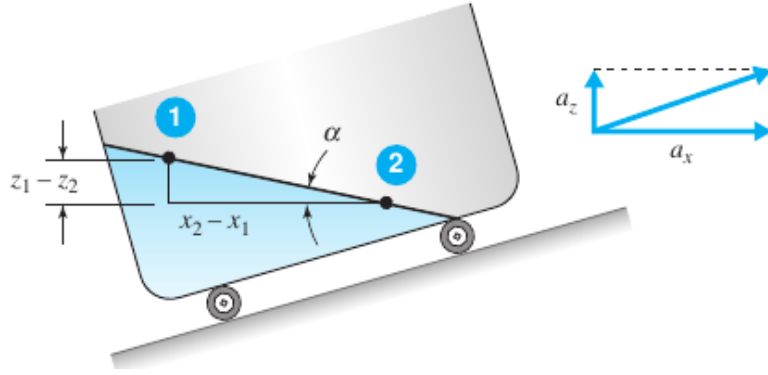


Figure 27: Linearly accelerating container. Figure taken from [4].

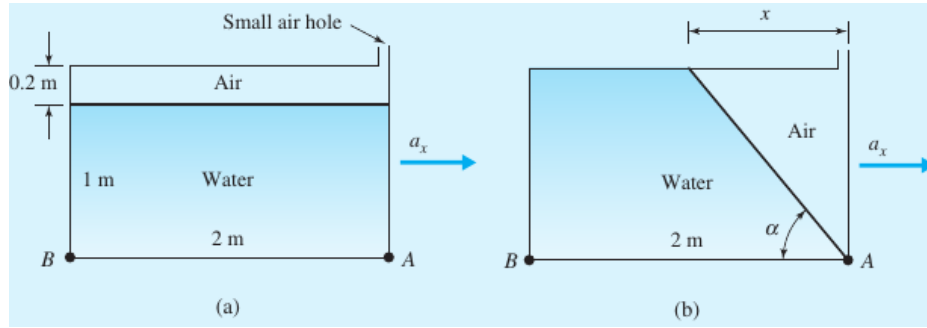


Figure 28: Linearly accelerating tank. Figure taken from [4].

Example. The rectangular, open tank in Fig. 28 (length 2 m, initial water depth 1.2 m, air gap 0.2 m) is accelerated to the right with horizontal acceleration a_x while $a_z = 0$. Find:

1. the acceleration a_x required for the free surface in Fig. E2.11b to pass through point A (the right top corner),
2. the pressure p_B at the left bottom corner B,
3. the total hydrostatic force on the tank bottom if the tank width is 1 m.

1) *Free-surface angle and required acceleration.* In a uniformly accelerated container, lines of constant pressure are orthogonal to the effective acceleration $\vec{g}_{\text{eff}} = (g + a_z)\hat{k} - a_x\hat{i}$, so the free surface (an isobar) satisfies

$$\tan \alpha = \frac{a_x}{g + a_z} \Rightarrow \tan \alpha = \frac{a_x}{g}.$$

The final free surface must rotate about the fluid volume so that the air *area* (since width is constant) is conserved. Before acceleration the air rectangle is

$$A_{\text{air, ini}} = 0.2 \times 2 = 0.4 \text{ m}^2.$$

After tilting, the air area becomes a right triangle of base 1.2 and height x at the right wall:

$$A_{\text{air, fin}} = \frac{1}{2}(1.2)x.$$

Conservation $A_{\text{air, ini}} = A_{\text{air, fin}}$ gives

$$0.2 \times 2 = \frac{1}{2}(1.2)x \quad \Rightarrow \quad x = 0.667 \text{ m.}$$

Hence the free-surface slope is

$$\tan \alpha = \frac{1.2}{0.667} = 1.8,$$

and therefore

$$a_x = g \tan \alpha = (9.81)(1.8) = 1.766 \times 10^1 \text{ m s}^{-2} \approx 17.66 \text{ m s}^{-2}.$$

2) *Pressure at B.* For hydrostatics in the accelerating frame (with $a_z = 0$) the pressure gradient is

$$\vec{\nabla} p = \rho(a_x \hat{i} - g \hat{k}).$$

Along the bottom ($z = \text{const}$) only the x -variation matters, so between points A (right bottom) and B (left bottom) separated by $\Delta x = x_B - x_A = -2 \text{ m}$,

$$p_B - p_A = -\rho a_x (x_B - x_A),$$

where p_A is the pressure at the bottom point vertically under A . The free surface passes through A , so $p_A = 0$ at A , and by vertical hydrostatic balance $p_A = 0$ at the bottom under A (same vertical line on an isobaric free surface). Hence, with $\rho = 1000 \text{ kg m}^{-3}$,

$$p_B = -\rho a_x (x_B - x_A) = -1000 (17.66) (-2) = 3.53 \times 10^4 \text{ Pa} = 35.3 \text{ kPa}.$$

3) *Resultant force on the bottom (width = 1 m).* Pressure varies linearly from $p = 35.3 \text{ kPa}$ at B to $p = 0$ at the bottom point beneath A . The average bottom pressure is thus

$$\bar{p} = \frac{p_B + p'_A}{2} = \frac{35.3 \text{ kPa} + 0}{2} = 17.65 \text{ kPa}.$$

With bottom area $A_{\text{bot}} = 2 \times 1 = 2 \text{ m}^2$, the total force is

$$F = \bar{p} A_{\text{bot}} = (17.65 \times 10^3) (2) = 3.53 \times 10^4 \text{ N} \approx 35\,300 \text{ N}.$$

Review question. A drag racer rests her coffee mug on a horizontal tray while she accelerates at 7 m/s^2 . The mug is 10 cm deep and 6 cm in diameter and contains coffee 7 cm deep at rest (see Fig. 29). Assuming rigid-body acceleration of the coffee, determine whether it will spill out of the mug.

4.4 Rotating containers

To study the statics of a fluid that rotates uniformly, consider the basic hydrostatic equation in a rotating frame. In addition to gravity (when present), the fictitious forces must be included. In cylindrical coordinates, the centrifugal body-force density is

$$\vec{f} = \frac{d\vec{F}}{dV} = \frac{dm}{dV} \Omega^2 r \hat{e}_r = \rho \Omega^2 r \hat{e}_r, \quad (233)$$

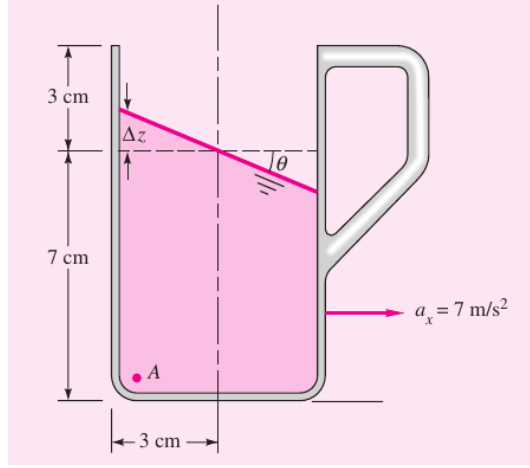


Figure 29: The coffee tilted during the acceleration. Figure taken from [5].

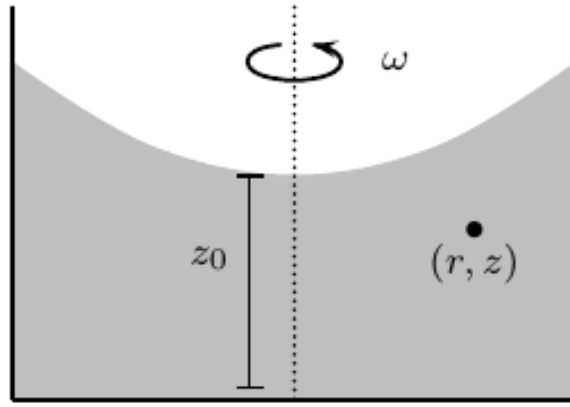


Figure 30: Liquid at rest in a rotating cylindrical container. Figure taken from [1].

so that the effective force density is

$$\vec{f}_e = \rho \vec{g} + \rho \Omega^2 r \hat{e}_r = \rho \vec{\nabla} \left[-\Phi + \frac{\Omega^2 r^2}{2} \right] = \vec{\nabla} p, \quad (234)$$

and therefore the hydrostatic balance can be written as

$$\frac{\vec{\nabla} p}{\rho} + \vec{\nabla} \left(\Phi - \frac{\Omega^2 r^2}{2} \right) = 0. \quad (235)$$

The quantity $\Phi - \Omega^2 r^2/2$ is called the *effective potential* Φ_{eff} . Equivalently, $-\rho \vec{\nabla} \Phi_{eff} = \vec{\nabla} p$.

4.4.1 Constant-density liquid

In this case, from the previous relation,

$$\vec{\nabla} \left[\frac{p}{\rho} + \Phi - \frac{\Omega^2 r^2}{2} \right] = \vec{0}, \quad \Rightarrow \quad p + \rho g z - \frac{\Omega^2}{2} \rho r^2 = C, \quad (236)$$

where we used $\Phi = gz$. Taking $p = p_0$ at $z = z_0$ and $r = 0$, one finds $p_0 + \rho g z_0 = C$, and thus

$$p = p_0 + \rho g (z - z_0) + \frac{\Omega^2}{2} \rho r^2 = p_0 + \rho g (z - z_0) + \frac{\Omega^2}{2} \rho (x^2 + y^2). \quad (237)$$

Isobars are therefore paraboloids of revolution. In particular, the free surface (where $p = p_0$, atmospheric pressure) satisfies

$$\Omega^2 r^2 = 2g (z - z_0). \quad (238)$$

Note that the shape of the surface does not depend on ρ , provided it is constant. In a bucket of diameter 20 cm rotating once per second the water stands 2 cm higher at the rim than in the centre.

As in the previous example of linear acceleration, the position of the free surface is found by conserving the volume of fluid. For a noncircular container with the axis of rotation off-center, a lot of laborious mensuration is required, and a single problem will take you all weekend. However, the calculation is easy for a cylinder rotating about its central axis, as in Fig. 31. Since the volume of a paraboloid is one-half the base area times its height, the still-water level is exactly halfway between the high and low points of the free surface. The center of the fluid drops an amount $h/2 = \Omega^2 R^2 / (4g)$, and the edges rise an equal amount.

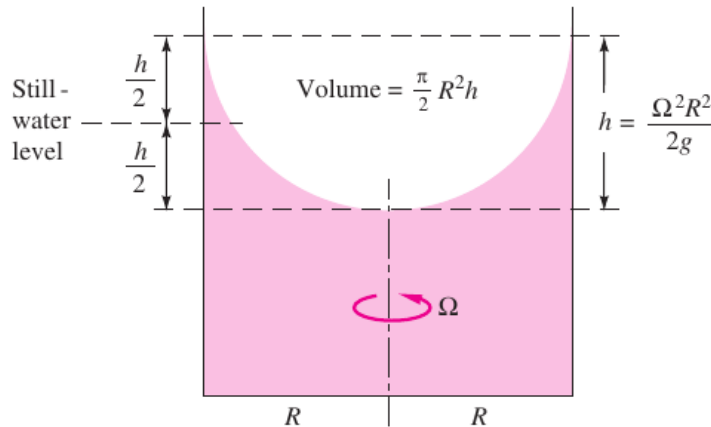


Figure 31: Determining the free-surface position for rotation of a cylinder of fluid about its central axis. Figure taken from [5].

Example. The coffee cup in previous example is removed from the drag racer, placed on a turntable, and rotated about its central axis until a rigid-body mode occurs (see Fig. 32). Find (a) the angular velocity that will cause the coffee to just reach the lip of the cup and (b) the gage pressure at point A for this condition.

Example: ultracentrifuge. An ultracentrifuge of radius 10 cm contains water (without an open surface) and rotates at $\Omega = 60000 \text{ rpm} \approx 6300 \text{ s}^{-1}$. The centrifugal acceleration at the rim becomes 400000 times standard gravity g and the pressure close to 2000 atm, which is double the pressure at the bottom of the deepest abyss in the sea. At such pressures, the water density is increased by 10%.

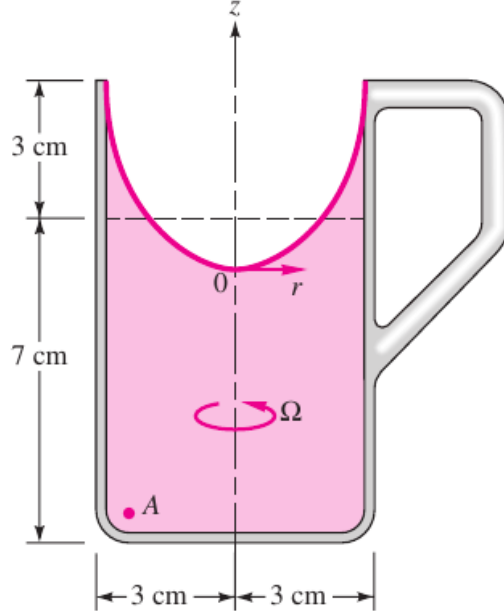


Figure 32: The coffee tilted during the rotation. Figure taken from [5].

4.4.2 Ideal gas in a closed container

For an ideal gas with constant temperature T and mean molecular weight μ ,

$$\frac{\vec{\nabla} p}{\rho} + \nabla \left(\Phi - \frac{\omega^2 r^2}{2} \right) = \vec{0} \quad \Longleftrightarrow \quad \vec{\nabla} p \frac{kT}{p\mu} + \nabla \left(\Phi - \frac{\omega^2 r^2}{2} \right) = \vec{0}. \quad (239)$$

Hence,

$$\vec{\nabla} \left[\frac{kT}{\mu} \ln p + gz - \frac{\omega^2 r^2}{2} \right] = \vec{0} \quad \Rightarrow \quad \frac{kT}{\mu} \ln p + gz - \frac{\omega^2 r^2}{2} = C. \quad (240)$$

With $p = p_0$ at $r = 0$, $z = 0$, we obtain

$$\frac{kT}{\mu} \ln \left(\frac{p}{p_0} \right) + gz - \frac{\omega^2 r^2}{2} = 0, \quad \text{or} \quad p = p_0 \exp \left[\frac{\mu}{kT} \left(\frac{\omega^2 r^2}{2} - gz \right) \right]. \quad (241)$$

The isobars $z = \omega^2 r^2 / (2g)$ are, again, paraboloids of revolution.

Problem.

1. Consider a polytropic fluid $p = \alpha \rho^n$, at rest in a rotating system and in the presence of a uniform gravitational field $\vec{g} = -\hat{k}g$. Evaluate $\rho(r)$ if $\rho = \rho_0$ at $r = z = 0$.

4.4.3 The figure of Earth

The rotation of a planet modifies its hydrostatic equilibrium. In a reference frame rotating with angular velocity $\vec{\Omega}$, a fluid element experiences, in addition to gravity, the centrifugal acceleration. If s denotes the distance from the rotation axis, its magnitude is

$$a_{\text{cf}} = \Omega^2 s. \quad (242)$$

At the equator, $s = a$, where a is the radius of the planet, whereas at the poles $s = 0$. A convenient measure of the importance of rotation is therefore the ratio between the centrifugal acceleration at the equator and the surface gravitational acceleration,

$$q \equiv \frac{\Omega^2 a}{g_0}. \quad (243)$$

For an approximately spherical planet,

$$g_0 \simeq \frac{GM}{a^2}, \quad (244)$$

and consequently

$$q \simeq \frac{\Omega^2 a^3}{GM}. \quad (245)$$

The parameter q measures the relative importance of centrifugal effects. For the Earth,

$$a \simeq 6371 \text{ km}, \quad g_0 \simeq 9.8 \text{ m s}^{-2}, \quad \Omega \simeq 7.3 \times 10^{-5} \text{ s}^{-1}, \quad (246)$$

which gives

$$q \simeq 3.5 \times 10^{-3}. \quad (247)$$

Thus, the centrifugal acceleration at the equator amounts to only about 0.35% of the gravitational acceleration.

Despite its small magnitude, this centrifugal effect has an observable consequence for the shape of the Earth. In the rotating frame the effective potential can be written as

$$\Phi_{\text{eff}} = \Phi_g + \Phi_{\text{cf}}, \quad (248)$$

where

$$\Phi_{\text{cf}} = -\frac{1}{2}\Omega^2 s^2. \quad (249)$$

For a perfectly spherical Earth, $s = a \sin \theta$, and therefore

$$\Phi_{\text{cf}} = -\frac{1}{2}\Omega^2 a^2 \sin^2 \theta. \quad (250)$$

The centrifugal contribution vanishes at the poles and reaches its minimum at the equator. Thus, on a hypothetical perfectly spherical Earth, the effective potential decreases slightly when moving from the pole toward the equator. In this sense, rotation produces a shallow *centrifugal valley* around the equator.

A simple estimate of the geometrical importance of this effect is obtained from

$$\frac{\Delta a}{a} \sim q. \quad (251)$$

For the Earth this gives

$$\Delta a \sim qa \simeq 3.5 \times 10^{-3}(6371 \text{ km}) \simeq 22 \text{ km}. \quad (252)$$

If such a centrifugal valley suddenly appeared on an otherwise spherical Earth, water would tend to flow toward the equator. The actual Earth, however, has had geological times to respond to rotation: not only the oceans but also the material inside the Earth has redistributed itself.

The result is an approximately oblate planet, with an equatorial radius larger than its polar radius. Observationally,

$$a_{\text{eq}} - a_{\text{pole}} \simeq 21.4 \text{ km}, \quad (253)$$

remarkably close to the simple estimate above. This agreement should not be interpreted as an exact derivation of the Earth's shape. The deformation changes the mass distribution of the planet and therefore also changes its own gravitational potential. A consistent treatment must include this *self-gravitational* effect.

Example. Rotating neutron star (pulsar). Neutron stars are among the most extreme objects in the Universe. A typical neutron star may have radius $a = 10 \text{ km}$ and mass $M \simeq 4 \times 10^{30} \text{ kg}$ (about twice the Sun's), and can rotate with periods down to 1.4 ms (i.e. $\omega \simeq 4500 \text{ s}^{-1}$). Then the surface gravity is $g_0 \simeq 2.7 \times 10^{12} \text{ m s}^{-2}$ and the equatorial centrifugal acceleration is $\omega^2 a \simeq 2 \times 10^{11} \text{ m s}^{-2}$. The “levitation parameter” is therefore $q = \omega^2 a / g_0 \simeq 0.076$: about 22 times Earth's value, but still well below unity.

4.4.4 Centrifugal deformation of a spherical planet

Let us consider more carefully an initially spherical, slowly rotating planet of radius a . Because the rotation is axially symmetric, the deformation can depend only on the polar angle θ , or equivalently on the distance

$$s = a \sin \theta \quad (254)$$

from the rotation axis.

Let

$$r(\theta) = a + h(\theta), \quad |h| \ll a, \quad (255)$$

where $h(\theta)$ is the radial displacement of the surface relative to the original sphere. Hydrostatic equilibrium requires the surface of the planet to be an equipotential of the total effective potential.

To first order in the deformation, the potential evaluated at the surface may be written as

$$\Phi(\theta) = g_0 h(\theta) - \frac{1}{2} \Omega^2 a^2 \sin^2 \theta + \Delta \Phi(\theta). \quad (256)$$

The three terms have different physical origins:

- $g_0 h(\theta)$ is the change in the original gravitational potential caused by moving the surface radially by h ;
- $-\frac{1}{2} \Omega^2 a^2 \sin^2 \theta$ is the centrifugal potential;
- $\Delta \Phi(\theta)$ is the change in gravitational potential produced by the redistribution of the planet's own mass.

The last contribution is the *self-potential* of the deformation.

Neglecting the self-potential

It is instructive to first neglect the gravitational effect of the redistributed mass and set

$$\Delta\Phi = 0. \quad (257)$$

Since the surface must be an equipotential, Eq. (256) gives

$$g_0 h(\theta) - \frac{1}{2} \Omega^2 a^2 \sin^2 \theta = \text{const.} \quad (258)$$

Using

$$q = \frac{\Omega^2 a}{g_0}, \quad (259)$$

we obtain

$$h(\theta) = \frac{1}{2} a q \sin^2 \theta + C. \quad (260)$$

The constant C follows from conservation of volume (actually, conservation of mass by imposing ρ_1 constant). To first order in h , the change in volume is

$$\Delta V = \int h(\theta) dS. \quad (261)$$

For the original spherical surface,

$$dS = a^2 \sin \theta d\theta d\phi, \quad (262)$$

and conservation of volume therefore requires

$$\int_0^\pi h(\theta) \sin \theta d\theta = 0. \quad (263)$$

Substitution of Eq. (260) gives

$$h(\theta) = h_0 \left(\sin^2 \theta - \frac{2}{3} \right), \quad h_0 = \frac{1}{2} a q. \quad (264)$$

At the equator,

$$h_{\text{eq}} = h\left(\frac{\pi}{2}\right) = \frac{h_0}{3}, \quad (265)$$

whereas at the poles,

$$h_{\text{pole}} = h(0) = -\frac{2h_0}{3}. \quad (266)$$

Consequently, the pole-to-equator difference in radius is

$$h_{\text{eq}} - h_{\text{pole}} = h_0. \quad (267)$$

For the Earth this approximation gives

$$h_0 = \frac{1}{2} a q \simeq 11 \text{ km.} \quad (268)$$

This is only about one half of the observed value,

$$a_{\text{eq}} - a_{\text{pole}} \simeq 21.4 \text{ km.} \quad (269)$$

The discrepancy reveals an important piece of physics that has been neglected: the gravitational field produced by the redistributed mass itself.

The self-gravitational correction

Although the mass displaced by the deformation is small compared with the total mass of the planet, it is located close to the surface. Consequently, its contribution to the gravitational potential cannot in general be neglected.

For the deformation considered above, the corresponding self-potential to lowest order can be written as

$$\Delta\Phi(\theta) = -\frac{3}{5}\frac{\rho_1}{\rho_0}g_0h(\theta), \quad (270)$$

where ρ_1 is the density of the material being displaced and

$$\rho_0 = \frac{M}{\frac{4}{3}\pi a^3} \quad (271)$$

is the mean density of the planet.

Substituting Eq. (270) into Eq. (256), the part of the gravitational potential associated with the deformation becomes

$$g_0h + \Delta\Phi = g_0h \left(1 - \frac{3}{5}\frac{\rho_1}{\rho_0}\right). \quad (272)$$

The self-gravitational contribution therefore reduces the effective gravitational cost of deforming the surface. A larger deformation is required to compensate the same centrifugal potential.

The resulting pole-to-equator height difference is

$$h_0 = \frac{1}{2}aq \left(1 - \frac{3}{5}\frac{\rho_1}{\rho_0}\right)^{-1}. \quad (273)$$

For the Earth, taking representative values

$$\rho_1 \simeq 4.5 \text{ g cm}^{-3}, \quad \rho_0 \simeq 5.5 \text{ g cm}^{-3}, \quad (274)$$

one finds

$$1 - \frac{3}{5}\frac{\rho_1}{\rho_0} \simeq 0.51. \quad (275)$$

Thus,

$$h_0 \simeq \frac{11 \text{ km}}{0.51} \simeq 21.9 \text{ km}, \quad (276)$$

which is remarkably close to the observed equatorial–polar radius difference,

$$a_{\text{eq}} - a_{\text{pole}} \simeq 21.4 \text{ km}. \quad (277)$$

The agreement should not be overinterpreted. A fully self-consistent description of the Earth's figure must calculate the gravitational potential generated by the actual redistribution of matter throughout the planet and account for variations in density and compressibility. Nevertheless, this simple calculation illustrates an important principle of self-gravitating continuous media: *when the matter distribution changes, the gravitational potential responsible for hydrostatic equilibrium changes with it.*

Thus, the deformation of a rotating planet results from the interplay between centrifugal support and self-gravity. This same competition is central to the equilibrium structure of many rotating astrophysical objects, including planets and stars.

4.4.5 Other applications

This framework is essential for describing the structure of rotating astrophysical systems. For instance, in rotating stars and planets, the equilibrium configuration is determined by surfaces of constant effective potential, leading to an oblate shape due to centrifugal support. This is directly analogous to the parabolic surface formed by a liquid in a rotating container.

In accretion disks around compact objects, vertical hydrostatic equilibrium in the rotating frame results from the balance between gravity and pressure gradients. This leads to a Gaussian density stratification,

$$\rho(z) \propto \exp\left(-\frac{z^2}{2H^2}\right), \quad H \sim \frac{c_s}{\Omega},$$

where H is the disk scale height and c_s the sound speed.

Hydrostatics in non-inertial frames is also central to binary systems. In a co-rotating frame, the effective potential defines the Roche geometry, and equipotential surfaces determine the regions gravitationally bound to each component. Mass transfer occurs when a star fills its Roche lobe, allowing material to flow through the inner Lagrange point.

Finally, rotation plays an important role in planetary atmospheres and giant planets, where it modifies large-scale structure and induces equatorial bulging. More generally, the concept of an effective potential provides a unifying description of equilibrium configurations in rotating astrophysical fluids.

4.5 The Earth, the Moon, and the tides

After Newton we know that the Moon’s gravity acts on everything on Earth, including the water in the oceans, and tends to distort it, thereby creating the tides. Yet high tides occur at roughly the same time at antipodal points on Earth—and thus twice per day—so the explanation cannot be merely that the Moon “pulls” the sea toward itself.

The rotation of the Earth, together with the orbital motion of both Earth and Moon about their common center of mass, makes tides an essentially *dynamic* phenomenon that cannot be treated accurately by the hydrostatic methods used in this chapter. Nevertheless, a number of basic features can be revealed with simple approximations (and careful wording of explanations).

4.5.1 The tidal field

The Moon is sufficiently small and distant that its gravitational potential across the Earth can be approximated by that of a point mass m located at position $\vec{x}_0$:

$$\Phi(\vec{x}) = -\frac{Gm}{|\vec{x} - \vec{x}_0|}. \quad (278)$$

Choose coordinates with origin at Earth’s center and the z -axis pointing toward the Moon, so that $\vec{x}_0 = (0, 0, D)$, with $D = |\vec{x}_0|$ the Earth–Moon distance (see Fig. 33).

For points with $r = |\vec{x}| \ll D$ the potential can be expanded in powers of r/D . Using

$$|\vec{x} - \vec{x}_0| = \sqrt{x^2 + y^2 + (z - D)^2} = D \sqrt{1 - \frac{2z}{D} + \frac{r^2}{D^2}}, \quad (279)$$

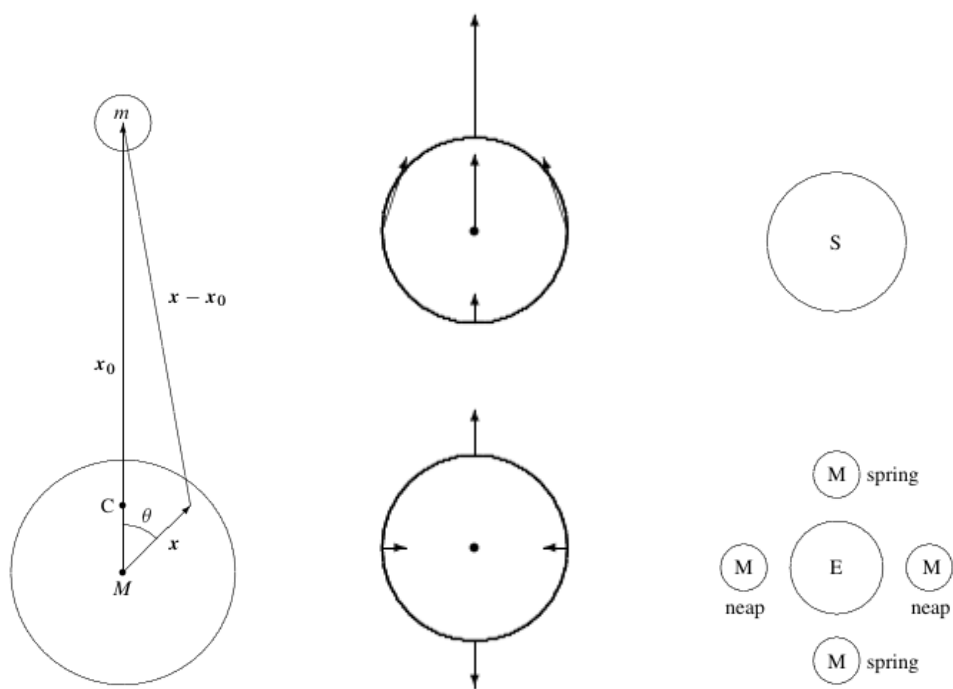


Figure 33: Left: geometry of the Earth and the Moon (not to scale). The point C is the center of mass of the system. Center: (top) how the Moon's gravity varies over the Earth (exaggerated); (bottom) the Moon's gravity with the unobservable constant acceleration canceled. This figure explains why the tides have a semi-diurnal period. Right: Earth and Sun with the Moon in extreme positions corresponding to spring tides and neap tides. The figure is (vastly) not to scale. Figures taken from Ref. [3].

and the binomial series to second order,

$$\frac{1}{|\vec{x} - \vec{x}_0|} = \frac{1}{D} \left(1 - \frac{2z}{D} + \frac{r^2}{D^2} \right)^{-1/2} \quad (280)$$

$$\approx \frac{1}{D} \left[1 + \frac{z}{D} + \frac{3z^2 - r^2}{2D^2} \right]. \quad (281)$$

The first term gives a constant $-Gm/D$, which can be dropped. The second term corresponds to a uniform field that keeps the Earth in orbit ($g_z = Gm/D^2 \approx 33\mu\text{m/s}^2$) and is also discarded when focusing on differential (“tidal”) effects. The leading nontrivial, spatially varying term is therefore the quadrupolar contribution, which defines the *tidal potential*

$$\Phi_{\text{tidal}} = -\frac{Gm}{D^3} \frac{1}{2} (3z^2 - r^2). \quad (282)$$

In spherical coordinates (r, θ, ϕ) with θ the angle from the z -axis ($z = r \cos \theta$), this can be written

$$\Phi_{\text{tidal}} = -\frac{1}{2} (3 \cos^2 \theta - 1) \frac{Gm}{D^3} r^2. \quad (283)$$

The Moon’s tidal *acceleration* is the gradient of the tidal potential. In spherical coordinates,

$$g_r = -\frac{\partial \Phi_{\text{tidal}}}{\partial r} = (3 \cos^2 \theta - 1) \frac{Gm}{D^3} r, \quad (284)$$

$$g_\theta = -\frac{1}{r} \frac{\partial \Phi_{\text{tidal}}}{\partial \theta} = -\frac{3}{2} \sin 2\theta \frac{Gm}{D^3} r, \quad (285)$$

$$g_\phi = -\frac{1}{r \sin \theta} \frac{\partial \Phi_{\text{tidal}}}{\partial \phi} = 0. \quad (286)$$

The radial component is largest at $\theta = 0^\circ$ and 180° (the sub-Moon and anti-Moon points) and smallest at $\theta = 90^\circ$ (the great circle 90° from the Moon), where it is negative and half in magnitude. The tangential component is largest in magnitude at $\theta = 45^\circ$ and 135° . On Earth’s surface the vertical part of the tidal acceleration is $\sim 3a Gm/D^3 \approx 1.65 \mu\text{m/s}^2$, where a is Earth’s radius.

4.5.2 The tides

If the Earth did not rotate and the Moon stood still above a particular spot, water would rush in to fill up these “valleys”, and the sea would come to equilibrium with its open surface at constant total gravitational potential. The total potential near the surface of the Earth is

$$\Phi = \Phi_{\text{Earth}} + \Phi_{\text{Moon}} = g_0 h - \frac{1}{2} (3 \cos^2 \theta - 1) \left(\frac{a}{D} \right)^2 \frac{Gm}{D}, \quad (287)$$

where g_0 the surface gravity and h a small (signed) height above the spherical surface of the Earth.

Requiring this potential to be constant gives the tidal height

$$h = h_0 + \frac{1}{2} (3 \cos^2 \theta - 1) \left(\frac{a}{D} \right)^2 \frac{Gm}{g_0 D}, \quad (288)$$

where h_0 is a constant. Since the average over the sphere of the second term vanishes,

$$\frac{1}{4\pi} \int_0^\pi d\theta \int_0^{2\pi} d\phi \sin\theta \phi (3\cos^2\theta - 1) = \frac{1}{2} \int_{-1}^{+1} (3z^2 - 1) dz = 0, \quad (289)$$

we conclude that h_0 is the average water depth.

The shape of the equipotential surface corresponding to $\Phi = 0$ becomes

$$h = \frac{1}{2}(3\cos^2\theta - 1) \left(\frac{a}{D}\right)^2 \frac{Gm}{g_0 D}. \quad (290)$$

4.5.3 Tidal range

The maximal difference between high and low tides (the tidal range) occurs between the extreme positions at $\theta = 0^\circ$ and $\theta = 90^\circ$:

$$H_0 = \frac{3}{2} \left(\frac{a}{D}\right)^2 \frac{Gm}{g_0 D} = \frac{3}{2} a \frac{m}{M} \left(\frac{a}{D}\right)^3, \quad (291)$$

where in the last equality we used $g_0 = GM/a^2$ with M the Earth's mass. Inserting the values for the Moon we get $H_0 \approx 54$ cm. Interestingly, the range of the tides due to the Sun turns out to be half as large, about 25 cm. This makes spring tides when the Sun and the Moon cooperate almost three times higher than neap tides when they do not. For the tides to reach full height, water must move in from huge areas of the Earth as is evident from the shallow shape of the potential. Where this is not possible, for example in lakes and enclosed seas, the tidal range becomes much smaller than in the open oceans. Local geography may also influence tides. In bays and river mouths funnelling can cause tides to build up to huge values. Spring tides in the range of 15 m have been measured in the Bay of Fundy in Canada.

4.6 Exercises

1. Estimate the change in sea level if the air pressure locally rises by $\Delta p = 20$ hPa and stays there (disregarding surface tension).
2. The tides raises the water above the average level at the Moon and anti-Moon positions. (a) How much water is found in each of these tidal bulges?

5 Buoyancy and stability

5.1 Archimedes' principle

The external force on a volume element inside a fluid is

$$\vec{F}_e = \int_V \vec{f}_e dV \quad (292)$$

and the buoyancy due to pressure acting on its surface,

$$\vec{F}_p = - \oint_S p d\vec{S} = - \oint_S p \hat{n} dS, \quad (293)$$

If the total $\vec{F} = \vec{F}_e + \vec{F}_p$ does not vanish, an unrestrained body will accelerate in the direction of $\vec{F}$ according to Newton's Second Law. Therefore, in mechanical equilibrium, weight and buoyancy must precisely cancel each other at all times to guarantee that the body will remain in place.

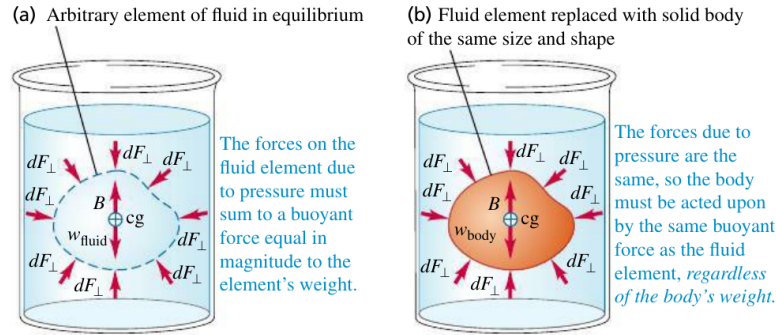


Figure 34: Archimedes's principle

This equation for the net pressure force holds whether the region is occupied by fluid or by some other body. In equilibrium the fluid volume is at rest, so

$$\vec{F}_p + \vec{F}_e = \vec{0}.$$

If the external force is gravity, $\vec{F}_e = \int_V \rho \vec{g} dV$ then the Archimedes' principle follows (see Fig. 34): *the force of buoyancy is equal and opposite to the weight of the displaced fluid.* That is,

$$\vec{F}_B = - \int_V \rho_{\text{fluid}} \vec{g} dV. \quad (294)$$

If a body submerged in the fluid has a weight greater than the weight of the displaced fluid, it sinks; otherwise it rises. The total force on the body may now be written

$$\vec{F} = \vec{F}_G + \vec{F}_B = \int_V (\rho_{\text{body}} - \rho_{\text{fluid}}) \vec{g} dV, \quad (295)$$

explicitly confirming that when the body is made from the same fluid as its surroundings —so that $\rho_{\text{body}} = \rho_{\text{fluid}}$ — the resultant force vanishes automatically. In general, however, the distributions of mass in the body and in the displaced fluid will be different.

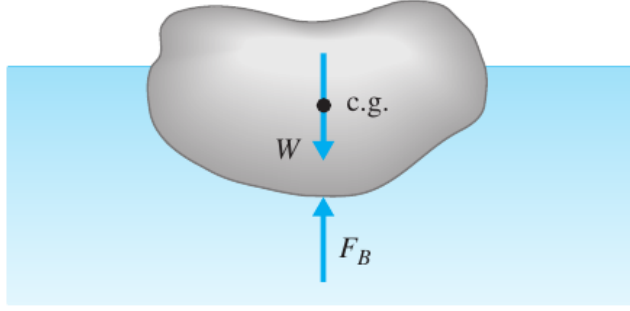


Figure 35: Forces on a floating object. Figure taken from Ref. [4].

An equivalent derivation of Archimedes' principle follows from considering a small vertical cylinder of height L and base area A within the fluid. The pressure on the lower face (p_2) exceeds that on the upper face (p_1), producing a net upward force

$$F = (p_2 - p_1)A. \quad (296)$$

Using the hydrostatic relation $p_2 - p_1 = \rho gL$, one obtains

$$F = \rho g(AL) = \rho gV = mg, \quad (297)$$

which is the weight of the displaced liquid (since ρ is the fluid density and $V = AL$ is the displaced volume).

5.1.1 Astrophysical applications

One of the most important applications is in *stellar convection*. In stellar interiors, if a fluid element is displaced upward and is less dense than its surroundings, it experiences a buoyant force that accelerates it further upward. This mechanism drives convective energy transport in regions where the temperature gradient exceeds the adiabatic gradient.

A related application is in the study of *stability of stratified atmospheres*, both in stars and planetary atmospheres. Small vertical displacements of fluid elements lead to restoring or amplifying forces depending on the density stratification. This is quantified by the Brunt–Väisälä frequency,

$$N^2 = \frac{g}{\rho} \frac{d\rho}{dz},$$

which determines whether the medium is stable ($N^2 > 0$) or unstable ($N^2 < 0$) to buoyant perturbations.

Archimedes' principle is also central to understanding *bubbles and cavities in astrophysical plasmas*, such as those observed in galaxy clusters. In these systems, regions filled with relativistic plasma (e.g., from active galactic nuclei) are less dense than the surrounding intracluster medium and rise buoyantly, transporting energy and redistributing material.

Another application appears in *proto-planetary and accretion disks*, where vertical stratification leads to buoyancy forces that influence turbulence and mixing. Buoyant forces can also play a role in dust settling and the vertical structure of the disk.

Finally, buoyancy is important in *core-collapse supernovae* and *neutron stars*, where convection driven by entropy and composition gradients affects the dynamics, neutrino transport, and overall evolution of the system.

In all these contexts, Archimedes' principle provides the physical basis for understanding how density differences in a gravitational field lead to motion, instability, and energy transport in astrophysical fluids.

5.1.2 Hydrometer

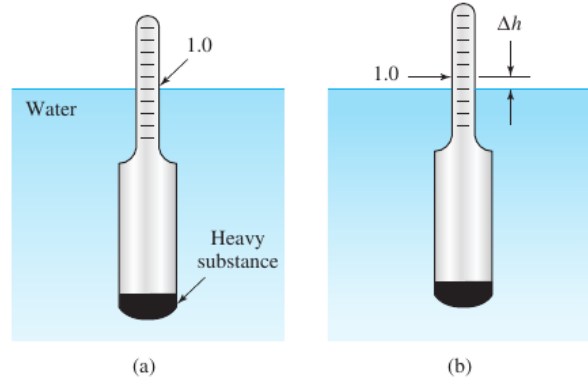


Figure 36: Hydrometer: (a) in water; (b) in an unknown liquid. Figure taken from Ref. [4].

A hydrometer, an instrument used to measure the specific gravity of liquids ($\gamma = \rho g$), operates on the principle of buoyancy. A sketch is shown in Fig. 36. The upper part (the stem) has a constant diameter. When placed in pure water the specific gravity is marked to read $S = \rho/\rho_{\text{water}} = 1.0$. The force balance is

$$W = \gamma_{\text{water}} V, \quad (298)$$

where W is the weight of the hydrometer and V is the submerged volume below the $S = 1.0$ line. In an unknown liquid of specific weight γ_x , a force balance gives

$$W = \gamma_x (V - A \Delta h), \quad (299)$$

where A is the cross-sectional area of the stem. Equating these expressions yields

$$\Delta h = \frac{V}{A} \left(1 - \frac{1}{S_x} \right), \quad (300)$$

where $S_x = \gamma_x/\gamma_{\text{water}}$ is the specific gravity of the unknown liquid relative to water. For a given hydrometer, V and A are fixed, so Δh depends only on S_x ; thus the stem can be calibrated to read S_x directly. Hydrometers are used, for example, to measure the amount of antifreeze in an automobile radiator or the charge in a lead-acid battery, since the density of the fluid changes as H_2SO_4 is consumed or produced.

5.1.3 Review questions

1. In hot-air ballooning, a large balloon is filled with air heated by a gas burner at the bottom. Why must the air be heated? How does the balloonist control ascent and descent?
2. You drop a solid sphere of aluminum in a bucket of water that sits on the ground. The buoyant force equals the weight of water displaced; this is less than the weight of the sphere,

so the sphere sinks to the bottom. If you take the bucket with you on an elevator that accelerates upward, the apparent weight of the water increases and the buoyant force on the sphere increases. Could the acceleration of the elevator be great enough to make the sphere pop up out of the water? Explain.

3. A rigid, lighter-than-air dirigible filled with helium cannot continue to rise indefinitely. Why? What determines the maximum height it can attain?
4. Which has a greater buoyant force on it: a 25 cm^3 piece of wood floating with part of its volume above water or a 25 cm^3 piece of submerged iron? Or, must you know their masses before you can answer? Explain.
5. A cargo ship travels from the Atlantic Ocean (salt water) to Lake Ontario (freshwater) via the St. Lawrence River. The ship rides several centimeters lower in the water in Lake Ontario than it did in the ocean. Explain.
6. You push a piece of wood under the surface of a swimming pool. After it is completely submerged, you keep pushing it deeper and deeper. As you do this, what will happen to the buoyant force on it? Will the force keep increasing, stay the same, or decrease? Why?
7. A piece of iron is glued to the top of a block of wood. When the block is placed in a bucket of water with the iron on top, the block floats. The block is now turned over so that the iron is submerged beneath the wood. Does the block float or sink? Does the water level in the bucket rise, drop, or stay the same? Explain.
8. You take an empty glass jar and push it into a tank of water with the open mouth of the jar downward, so that the air inside the jar is trapped and cannot get out. If you push the jar deeper into the water, does the buoyant force on the jar stay the same? If not, does it increase or decrease? Explain.
9. An ice cube floats in a glass of water. When the ice melts, will the water level in the glass rise, fall, or remain unchanged? Explain.

5.1.4 Problems

1. You are preparing some apparatus for a visit to a newly discovered planet Caasi having oceans of glycerine and a surface acceleration due to gravity of 5.40 m/s^2 . If your apparatus floats in the oceans on Earth with 25.0% of its volume submerged, what percentage will be submerged in the glycerine oceans of Caasi?
2. A rock with density 1200 kg/m^3 is suspended from the lower end of a light string. When the rock is in air, the tension in the string is 28.0 N . What is the tension in the string when the rock is totally immersed in a liquid with density 750 kg/m^3 ?
3. A cubical block of wood, 10.0 cm on a side, floats at the interface between oil and water with its lower surface 1.50 cm below the interface (Fig. 37). The density of the oil is 790 kg/m^3 . (a) What is the gauge pressure at the upper face of the block? (b) What is the gauge pressure at the lower face of the block? (c) What are the mass and density of the block?

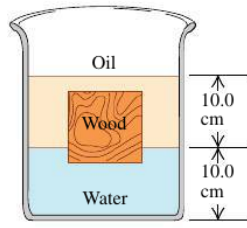


Figure 37: Problem.

4. A rock is suspended by a light string. When the rock is in air, the tension in the string is 39.2 N. When the rock is totally immersed in water, the tension is 28.4 N. When the rock is totally immersed in an unknown liquid, the tension is 21.5 N. What is the density of the unknown liquid?
5. A block composed of two materials with densities ρ_1 and ρ_2 (both less than that of water) and heights h_1 and h_2 is placed in a container with liquid (see left panel of Fig. 38). By what distance a is the block submerged?
6. A uniform cylindrical log of length L and radius R floats in water (see right panel of Fig. 38; the log is viewed along its axis). The density of the log is 0.509 g cm^{-3} . Determine h .

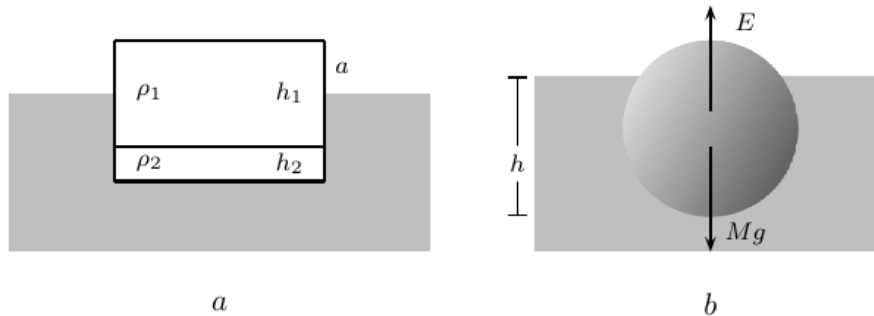


Figure 38: Problem.

7. A piece of wood is 0.600 m long, 0.250 m wide, and 0.080 m thick. Its density is 700 kg/m^3 . What volume of lead must be fastened underneath it to sink the wood in calm water so that its top is just even with the water level? What is the mass of this volume of lead?
8. A closed and elevated vertical cylindrical tank with diameter 2.00 m contains water to a depth of 0.800 m. A worker accidentally pokes a circular hole with diameter 0.0200 m in the bottom of the tank. As the water drains from the tank, compressed air above the water in the tank maintains a gauge pressure of $5.00 \times 10^3 \text{ Pa}$ at the surface of the water. Ignore any effects of viscosity.
 - (a) Just after the hole is made, what is the speed of the water as it emerges from the hole? What is the ratio of this speed to the efflux speed if the top of the tank is open to the air?
 - (b) How much time does it take for all the water to drain from the tank? What is the ratio of this time to the time it takes for the tank to drain if the top of the tank is open to the air?

9. A cubical block of density ρ_B and with sides of length L floats in a liquid of greater density ρ_L .
- What fraction of the block's volume is above the surface of the liquid?
 - The liquid is denser than water (density ρ_W) and does not mix with it. If water is poured on the surface of that liquid, how deep must the water layer be so that the water surface just rises to the top of the block? Express your answer in terms of L , ρ_B , ρ_L , and ρ_W .
 - Find the depth of the water layer in part (b) if the liquid is mercury, the block is made of iron, and $L = 10.0$ cm.
10. The hydrometer shown in Fig. 39 with no mercury has a mass of 0.01 kg. It is designed to float at the midpoint of the 12 cm-long stem in pure water.
- Calculate the mass of mercury needed.
 - What is the specific gravity of the liquid if the hydrometer is just submerged?
 - What is the specific gravity of the liquid if the stem of the hydrometer is completely exposed?

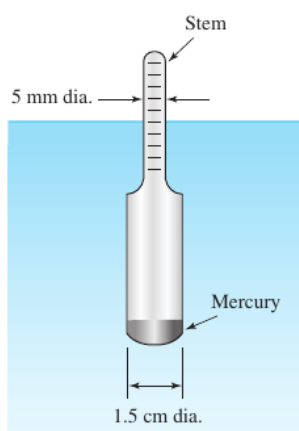


Figure 39: Problem.

11. The uniform 5 m-long round wooden rod in Fig. 40 is tied to the bottom by a string. Determine (a) the tension in the string and (b) the specific gravity of the wood. Is it possible, with the given information, to determine the inclination angle θ ? Explain.

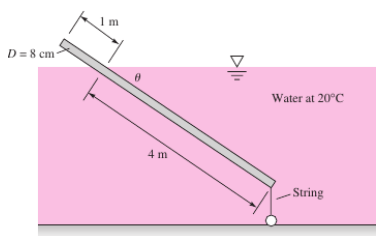


Figure 40: Problem.

12. The uniform rod in Fig. 41 is hinged at point B on the waterline and is in static equilibrium as shown when 2 kg of lead (SG = 11.4) are attached to its end. What is the specific gravity of the rod material? What is peculiar about the rest angle $\theta = 30^\circ$?

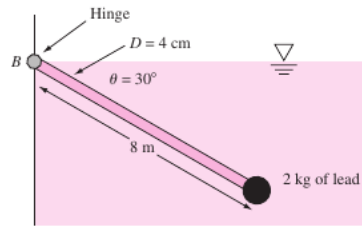


Figure 41: Problem.

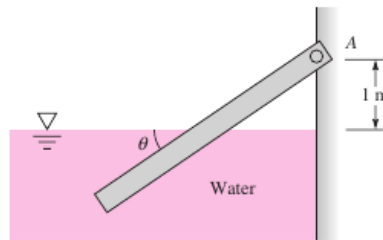


Figure 42: Problem.

13. A uniform wooden beam (SG = 0.65) is $10\text{ cm} \times 10\text{ cm} \times 3\text{ m}$ and is hinged at A , as in Fig. 42. At what angle θ will the beam float in the 20°C water?

5.2 Stability of floating bodies

5.2.1 Vertical stability

The notion of stability can be demonstrated by considering the vertical stability of a floating object. If the object is raised a small distance, the buoyant force decreases and the object's weight returns the object to its original position. Conversely, if a floating object is lowered slightly, the buoyant force increases and the larger buoyant force returns the object to its original position. Thus a floating object has vertical stability, since a small departure from equilibrium results in a restoring force.

5.2.2 Rotational stability

Consider now the rotational stability of a submerged body, shown in Fig. 43. In part (a) the center of gravity G of the body is above the centroid C (also referred to as the *center of buoyancy*) of the displaced volume, and a small angular rotation results in a moment that will continue to increase the rotation; hence the body is unstable and overturning would result. If the center of gravity is below the centroid, as in part (c), a small angular rotation provides a restoring moment and the body is stable. Part (b) shows neutral stability for a body in which the center of gravity and the centroid coincide, a situation encountered whenever the density is constant throughout the submerged body.

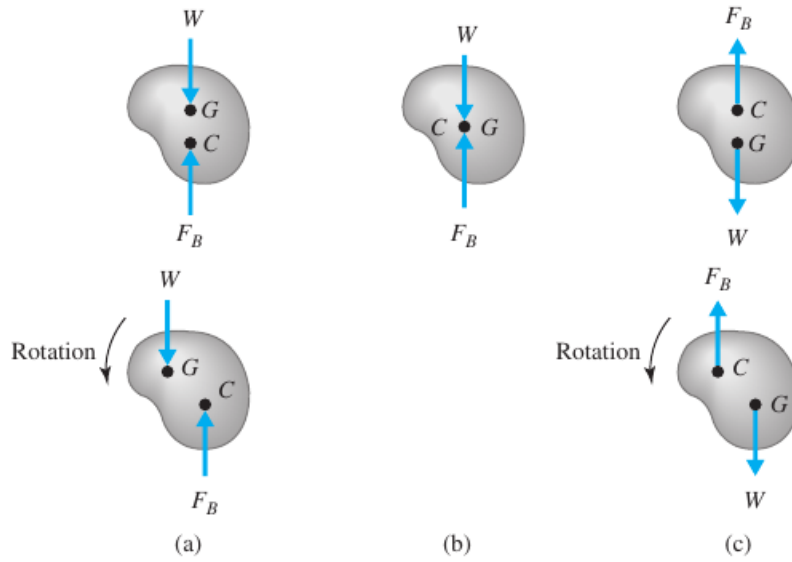


Figure 43: Stability of a submerged body: (a) unstable; (b) neutral; (c) stable. Figure taken from Ref. [4].

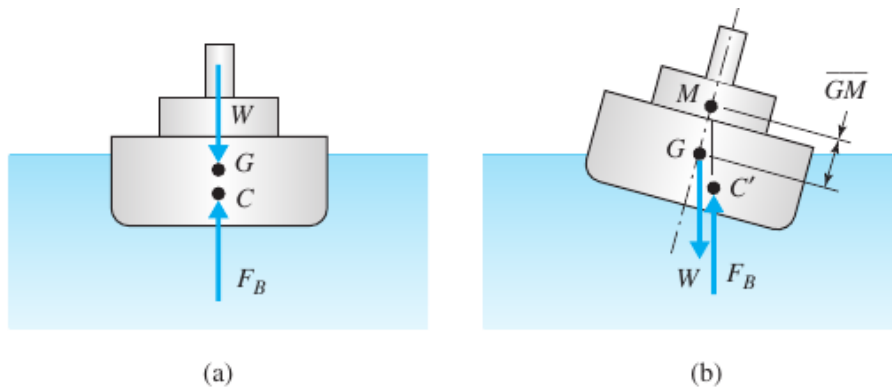


Figure 44: Stability of a floating body: (a) equilibrium position; (b) rotated position. Figure taken from Ref. [4].

Next, consider the rotational stability of a floating body. If the center of gravity is below the centroid, the body is always stable, as with the submerged body of Fig. 43 (c). The body may be stable, though, even if the center of gravity is above the centroid, as sketched in Fig. 44 (a). When the body rotates the centroid of the volume of displaced liquid moves to the new location C' , shown in part (b). If the centroid C' moves sufficiently far, a restoring moment develops and the body is stable, as shown. This is determined by the *metacentric height* GM defined as the distance from G to the point of intersection of the buoyant force before rotation with the buoyant force after rotation. If $\overline{GM}$ is positive, as shown, the body is stable; if $\overline{GM}$ is negative (M lies below G), the body is unstable.

6 Surface tension

6.1 Basic physics of surface tension

From Archimedes's principle we know that if an object is less dense than water, it will float partially submerged. But a paper clip can rest atop a water surface even though its density is several times that of water. This is an example of surface tension: The surface of the liquid behaves like a membrane under tension (see Fig. 45).



Figure 45: The surface of the water acts like a membrane under tension, allowing this water strider to “walk on water”. Figure taken from Ref. [8].

A liquid, being unable to expand freely, will form an interface with a second liquid or gas. Surface tension arises because the molecules of the liquid exert attractive forces on each other. There is zero net force on a molecule within the interior of the liquid, but a surface molecule is drawn into the interior (see Fig. 46). Thus the liquid tends to minimize its surface area, just as a stretched membrane does. As such, surface tension manifests itself only in liquids at an interface, usually a liquid-gas interface.

- Surface tension explains why raindrops are spherical (not teardrop-shaped): a sphere has a smaller surface area for its volume than any other shape. This force holds a drop of water suspended on a rod and limits the size of the drop that may be held.
- It also explains why hot, soapy water is used for washing. To wash clothing thoroughly, water must be forced through the tiny spaces between the fibers. This requires increasing the surface area of the water, which is difficult to achieve because of surface tension. The job is made easier by increasing the temperature of the water and adding soap, both of which decrease the surface tension.
- Surface tension is important for a millimeter-sized water drop, which has a relatively large surface area for its volume. (A sphere of radius r has surface area $4\pi r^2$ and volume $(4\pi/3)r^3$. The ratio of surface area to volume is $3/r$, which increases with decreasing radius.) But for large quantities of liquid, the ratio of surface area to volume is relatively small, and surface tension is negligible compared to pressure forces.
- It may also play a significant role when two immiscible liquids (e.g., oil and water) are in contact with each other.
- Although surface tension is present at all interfaces, it is most important for small fluid bodies (at the human scale). At length scales where surface tension does come into play, the shape of an interface bears little relation to the gravitational equipotential surfaces that dominate the static shapes of large-scale systems.

Molecules in a liquid are attracted by neighboring molecules.

At the surface, the unbalanced attractions cause the surface to resist being stretched.

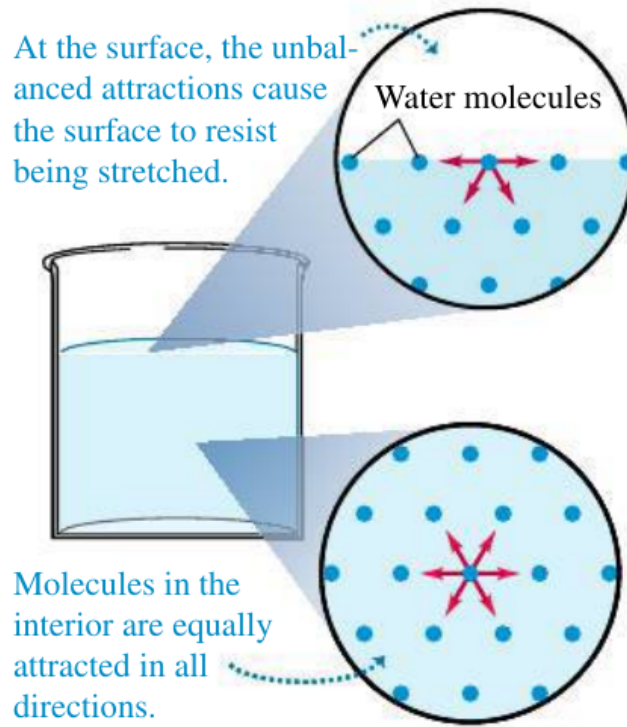


Figure 46: A molecule at the surface of a liquid is attracted into the bulk liquid, which tends to reduce the liquid's surface area. Figure taken from Ref. [8].

Surface tension has units of force per unit length, N/m (notice that $\text{J m}^{-2} = \text{kg s}^{-2}$). The two most common interfaces are water–air and mercury–air. For a clean surface at $20^\circ\text{C} = 68^\circ\text{F}$, the measured surface tension is

$$\alpha = \left\{ \begin{array}{ll} 2.13 \times 10^{-2} \text{ N/m}, & \text{Ammonia,} \\ 2.88 \times 10^{-2} \text{ N/m}, & \text{Benzene,} \\ 2.28 \times 10^{-2} \text{ N/m}, & \text{Ethanol,} \\ 4.84 \times 10^{-2} \text{ N/m}, & \text{Ethylene glycol,} \\ 2.16 \times 10^{-2} \text{ N/m}, & \text{Gasoline,} \\ 6.30 \times 10^{-2} \text{ N/m}, & \text{Glycerin,} \\ 4.84 \times 10^{-1} \text{ N/m}, & \text{Mercury,} \\ 2.23 \times 10^{-2} \text{ N/m}, & \text{Methanol,} \\ 7.28 \times 10^{-2} \text{ N/m}, & \text{Water,} \\ 7.28 \times 10^{-2} \text{ N/m}, & \text{Seawater (30\%).} \end{array} \right. \quad (301)$$

These are design values and can change considerably if the surface contains contaminants like detergents or slicks. Generally α decreases with liquid temperature and is zero at the critical

point (see Fig. 47).

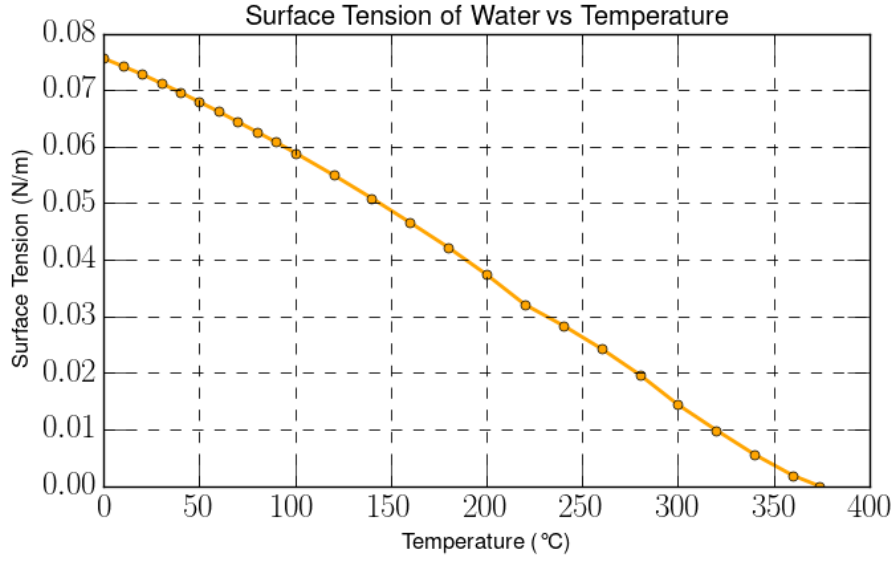


Figure 47: Surface tension of a clean air–water interface. Data from Table 7.

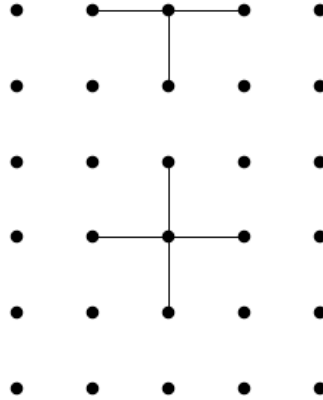


Figure 48: Cross-section of a primitive three-dimensional model of a material interfacing to vacuum. A molecule at the surface has only five bonds compared to the six that link a molecule in the interior. The missing bond energy at the surface gives rise to a positive surface energy density, which is the cause of surface tension. Figure taken from Ref. [3].

6.1.1 Molecular origin of surface energy

Let's consider a primitive three-dimensional model of a material in which the molecules are placed in a cubic grid with grid length L_{mol} (see Fig. 48). Each molecule in the interior has six bonds to its neighbours with a total binding energy of $-\epsilon$, but a surface molecule will only have five bonds when the material interfaces with a vacuum. The (negative) binding energy of the missing bond is equivalent to an extra positive energy $\epsilon/6$ for a surface molecule relative to an interior molecule, and thus an extra surface energy density,

$$\alpha \approx \frac{1}{6} \frac{\epsilon}{L_{\text{mol}}^2}. \quad (302)$$

The binding energy may be estimated from the specific enthalpy of evaporation h of the material as $\epsilon \approx hM_{\text{mol}}/N_A$. For water, the specific evaporation enthalpy is $h \approx 2.2 \times 10^6 \text{ J kg}^{-1}$, leading to the estimate $\alpha \approx 0.12 \text{ J m}^{-2}$. The measured value of the surface energy for the water/air interface is in fact $\alpha \approx 0.073 \text{ J m}^{-2}$ at room temperature.

6.2 Pressure change across a curved interface due to surface tension

The force due to surface tension results from a length multiplied by the surface tension; the length to use is the length of fluid in contact with a solid. For instance, the circumference in the case of a bubble.

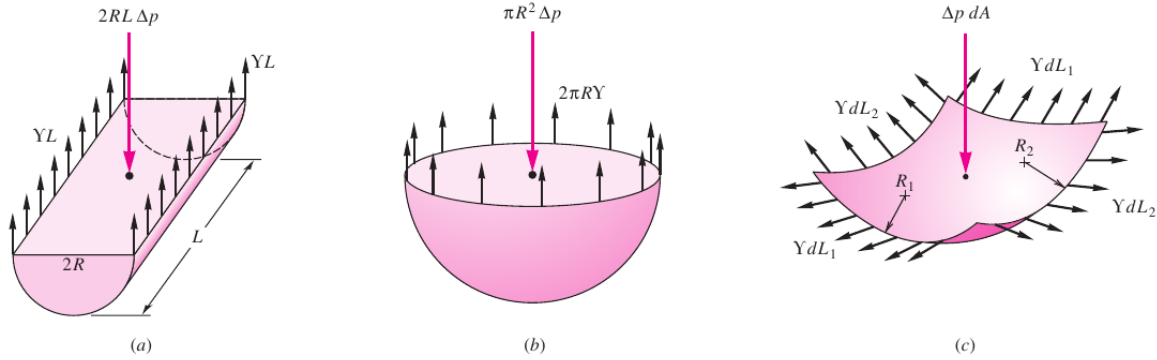


Figure 49: Pressure change across a curved interface due to surface tension (here denoted as Υ): (a) interior of a liquid cylinder; (b) interior of a spherical droplet; (c) general curved interface. Figure taken from Ref. [5].

If the interface is curved, a mechanical balance shows that there is a pressure difference across the interface, the pressure being higher on the concave side as illustrated in Fig. 49. For a liquid cylinder, the pressure increase in the interior is balanced by two surface-tension forces (we are not considering the weight of the liquid in this calculation):

$$2RL \Delta p = 2\alpha L \quad (303)$$

or

$$\Delta p = \frac{\alpha}{R}. \quad (304)$$

For a spherical droplet, the pressure increase in the interior balances a ring of surface-tension force:

$$\pi R^2 \Delta p = 2\pi R \alpha \quad (305)$$

or

$$\Delta p = \frac{2\alpha}{R}. \quad (306)$$

Review question. From the result for a spherical droplet we can predict the pressure increase inside a soap bubble of radius R :

- a) $\Delta p_{\text{bubble}} \approx \Delta p_{\text{droplet}} = \frac{2\alpha}{R}$.
- b) $\Delta p_{\text{bubble}} \approx 2 \Delta p_{\text{droplet}} = \frac{4\alpha}{R}$.
- c) $\Delta p_{\text{bubble}} \approx \frac{1}{2} \Delta p_{\text{droplet}} = \frac{\alpha}{R}$.
- d) $\Delta p_{\text{bubble}} \approx 4 \Delta p_{\text{droplet}} = \frac{8\alpha}{R}$.

Panel c) of Fig. 49 shows the general case of an arbitrarily curved interface whose principal radii of curvature are R_1 and R_2 . A force balance normal to the surface will show that the pressure increase on the concave side is

$$\Delta p = \alpha \left(\frac{1}{R_1} + \frac{1}{R_2} \right). \quad (307)$$

Previous results can all be derived from this general relation; for example, in the cylinder case $R_1 = R$ and $R_2 = \infty$.

6.3 Work and surface energy density

Increasing the area of the interface by a tiny amount dA takes an amount of work equal to the surface energy contained in the extra piece of surface,

$$dW = \alpha dA. \quad (308)$$

This is quite analogous to the mechanical work $dW = -p dV$ performed against pressure when the volume of the system is expanded by dV . But whereas a volume expansion under positive pressure takes negative work, increasing the surface area takes positive work. This resistance against extension of the surface shows that the interface has a permanent internal tension, called *surface tension*, which we shall now see equals the energy density α .

Formally, surface tension is defined as the force per unit of length that acts orthogonally to an imaginary line drawn on the interface. Suppose we wish to stretch the interface along a straight line of length L by a uniform amount ds (see Fig. 50). Since the area is increased by $dA = L ds$, it takes an amount of work $dW = \alpha L ds$, implying that the force acting orthogonally to the line is $\mathcal{F} = \alpha L$, or $\mathcal{F}/L = \alpha$. Surface tension is thus identical to the surface energy density. This is also reflected in the equality of the natural units for the two quantities, $\text{N m}^{-1} = \text{J m}^{-2}$.

Formally, surface tension is defined in much the same way as pressure was defined. Let an oriented open surface be divided into two parts by an oriented curve, such that the surface has a uniquely defined left- and right-hand side with respect to the curve (see Fig. 51). If $\hat{n}$ denotes the normal to the surface, then

$$d\vec{\mathcal{F}} = \alpha d\vec{\ell} \times \hat{n} \quad (309)$$

is the force that the right-hand side of the surface exerts on the left-hand side through the curve element $d\vec{\ell}$. The total force is obtained by integrating this expression along the curve. It follows immediately that to shift a piece of the curve of length L by the amount ds orthogonally to the curve, the work is given by $dW = \alpha dA$ with $dA = L ds$.

- The surface energy density associated with a liquid or solid interface against vacuum (or gas) is always positive because of the missing negative binding energy of the surface molecules.

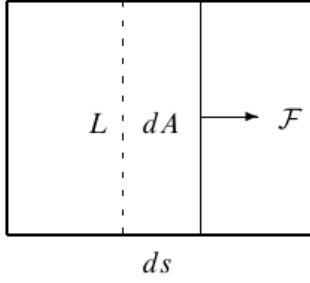


Figure 50: An external force $\mathcal{F}$ performs the work $dW = \mathcal{F}ds$ to stretch the surface by ds . Since the area increase is $dA = Lds$, the force is $\mathcal{F} = \alpha L$. The force per unit of length, $\alpha = \mathcal{F}/L$, is the surface tension. Figure taken from Ref. [3].

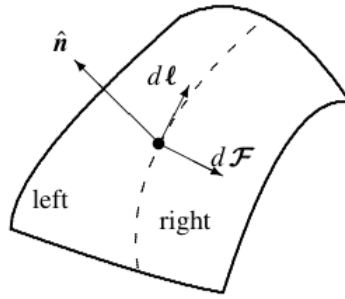


Figure 51: A piece of an oriented surface divided by an oriented curve (dashed). The cross-product of the curve element $d\vec{\ell}$ and the surface normal $\hat{n}$ determines the direction of the force with which the piece of the surface on the right acts on the piece on the left across the curve element. Figure taken from Ref. [3].

- Interfaces between solids and liquids or between liquids and liquids are not required to have positive interfacial energy density. The sign depends on the strength of the cohesive forces holding molecules of a material together compared to the strength of the adhesive forces between the opposing molecules of the interfacing materials.
- The interface between a solid and liquid is normally not deformable, and a negative interfacial energy density has no dramatic effect.
- If, on the other hand, the interfacial energy density between two liquids is negative, a large amount of energy can be released by maximizing the area of the interface. Folding the interface like crumpled paper, the two fluids are mixed thoroughly instead of being kept separate.
- Fluids that readily mix with each other, such as alcohol and water, may be viewed as having negative interfacial energy density, although the concept is not particularly well-defined in this case.
- Immiscible fluids like oil and water must, on the other hand, have positive interfacial energy density, which makes them seek toward minimal interface area with maximal smoothness.

- Since an interface has no macroscopic thickness, it may be viewed as being locally flat everywhere, implying that the surface energy density cannot depend on the curvature of the interface, but only on the properties of the interfacing materials. If these are homogeneous and isotropic—as they often are—the value of the energy density will be the same everywhere on the interface. We focus on hydrostatics of homogeneous materials and assume that surface tension takes the same value everywhere on a static interface (with the exception of soap films).
- At an interface between homogeneous materials, surface tension does not depend on how much the interface has already been stretched, and this makes the interface quite different from an elastic membrane that like the skin of a balloon, resists stretching with increasing force because elastic tension increases as the deformation grows. Soap films and biological membranes are exceptions and behave like elastic membranes because of their peculiar surfactant molecules. That is in fact what accounts for their great stability.

6.3.1 Pressure excess in a sphere

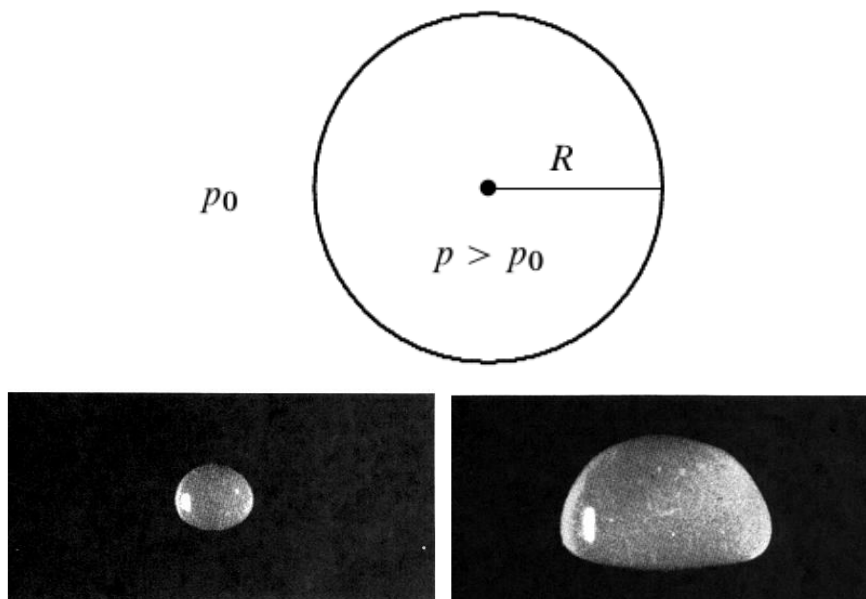


Figure 52: Top: Surface tension makes the pressure higher inside a spherical droplet or bubble. Bottom: Steadily falling raindrops are subject to the combined effects of surface tension, gravity, friction, and air currents. Left: Small raindrops are dominated by surface tension and therefore nearly spherical. Here the drop radius is slightly larger than the capillary length of 2.7 mm and the drop is slightly oval. Right: Larger raindrops assume a typical “hamburger” shape. Figure taken from Ref. [3].

Consider a spherical ball of liquid of radius a , for example hovering weightlessly in a spacecraft. Surface tension will attempt to contract the ball but is stopped by the build-up of an extra pressure Δp inside the liquid. If we increase the radius by an amount da , we must perform the work $dW_1 = \alpha dA = \alpha d(4\pi a^2) = \alpha 8\pi a da$ against surface tension. This work is compensated for by the thermodynamic work against the pressure excess $dW_2 = -\Delta p dV = -\Delta p 4\pi a^2 da$. In equilibrium

there should be nothing to gain, $dW_1 + dW_2 = 0$, leading to

$$\Delta p = \frac{2\alpha}{a}. \quad (310)$$

The pressure excess is inversely proportional to the radius of the sphere. It should be emphasized that the pressure excess is equally valid for a spherical raindrop in air and a spherical air bubble in water.

Example: Raindrops. A spherical raindrop of diameter 1 mm has an excess pressure of only about 300 Pa, which is tiny compared to atmospheric pressure (10^5 Pa). In clouds, raindrops are tiny with diameter about $1 \mu\text{m}$, implying a pressure excess a thousand times larger, about 3 atm. Falling raindrops are deformed by the interaction with the air and never take the familiar teardrop shape with a pointed top and a rounded bottom. Small raindrops with radius up to about 2 mm are nearly perfect spheres, but for larger radius they become increasingly flattened (see Fig. 52). Eventually, when the radius exceeds 4.5 mm, a raindrop will break up into smaller drops due to the interaction with the air.

6.3.2 Capillary length

When can we disregard the influence of gravity on the shape of a raindrop? For a spherical air bubble or raindrop of radius a , the condition is that the change in hydrostatic pressure across the drop should be negligible compared to the pressure excess due to surface tension, i.e. $\rho_0 g_0 2a \ll 2\alpha/a$, where ρ_0 is the density of water (minus the negligible density of air). Consequently, we require

$$a \ll R_c = \sqrt{\frac{\alpha}{\rho_0 g_0}}, \quad (311)$$

where the critical radius R_c is called the *capillary length*. It equals 2.7 mm for an air–water interface at 25°C and 1.9 mm for mercury.

6.4 Young-Laplace law

A smooth surface may, at a given point, be intersected with an infinite number of planes containing the normal to the surface (see Fig. 53). In each normal plane the intersection is a smooth planar curve which at the given point may be approximated by a circle centred on the normal. The centre of this circle is called the *centre of curvature* and its radius the *radius of curvature* of the intersection. Usually the radius of curvature is given a sign depending on which side of the surface the centre of curvature is situated. As the intersection plane is rotated, the centre of curvature moves up and down the normal between extreme values R_1 and R_2 of the signed radius of curvature, called the *principal radii of curvature*. The corresponding principal intersection planes are orthogonal and the radius of curvature along any other normal intersection can be calculated from the principal radii.

Consider now a small rectangle with sides $d\ell_1$ and $d\ell_2$ aligned with the principal directions in a given point of the surface, and let to begin with the principal radii of curvature, R_1 and R_2 , be both positive (see Fig. 54). Surface tension pulls at this rectangle from all four sides, but the curvature of the surface makes each of these forces not quite parallel with the tangent plane. In the 1-direction, surface tension acts with two nearly opposite forces of magnitude $\alpha d\ell_2$, each forming a tiny angle of magnitude $\frac{1}{2}d\ell_1/R_1$ with the tangent to the surface in the 1-direction.

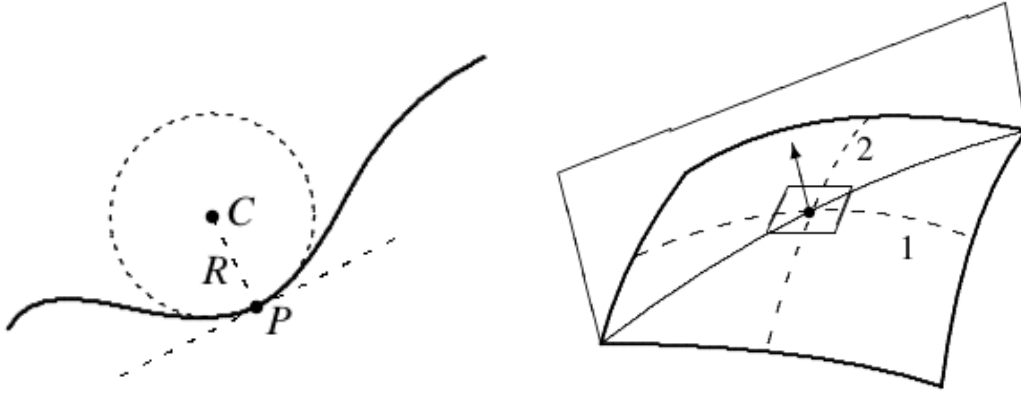


Figure 53: Left: The local geometric properties of a smooth planar curve (fully drawn) near a given point are approximated by the tangent (large dashes) and the osculating circle (small dashes) having the same tangent as the curve. Right: A plane containing the normal in a point of a surface intersects the surface in a planar curve. The extreme values of the signed radius of curvature define the principal directions (dashed). The small rectangle has sides parallel with the principal directions. Figure taken from Ref. [3].

Projecting both of these forces on the normal, we calculate the total force in the direction of the center of curvature C_1 as $dF = 2 \times \alpha dl_2 \times \frac{1}{2} dl_1 / R_1 = \alpha dA / R_1$, where $dA = dl_1 dl_2$ is the area of the rectangle. Dividing by dA we obtain the excess in pressure $\Delta p = \alpha / R_1$ on the side of the surface containing the center of curvature. Finally, adding the contribution from the 2-direction we arrive at the *Young-Laplace law* for the pressure discontinuity due to surface tension:

$$\Delta p = \alpha \left(\frac{1}{R_1} + \frac{1}{R_2} \right). \quad (312)$$

Exercise. Consider a quadratic surface

$$z = ax^2 + by^2 + 2cxy$$

with a unique extremum at $x = y = 0$. For a suitable choice of coordinates, a smooth function may always be approximated with such a function at any given point.

- Determine the radius of curvature of the surface along a line in the xy -plane forming an angle ϕ with the x -axis.
- Determine the extrema of the radius of curvature as a function of ϕ and show that they correspond to orthogonal directions.

6.5 Contact angle

A second important surface effect is the *contact angle* θ , which appears when a liquid interface intersects with a solid surface, as in Fig. 55. The force balance would then involve both α and θ . If the contact angle is less than 90° , the liquid is said to *wet* the solid; if $\theta > 90^\circ$, the liquid is termed *nonwetting*.

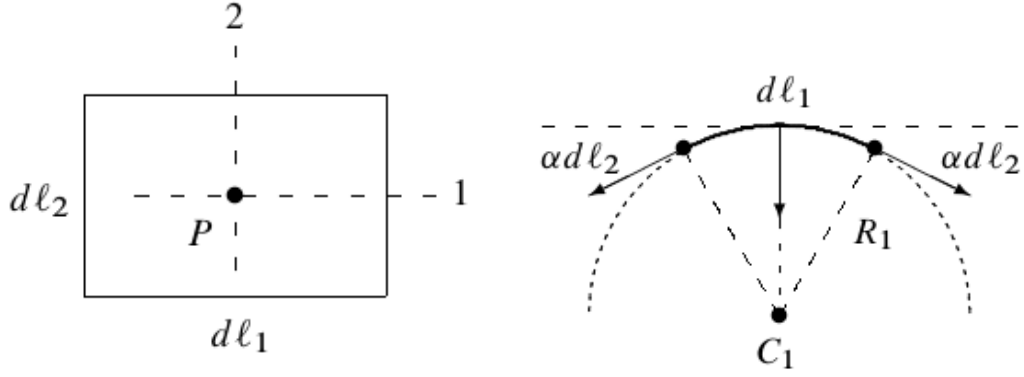


Figure 54: Left: Tiny rectangle aligned with the principal directions, seen from “above”. The sides are $d\ell_1$ and $d\ell_2$. Right: The rectangular piece of the surface with sides $d\ell_1$ and $d\ell_2$ is subject to two tension forces along the 1-direction resulting in a normal force pointing toward the center of curvature C_1 . The tension forces in the 2-direction contribute analogously to the normal force. Figure taken from Ref. [3].

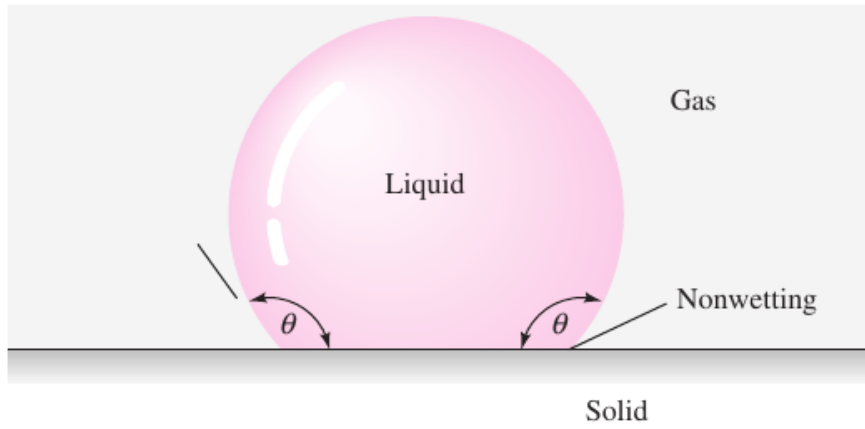


Figure 55: Contact-angle effects at liquid–gas–solid interface. If $\theta < 90^\circ$, the liquid “wets” the solid; if $\theta > 90^\circ$, the liquid is nonwetting. Figure taken from Ref. [5].

- The contact angle depends strongly on surface properties, fluid composition, and additives.
- Surfactants reduce both surface tension and contact angle, improving wetting (e.g., dish-washing liquids).
- Waxes increase the contact angle, making water bead up and reducing corrosion (as in car surfaces).

For example, water wets soap but does not wet wax. Water is extremely wetting to a clean glass surface, with $\theta \approx 0^\circ$. Like α , the contact angle θ is sensitive to the actual physico-chemical conditions of the solid–liquid interface. For a clean mercury–air–glass interface, $\theta \approx 140^\circ$.

According to **Young’s law**, the equilibrium at the contact line (see Fig. 56) satisfies:

$$\alpha_{sg} = \alpha_{sl} + \alpha \cos \chi, \quad (313)$$

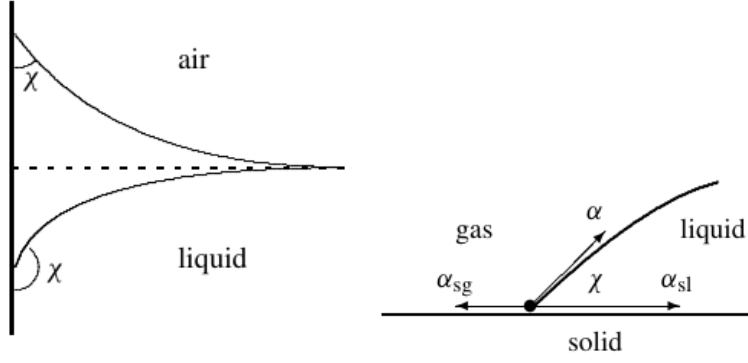


Figure 56: Fig. 1.12 Contact-angle effects at liquid–gas–solid interface. If $\theta < 90^\circ$, the liquid “wets” the solid; if $\theta > 90^\circ$, the liquid is nonwetting. Figure taken from Ref. [5].

leading to

$$\cos \chi = \frac{\alpha_{sg} - \alpha_{sl}}{\alpha}, \quad (314)$$

where α_{sg} and α_{sl} are the solid/gas and solid/liquid surface tensions, respectively.

For equilibrium contact angles:

- $0 < \alpha_{sg} - \alpha_{sl} < \alpha \Rightarrow$ mostly wetting ($0 < \chi < 90^\circ$),
- $-\alpha < \alpha_{sg} - \alpha_{sl} < 0 \Rightarrow$ mostly non-wetting ($90^\circ < \chi \leq 180^\circ$),
- $\alpha_{sg} - \alpha_{sl} > \alpha \Rightarrow$ completely wetting,
- $\alpha_{sg} - \alpha_{sl} < -\alpha \Rightarrow$ completely non-wetting.

6.5.1 Capillary effect

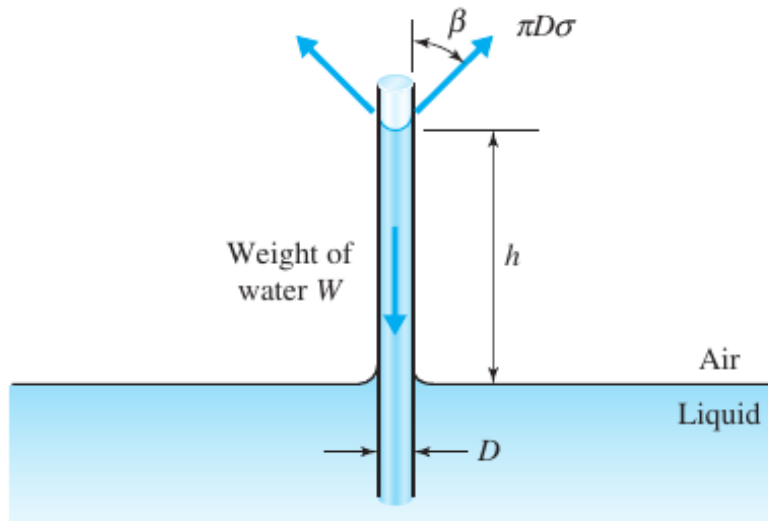


Figure 57: Rise in a capillary tube. Figure taken from Ref. [4].

Figure 57 shows the rise of a liquid in a clean glass capillary tube due to surface tension. The liquid makes a contact angle β with the glass tube. Experiments have shown that this angle for water and most liquids in a clean glass tube is zero. There are also cases for which this angle is greater than 90° (e.g., mercury); such liquids have a *capillary drop*. If h is the capillary rise, D the diameter, ρ the density, and σ the surface tension, h can be determined from equating the vertical component of the surface tension force to the weight of the liquid column.

Water has a well-known tendency to rise above the ambient level in a narrow vertical glass tube which is lowered into the liquid. Closer inspection reveals that the surface inside the tube is concave. This is called the *capillary effect* and is caused by the acute contact angle of water in conjunction with its surface tension which creates a negative pressure just below the liquid surface, balancing the weight of the raised water column. Mercury with its obtuse contact angle displays instead a convex surface shape, creating a positive pressure just below the surface which forces the liquid down to a level where the pressure equals that at the ambient level.

Let us first calculate the effect for an acute angle of contact. At the centre of the tube the radii of curvature are equal, and since the centre of curvature lies outside the liquid, they are also negative, $R_1 = R_2 = -R_0$ where R_0 is positive. Hydrostatic balance at the centre of the tube then takes the form $\rho_0 g_0 h = 2\alpha/R_0$ where h is the central height, such that

$$h = \frac{2\alpha}{\rho_0 g_0 R_0} = 2 \frac{R_c^2}{R_0}. \quad (315)$$

It should be noted that this is an exact relation which does not depend on the surface being spherical. It also covers the case of an obtuse contact angle by taking R_0 to be negative.

When the tube radius a is small compared to R_c , gravity has no effect on the shape, and the surface may be assumed to be a sphere of radius R_0 . A simple geometric construction shows that $a = R_0 \cos \chi$, and thus,

$$h = 2 \frac{R_c^2}{a} \cos \chi. \quad (316)$$

It is as expected positive for acute and negative for obtuse contact angles. From the same geometry it also follows that the depth of the central point of the surface is $d = R_0(1 - \sin \chi)$, or

$$d = a \frac{1 - \sin \chi}{\cos \chi}. \quad (317)$$

Both of these expressions are modified for larger radius, $a \gtrsim R_c$, where the surface flattens.

Example. A capillary tube has diameter $2a = 1$ mm. Water with $\chi \approx 0$ rises $h = +30$ mm with a surface depth $d = +0.5$ mm. Mercury with contact angle $\chi \approx 140^\circ$ sinks on the other hand to $h = -11$ mm and $d = -0.2$ mm under the same conditions.

6.5.2 Capillary Action in Microgravity

On Earth, gravity dominates over many fluid effects, but in microgravity a different picture emerges. This animation ([Surface Tension Demonstration Video](#)) shows a two-channel apparatus partially filled with silicone oil being dropped. While in free-fall, the liquid experiences microgravity conditions and the height of the fluid in the two connected channels changes. The oil meniscus climbs up the walls of the tubes thanks to capillary action. This is the result of intermolecular forces between the liquid and solid walls. Capillary action is most effective in narrow

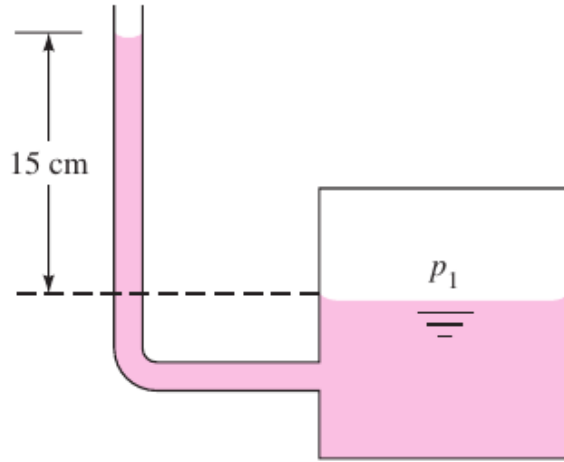


Figure 58: Problem.

tubes where surface tension and the adhesion between the liquid and solid can actually propel liquid up the walls, as seen here. On Earth we mostly ignore capillary action, except in very small spaces, but for space systems, it is a major force to reckon with in designing flows.

6.5.3 Problems

1. The system in Fig. P1.65 is used to calculate the pressure p_1 in the tank by measuring the 15-cm height of liquid in the 1-mm-diameter tube. The fluid is at 60°C . Calculate the true fluid height in the tube and the percentage error due to capillarity if the fluid is (a) water or (b) mercury.
2. A thin wire ring, 3 cm in diameter, is lifted from a water surface at 20°C . Neglecting the wire weight, what is the force required to lift the ring? Is this a good way to measure surface tension? Should the wire be made of any particular material?
3. A vertical concentric annulus, with outer radius r_o and inner radius r_i , is lowered into a fluid of surface tension Υ and contact angle $\theta < 90^\circ$. Derive an expression for the capillary rise h in the annular gap if the gap is very narrow.
4. Make an analysis of the shape $\eta(x)$ of the water–air interface near a plane wall, as in Fig. 59, assuming that the slope is small, $R^{-1} \approx d^2\eta/dx^2$. Also assume that the pressure difference across the interface is balanced by the specific weight and the interface height, $\Delta p \approx \rho g \eta$. The boundary conditions are a wetting contact angle θ at $x = 0$ and a horizontal surface $\eta = 0$ as $x \rightarrow \infty$. What is the maximum height h at the wall?
5. A solid cylindrical needle of diameter d , length L , and density ρ_n may float in liquid of surface tension α . Neglect buoyancy and assume a contact angle of 0° . Derive a formula for the maximum diameter $d_{\max}$ able to float in the liquid. Calculate $d_{\max}$ for a steel needle (SG = 7.84) in water at 20°C .
6. Derive an expression for the capillary height change h for a fluid of surface tension α and contact angle θ between two vertical parallel plates a distance W apart, as in Fig. 60. What will h be for water at 20°C if $W = 0.5$ mm?

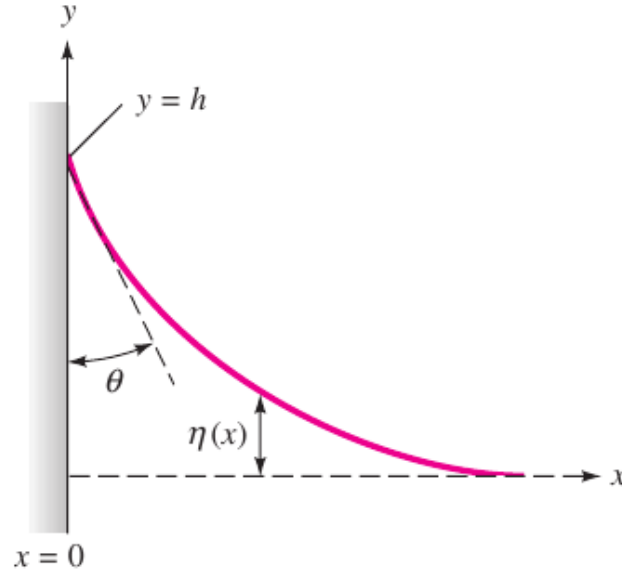


Figure 59: Problem.

7. A soap bubble of diameter D_1 coalesces with another bubble of diameter D_2 to form a single bubble D_3 with the same amount of air. Assuming an isothermal process, derive an expression for finding D_3 as a function of D_1 , D_2 , p_{atm} , and α .
8. Find an expression for the rise of liquid between two parallel plates a distance t apart. Use a contact angle β and surface tension α .
9. Write an expression for the maximum diameter d of a needle of length L that can float in a liquid with surface tension α . The density of the needle is ρ .
10. Would a 7-cm-long 4-mm-diameter steel needle be able to float in 15°C water? Use $\rho_{\text{steel}} = 7850 \text{ kg/m}^3$.
11. Find an expression for the maximum vertical force F needed to lift a thin wire ring of diameter D slowly from a liquid with surface tension α .

Appendix

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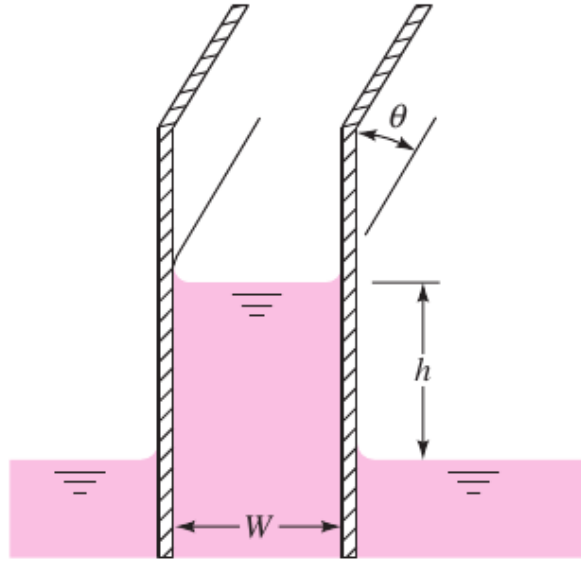


Figure 60: Problem.

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Table 6: Density ρ , pressure p , and gravitational acceleration g as a function of depth within the Earth (after Anderson & Hart, 1976; Carmichael, 1989).

Depth (km)	ρ (g cm ⁻³)	p (kbar)	g (cm s ⁻²)
Crust			
0	1.02	0	981
3	1.02	3	982
21	2.80	5	983
Mantle (solid)			
21	3.49	5	983
41	3.51	12	983
61	3.52	19	984
81	3.48	26	984
101	3.44	33	984
121	3.40	39	985
171	3.37	56	987
221	3.34	73	989
271	3.37	89	991
321	3.47	106	993
371	3.59	124	994
571	3.95	199	999
871	4.54	328	997
1171	4.67	466	992
1471	4.81	607	991
1771	4.96	752	994
2071	5.12	903	1002
2371	5.31	1061	1017
2671	5.45	1227	1042
2886	5.53	1352	1069
Outer core (liquid)			
2886	9.96	1352	1069
2971	10.09	1442	1050
3371	10.63	1858	953
3671	11.00	2154	874
4071	11.36	2520	760
4471	11.69	2844	641
4871	11.99	3116	517
5156	12.12	3281	427
Inner core (solid)			
5156	12.30	3281	427
5371	12.48	3385	355
5771	12.52	3529	218
6071	12.53	3592	122
6371	12.58	3617	0

Table 7: Surface tension, vapor pressure, and sound speed of water. Data taken from Ref. [5].

$T, ^\circ\text{C}$	$\sigma, \text{N/m}$	p_v, kPa	$a, \text{m/s}$
0	0.0756	0.611	1402
10	0.0742	1.227	1447
20	0.0728	2.337	1482
30	0.0712	4.242	1509
40	0.0696	7.375	1529
50	0.0679	12.34	1542
60	0.0662	19.92	1551
70	0.0644	31.16	1554
80	0.0626	47.35	1554
90	0.0608	70.11	1543
100	0.0589	101.3	1543
120	0.0550	198.5	1518
140	0.0509	361.3	1483
160	0.0466	617.8	1440
180	0.0422	1002	1389
200	0.0374	1554	1338
220	0.0321	2318	1268
240	0.0284	3344	1190
260	0.0243	4683	1112
280	0.0197	6412	1020
300	0.0144	8851	920
320	0.0099	11,274	820
340	0.0056	14,586	630
360	0.0019	18,651	370
374*	0.0*	22,090*	0*

*Critical point.